

Phenix Overview and General Reference (Part 2 of 5)

20 documentation pages

Generated from local Phenix documentation

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Merging two models with combine_models

combine_models.html

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Purpose

This routine takes a model and finds pieces of a second model file that improve its fit to density when they replace the corresponding pieces in the first model.

Usage

The main uses of phenix.combine_models are:

- Taking the best parts of 2 models and merging them to make a new model.
- Replacing a segment in one model with a corresponding segment from another.

How combine_models works:

The phenix.combine_models tool can combine the best parts of two or more models. You can use it in either of two different ways. One approach keeps segments in the models intact, and the second applies crossovers within segments.

In the first approach (`merge_by_segment_correlation=True`), each fragment in each model is scored based on correlation to the map. The scoring also includes a weight based on the square root of the length of the segment. It also includes weights based on whether a segment has been assigned to the sequence and the number of secondary structure hydrogen bonds in that segment. This approach can combine chains of different types as well. A final increment is added to the score for a segment if it is considered very likely to be correct with different thresholds for chains of different types.

Then the segments are picked in rank order to create a composite model. Any part of a segment that does not overlap an existing part of the composite model is kept. If symmetry is present, overlaps include that symmetry.

In the second approach, (`merge_by_segment_correlation=False`), a working model is created by taking all of the first model supplied and filling in any empty regions with fragments from other models. Then one by one, the segments in the working model are recombined with all other segments. To carry out recombination between two chains, residues that match in the two chains are identified. Then for each segment between matching pairs of residues, whichever chain has the higher correlation to the map is kept to create a composite model.

This second approach to combine_models starts with two input models. The first model is used as the default; if nothing can be found in the second model that is better than what is in the first model, then that part of the first model is kept. The second model is used as a template for improving the first model. Fragments of the second model are considered as alternatives for corresponding segments in the first model.

The fit of the models to density is used to decide which of a pair of fragments is best. In general, the correlation of model density with the map is used as the criterion. In cases where unequal numbers of residues are considered, then this correlation is weighted by the square root of the number of residues in each

case. During the optional `merge_second_model` step, the scoring is optionally based on correlation of density, or by default, based on density at the positions of the main-chain atoms in the model.

If the two input models are not in the same asymmetric unit of the crystal, then `combine_models` will move the pieces from the second model to the corresponding locations in the first model. In this way the final model has all its parts in the same place.

The second approach in `combine_models` has four main steps, each of which is optional:

- Selecting parts of the input models to consider, using atom selections. This step allows you to do things such as removing a segment from the first model, then crossing with the corresponding segment from the second model, effectively replacing that segment in a way that you specify.
- Creating as complete a starting model as possible (`merge_second_model=True`). This is done by cutting the second model into small pieces and reassembling the first model, considering these pieces along with all the segments in the first model as potential pieces of the reassembled model. The result is a new working model.
- Crossing the working model with the second input model, allowing only equal-length crossovers (`matching=True`). In this step, pairs of segments in the working model and the second input model that have overlapping residues are considered, one at a time. The local map correlation is calculated for each residue in each fragment. Then at each position, the coordinates from one fragment or the other are chosen, with the choice crossing over between one fragment and the other only at positions where the main-chain atoms in the residue are within `match_distance` (default 0.5 Å) of each other.
- Crossing the working model with the second input model, allowing only unequal-length crossovers (`non_matching=True`). This step is just like the previous one, except that only unequal-length crossovers are considered. If this method is included, then it may be necessary to reassign the sequence afterwards as the alignment may change
- Checking the sequence. If unequal-length crossovers are made, then the sequence alignment to the model may need to be changed. This step carries that out.

Output files from `combine_models`

`combine_models.pdb`: A PDB file with your combined model.

Examples

Standard run of `combine_models`:

Running `combine_models` is easy. From the command-line you can type:

```
phenix.combine_models pdb_in=first.pdb \
  second_pdb_in=second.pdb \
  seq.dat \
  map_coeffs.mtz
```

This will combine `first.pdb` and `second.pdb` based on fit to the map from `map_coeffs.mtz`, recheck the sequence alignment to `seq.dat`, and write out the resulting model.

Ranking all fragments and picking the best ones:

```
phenix.combine_models pdb_in=model.pdb seq.dat map_coeffs.mtz merge_by_segment_correlation=True
phenix.combine_models pdb_in=model.pdb
```

```
seq.dat map_coeffs.mtz merge_by_segment_correlation=True
```

This will score each segment (fragment), then work down from best to worst, keeping any part of any segment that does not overlap with a better-scoring segment.

Selecting pieces from the two models:

To take first.pdb and then see if residues A21-A30 of second.pdb can improve it, you can type:

```
phenix.combine_models pdb_in=first.pdb \  
  second_pdb_in=second.pdb \  
  seq.dat \  
  map_coeffs.mtz \  
  second_pdb_in_atom_selection="(chain A and resid 21:30)" \  
  \
```

Replacing a specific segment:

To take first.pdb and then see if residues A21-A30 and B21-B30 can be improved by replacing them with residues C10-C20 and D10-D20 of second.pdb, you can tell combine_models to ignore residues A22-A29 and B22-B29 and to consider only residues C10-C20 and D10-D20 of second.pdb:

```
phenix.combine_models pdb_in=first.pdb \  
  second_pdb_in=second.pdb \  
  seq.dat \  
  map_coeffs.mtz \  
  pdb_in_atom_selection="(not ( (chain A or chain B) and resid 22:29) )" \  
  second_pdb_in_atom_selection="( (chain C or chain D) and resid 10:20)" \  
  \
```

Crossing two models that have entirely matched residues:

If your first.pdb and second.pdb have exactly the same residues present, and just differ in coordinates, then you might want to preserve all the connectivity by skipping the merge_second_model step, and by skipping the non_matching crossover step, and by skipping the reassignment of sequence. You can type preserve_connectivity=True as a shortcut for this:

```
phenix.combine_models pdb_in=first.pdb \  
  second_pdb_in=second.pdb \  
  seq.dat \  
  map_coeffs.mtz \  
  preserve_connectivity=True
```

Possible Problems

If you are supplying a very big map, running combine_models can require a lot of memory. In this case, you may want to run map_box with your map and a model file that contains all your models. This will box the map around the region where your models are located and make a much smaller map that will require less memory.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

input_files pdb_in = None Input starting PDB file. This model will be the default output model. Parts of the model that are better in the second model be replaced by the second model. pdb_in_atom_selection = None Any selection specified with pdb_in_atom_selection is applied to the pdb_in input model before using it. second_pdb_in = None Input second PDB file. Parts that are better in this model will replace corresponding parts of the first model. second_pdb_in_atom_selection = None Any selection specified with second_pdb_in_atom_selection is applied to the second_pdb_in input model before using it . If multiple second_pdb_in files are supplied, the same number of atom selections are expected. map_in = None CCP4 or MRC-style map file (alternative to mtz_in) mtz_in = None Input MTZ file with map coefficients map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM) seq_file = None Sequence file for sequence alignment ncs_file = None Input NCS spec file for use with segment_and_split_map. output target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb. output_files pdb_out = combine_models.pdb Output PDB file log = combine_models.log Output log file params_out = combine_models.params.eff Parameters file to rerun combine_models linkage_output_file = linkage_output.pkl Linkage dictionary output_directory = None Not used (output directory for passing to map_to_model) crystal_info chain_type = *PROTEIN DNA RNA You can specify whether to include protein, DNA, or RNA If you use merge_by_segment_correlation you can include multiple chain types and chain_type is ignored resolution = None High-resolution limit solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1. space_group = None Space group (normally read from the data file) unit_cell = None Unit Cell (normally read from the data file) map_inside_box = True Place centers of all chains inside (0,1). This is required during normal operation. use_sg_symmetry = None Not used directories output_dir = "" Output directory where files are to be written crossing_high_resolution = None Resolution used in map calculation match_distance = 1.5 Residue pairs must have rmsd of match_distance (Å) or lower to be crossed. A value between 0.5 Å and 1.5 Å is generally best. merge_by_segment_correlation = False Split all models into segments (breaking where residue numbers are not sequential), rank segments by correlation to map weighted by number of atoms, take highest-scoring non-overlapping segments. Followed by standard merge with all unused segments. keep_all_of_first_model = None Try to keep all of first model. Applies to merge_by_segment_correlation only. Does not work fully if trim_to_au is used. optimize_ncs = False Optimize NCS after merge_by_segment_correlation. NOTE: assumes that all NCS operators are allowed and may not place the resulting model in density if the NCS is not present in the density. trim_to_au = True Split up any chains that are in multiple NCS au weight_rms = 1 Weight on closest approach of fragments scc_cut = 0.90 Eliminate NCS copies with CC less than cc_cut of max weight_cc = 0.1 Weight on CC in optimizing NCS (typically 10) weight_rad_gyr = 0.1 Weight on radius of gyration in optimize_ncs n_substitutions = 3 Substitutions in optimization n_consider = 20 Substitutions in optimization applied to last n_consider fragments n_random_op...

- input_files pdb_in = None Input starting PDB file. This model will be the default output model. Parts of the model that are better in the second model be replaced by the second model. pdb_in_atom_selection = None Any selection specified with pdb_in_atom_selection is applied to the pdb_in input model before using it. second_pdb_in = None Input second PDB file. Parts that are better in this model will replace corresponding parts of the first model. second_pdb_in_atom_selection = None Any selection specified with second_pdb_in_atom_selection is applied to the second_pdb_in input model before using it . If multiple second_pdb_in files are supplied, the same number of atom selections are expected. map_in = None CCP4 or MRC-style map file (alternative to mtz_in) mtz_in = None Input MTZ file with map

coefficientsmap_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)seq_file = None Sequence file for sequence alignment ncs_file = None Input NCS spec file for use with segment_and_split_map.

- pdb_in = None Input starting PDB file. This model will be the default output model. Parts of the model that are better in the second model be replaced by the second model.

- pdb_in_atom_selection = None Any selection specified with pdb_in_atom_selection is applied to the pdb_in input model before using it.

- second_pdb_in = None Input second PDB file. Parts that are better in this model will replace corresponding parts of the first model.

- second_pdb_in_atom_selection = None Any selection specified with second_pdb_in_atom_selection is applied to the second_pdb_in input model before using it . If multiple second_pdb_in files are supplied, the same number of atom selections are expected.

- map_in = None CCP4 or MRC-style map file (alternative to mtz_in)

- mtz_in = None Input MTZ file with map coefficients

- map_coeff_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeff_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)

- seq_file = None Sequence file for sequence alignment

- ncs_file = None Input NCS spec file for use with segment_and_split_map.

- outputtarget_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- target_output_format = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- output_files pdb_out = combine_models.pdb Output PDB file log = combine_models.log Output log file params_out = combine_models_params.eff Parameters file to rerun

- combine_models linkage_output_file = linkage_output.pkl Linkage dictionary output_directory = None Not used (output directory for passing to map_to_model)

- pdb_out = combine_models.pdb Output PDB file

- log = combine_models.log Output log file

- params_out = combine_models_params.eff Parameters file to rerun combine_models

- linkage_output_file = linkage_output.pkl Linkage dictionary

- output_directory = None Not used (output directory for passing to map_to_model)

- crystal_info chain_type = *PROTEIN DNA RNA You can specify whether to include protein, DNA, or RNA If you use merge_by_segment_correlation you can include multiple chain types and chain_type is ignored resolution = None High-resolution limit solvent_content = None Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by

the macromolecule. Normally set automatically. Values go from 0 to 1. `space_group = None` Space group (normally read from the data file) `unit_cell = None` Unit Cell (normally read from the data file) `map_inside_box = True` Place centers of all chains inside (0,1). This is required during normal operation. `use_sg_symmetry = None` Not used

- `chain_type = *PROTEIN DNA RNA` You can specify whether to include protein, DNA, or RNA If you use `merge_by_segment_correlation` you can include multiple chain types and `chain_type` is ignored
- `resolution = None` High-resolution limit
- `solvent_content = None` Solvent fraction of the cell. If this is density cut out from a bigger cell, you can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- `space_group = None` Space group (normally read from the data file)
- `unit_cell = None` Unit Cell (normally read from the data file)
- `map_inside_box = True` Place centers of all chains inside (0,1). This is required during normal operation.
- `use_sg_symmetry = None` Not used
- `directoriesoutput_dir = ""` Output directory where files are to be written
- `output_dir = ""` Output directory where files are to be written
- `crossinghigh_resolution = None` Resolution used in map calculation `match_distance = 1.5` Residue pairs must have rmsd of `match_distance` (Å) or lower to be crossed. A value between 0.5 Å and 1.5 Å is generally best. `merge_by_segment_correlation = False` Split all models into segments (breaking where residue numbers are not sequential), rank segments by correlation to map weighted by number of atoms, take highest-scoring non-overlapping segments. Followed by standard merge with all unused segments. `keep_all_of_first_model = None` Try to keep all of first model. Applies to `merge_by_segment_correlation` only. Does not work fully if `trim_to_au` is used. `optimize_ncs = False` Optimize NCS after `merge_by_segment_correlation`. NOTE: assumes that all NCS operators are allowed and may not place the resulting model in density if the NCS is not present in the density. `trim_to_au = True` Split up any chains that are in multiple NCS `auweight_rms = 1` Weight on closest approach of fragments `cc_cut = 0.90` Eliminate NCS copies with CC less than `cc_cut` of `maxweight_cc = 0.1` Weight on CC in optimizing NCS (typically 10) `weight_rad_gyr = 0.1` Weight on radius of gyration in `optimize_ncsn_substitutions = 3` Substitutions in optimization `n_consider = 20` Substitutions in optimization applied to last `n_consider` fragments `sn_random_opt = 0` Try factor `max_optimize_tries = 500` Maximum optimization steps `overlap_tolerance = None` Minimum separation between N/C1'/P atoms in separate chains. Applies to `merge_by_segment_correlation` only. Default is 3 Å for C1'/P and 2 Å for N `ncs_overlap_tolerance_factor = 1.5` Overlap tolerance required for NCS-related chains. Typically larger than for chains in same NCS `au` to ensure they do not overlap `very_good_protein_score = 20.0` Score indicating in combination with `very_good_protein_cc` that an PROTEIN fragment is almost certainly correct `very_good_protein_cc = 0.75` CC indicating in combination with `very_good_protein_score` that an PROTEIN fragment is almost certainly correct `very_good_rna_score = 7.0` Score indicating in combination with `very_good_rna_cc` that an RNA fragment is almost certainly correct `very_good_rna_cc = 0.60` CC indicating in combination with `very_good_rna_score` that an RNA fragment is almost certainly correct `min_fractional = 0.05` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain `min_fractional_multiple_chain_types = 0.35` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain. Applies if multiple chain types are present (otherwise `min_fractional` applies) `max_residues_in_score = 20` Maximum number of residues to use when increasing score by `sqrt(n_residues)` `weight_by_ss = False` Apply extra weight on residues with

H-bondsassignment_weight = None Apply extra weight on residues assigned to sequenceextra_weight_h_bonds = 1. Extra weight on residues with H-bondsmin_absolute = 5 Minimum residues in a chain remaining after removing parts occupied by a better-scoring chainmerge_remainder = True Merge remainder in merge_by_segment_correlationdump_remainder = False If merge_remainder is False and dump_remainder is True, dump the remainder pdb to remainder.pdbncs_in_cell_only = True Only consider NCS-related molecules that are in cellremove_only_if_centers_outside = False Remove outside cell only if centers outside cell. Modifies remove_outside_cell. Must set remove_outside_cell as well to apply.remove_outside_cell = False Remove chains with atoms outside cellmerge_segments = False Identify segments in first and second models For each gap or end in first model, try to insert or attach a segment from the second model. If selected, no other merging is done.min_segment_length = 5 Segments shorter than this will be ignored in merge_segments.merge_second_model = True Cut second model into pieces and merge them into first model. (Useful for filling in gaps in first model.)merge_both_models = False Cut first and second models into pieces and merge them (useful for combining ...

- high_resolution = None Resolution used in map calculation
- match_distance = 1.5 Residue pairs must have rmsd of match_distance (Å) or lower to be crossed. A value between 0.5 Å and 1.5 Å is generally best.
- merge_by_segment_correlation = False Split all models into segments (breaking where residue numbers are not sequential), rank segments by correlation to map weighted by number of atoms, take highest-scoring non-overlapping segments. Followed by standard merge with all unused segments.
- keep_all_of_first_model = None Try to keep all of first model. Applies to merge_by_segment_correlation only. Does not work fully if trim_to_au is used.
- optimize_ncs = False Optimize NCS after merge_by_segment_correlation. NOTE: assumes that all NCS operators are allowed and may not place the resulting model in density if the NCS is not present in the density.
- trim_to_au = True Split up any chains that are in multiple NCS au
- weight_rms = 1 Weight on closest approach of fragments
- cc_cut = 0.90 Eliminate NCS copies with CC less than cc_cut of max
- weight_cc = 0.1 Weight on CC in optimizing NCS (typically 10)
- weight_rad_gyr = 0.1 Weight on radius of gyration in optimize_ncs
- n_substitutions = 3 Substitutions in optimization
- n_consider = 20 Substitutions in optimization applied to last n_consider fragments
- n_random_opt = 0 Try factor
- max_optimize_tries = 500 Maximum optimization steps
- overlap_tolerance = None Minimum separation between N/C1'/P atoms in separate chains. Applies to merge_by_segment_correlation only. Default is 3 Å for C1'/P and 2 Å for N
- ncs_overlap_tolerance_factor = 1.5 Overlap tolerance required for NCS-related chains. Typically larger than for chains in same NCS au to ensure they do not overlap
- very_good_protein_score = 20.0 Score indicating in combination with very_good_protein_cc that an PROTEIN fragment is almost certainly correct
- very_good_protein_cc = 0.75 CC indicating in combination with very_good_protein_score that an PROTEIN fragment is almost certainly correct

- `very_good_rna_score = 7.0` Score indicating in combination with `very_good_rna_cc` that an RNA fragment is almost certainly correct
- `very_good_rna_cc = 0.60` CC indicating in combination with `very_good_rna_score` that an RNA fragment is almost certainly correct
- `min_fractional = 0.05` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain
- `min_fractional_multiple_chain_types = 0.35` Minimum fraction of a chain remaining after removing parts occupied by a better-scoring chain. Applies if multiple chain types are present (otherwise `min_fractional` applies)
- `max_residues_in_score = 20` Maximum number of residues to use when increasing score by $\sqrt{n_residues}$
- `weight_by_ss = False` Apply extra weight on residues with H-bonds
- `assignment_weight = None` Apply extra weight on residues assigned to sequence
- `extra_weight_h_bonds = 1` Extra weight on residues with H-bonds
- `min_absolute = 5` Minimum residues in a chain remaining after removing parts occupied by a better-scoring chain
- `merge_remainder = True` Merge remainder in `merge_by_segment_correlation`
- `dump_remainder = False` If `merge_remainder` is False and `dump_remainder` is True, dump the remainder pdb to `remainder.pdb`
- `ncs_in_cell_only = True` Only consider NCS-related molecules that are in cell
- `remove_only_if_centers_outside = False` Remove outside cell only if centers outside cell. Modifies `remove_outside_cell`. Must set `remove_outside_cell` as well to apply.
- `remove_outside_cell = False` Remove chains with atoms outside cell
- `merge_segments = False` Identify segments in first and second models For each gap or end in first model, try to insert or attach a segment from the second model. If selected, no other merging is done.
- `min_segment_length = 5` Segments shorter than this will be ignored in `merge_segments`.
- `merge_second_model = True` Cut second model into pieces and merge them into first model. (Useful for filling in gaps in first model.)
- `merge_both_models = False` Cut first and second models into pieces and merge them (useful for combining all pieces of both models). Alternative to `merge_second_model`.
- `trim_in_merge = False` Trim all the fragments of first and second model back to match density before merging of models. See also `remove_bad_fragments` in `merge` which removes bad fragments (after any trimming is done).
- `remove_bad_fragments_in_merge = True` Remove bad fragments in `merge`. See also `trim_in_merge` which trims back fragments before trying to merge them.
- `extend_in_merge = False` Extend fragments of first and second model during merging of models
- `fragment_length = 10` Length of pieces that second model will be cut down to in `merge_second_model` or `merge_both_models`
- `use_cc_in_combine_extend = None` You can choose to use the correlation of density rather than density at atomic positions to score models in the `merge_second_model` or `merge_both_models` step. This may be useful at lower resolution (> 3 Å)

- `matching = True` Carry out crossover using segments that match (have the same number of residues)
- `non_matching = True` Carry out crossover using segments that do not match (do not have the same number of residues)
- `check_sequence = None` After running all other procedures, redo the sequence assignment. This may be necessary if the models did not have the same sequence assignments or if unassigned pieces are now put together and can be assigned. If `None`, then `check_sequence` will be set to `True` if `non_matching=True`.
- `preserve_connectivity = None` This is a shortcut for turning off `merge_second_model`, `merge_both_models`, `non_matching`, and `check_sequence`. It is useful if your two models have the same residues, just with different coordinates, and you want to maintain the connectivity.
- `merge_only = None` This is a shortcut for turning off everything except `merge_second_model` and `merge_both_models`.
- `standard_merge = None` Merge by reading all chains and running resolve merging.
- `wang_radius = 5` Smoothing radius for solvent identification
- `cc_min = 0.20` Minimum map-model correlation to keep a segment after refinement
- `cc_ratio_min = None` Minimum map-model correlation relative to maximum to keep a segment after refinement
- `combine_models_score_min = None` Minimum score for a chain in `combine_models` (after `helices_strands`). Score is mostly defined by $\sqrt{n_{\text{atoms}}} \times \text{overall map CC}$.
- `cc_min_rna = 0.25` Minimum map-model correlation (RNA building)
- `map_inside_box = None` Map final chains inside unit cell
- `controlverbose = False` Verbose output
`nproc = 1` Number of processors to use
`*multiprocessing sge lsf pbs condor pbspro slurm` Choices are multiprocessing (single machine) or queuing systems
`queue_run_command = None` run command for queue jobs. For example `qsub`.
`raise_sorry = False` Raise sorry if problems
`return_on_failure = False` Just return on failure
`debug = False` Debugging output
`dry_run = False` Just read in and check parameter names
`i_ran_seed = 351219` random seed
`resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`;
- `verbose = False` Verbose output
- `nproc = 1` Number of processors to use
- `*multiprocessing sge lsf pbs condor pbspro slurm` Choices are multiprocessing (single machine) or queuing systems
- `queue_run_command = None` run command for queue jobs. For example `qsub`.
- `raise_sorry = False` Raise sorry if problems
- `return_on_failure = False` Just return on failure
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `i_ran_seed = 351219` random seed
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual

command in single quotes (') like this: `resolve_command_list="'no_build' 'b_overall 23' "`;

Generating alternative conformations matching a map with `create_alt_conf`

`create_alt_conf.html`

Author(s)

- `create_alt_conf`: Tom Terwilliger, Pavel Afonine, Nigel Moriarty, Dorothee Liebschner

`create_alt_conf`: Tom Terwilliger, Pavel Afonine,
Nigel Moriarty, Dorothee Liebschner

Purpose

The purpose of `create_alt_conf` is to create alternative side-chain conformations for a structure using very high-resolution X-ray data or a very high-resolution cryo-EM map as a guide.

The problem that it solves is that if you have multiple conformations present in a structure, residues that have the same altloc (A or B or C etc) should have plausible relationships, and in particular, they should not clash. Residues with different altloc (A vs B) can have any relationship as they are in different conformers. Finding the assignment of side chain conformations to conformers A, B, C etc that yields good geometry for all the conformers can be difficult.

Note that the model that is produced is not intended to be a final model, rather it is a model that has some plausible alternative conformations. It is recommended that you use the conformers in the output model as suggestions for your model.

How `create_alt_conf` works:

Generating alternative conformations:

The starting point for `create_alt_conf` is normally a model with a single conformation (no A or B altlocs).

The first step is to generate a set of plausible conformations for each side chain based on the current model and the X-ray data or cryo-EM map.

For X-ray data, a 4mFo-3DFc map is created. This map is expected to show conformations that are not represented in the model. For cryo-EM data, the input map is used.

In either case, plausible conformations for at each side chain position are found by testing each rotamer in a rotamer library for the side chain type present at that position and choosing those that fit the density. The conformer that best fits the density and that is different from the current conformer at that position is chosen. If no suitable alternative is found, the original is used. Conformers that clash with all conformers at any other position are discarded. The new side chain conformers are then used to create a new model with side chains that match the density map and that are as different as possible from the starting side chains. This model is refined using the X-ray data or cryo-EM map. The result of this step is a pair of models, the original and one that has alternative side-chain conformations. Each model has just one conformer.

If you want, you can instead supply a model with multiple conformations and it will just split that model into separate models, one for each conformation (each altloc A or B).

Optimizing assignment of side-chain conformers to models:

The key step in `create_alt_conf` is creating a set of models in which each model has good geometry and the models collectively fit the density. The starting point for this step is a set of models that have different side-chain conformations at some or all positions. The challenging part of this step is arranging the side-chain conformers in a way that minimizes clashes but that uses all the supplied side chain conformers.

The procedure used is to generate a diverse set of refined multi-conformer models, to score each model, then to find an optimized model by recombination among the multi-conformer models. The procedure is complicated somewhat by the need to select a small number of test models for scoring, as scoring requires refinement and is therefore a slow process.

Scoring function:

The scoring function used to group side-chain conformers is the Holton geometry validation score. This score is a composite of geometry restraints used in refinement and validation metrics and it is a rather good indication of the overall geometric quality of a model. The scoring function requires a refined model, as unrefined models will generally have very poor geometry.

The scoring function is also calculated on a per-residue basis. This per-residue score is the Holton geometry validation score, only including interactions that involve this residue. Note that the sum of per-residue scores calculated in this way does not equal the total score, both because of the way in which scores are calculated, and because some interactions are only within a residue and others are between residues. Even so, the sum of per-residue scores is closely related to the total score. This allows estimation of the total score from per-residue scores.

Probabilistic estimation of score for a new arrangement of side chains:

To reduce the number of models that need to be fully scored, a procedure for probabilistic estimation of the expected score for a model with a new arrangement of side chains is used. In this estimation process, it is assumed that many of the most significant contacts between residues will be with nearby residues (segments of length `group_length` are considered).

The starting point for this estimation is a set of scored models with varying arrangements of alternative conformations. These arrangements of alternative conformations are set up so that for any stretch of `group_length` residues, there will be models that match at all other positions but that differ in all possible ways at this set of `group_length` residues. The differences in per-residue scores among these models are then used to estimate the expected effect of changing the arrangement of conformers for any one residue, in the context of the conformers surrounding that residue. Additionally, an error estimate in this expected effect is estimated.

With these predictors for the effects of changes in alternative conformations at any one residue in the context of its neighbors, an estimate can be made for the expected score for a model with any arrangement of conformations. The score for the most similar model is taken as a starting value, then the minimal set of changes in arrangement is applied and the expected change in score for each change is noted, along with its uncertainty. This yields a probabilistic estimate of the expected score for this arrangement.

Recombination procedure:

The recombination procedure is targeted (not random recombination). The reason for this is that scoring requires refinement, a relatively slow process. The current set of scored models is used as described above to create a probabilistic estimate of the score that would be found for any new arrangement of side chains as

conformers in a model. A simple recombination and mutation procedure is used to find arrangements that are predicted to have good Holton geometry validation scores. The arrangements with good predicted scores are then generated, refined and scored to identify optimized arrangements that actually do have good scores.

Quick run:

You can skip the extensive optimization procedure and use a shorter one instead with the keyword `quick=True`. This is still not that fast, but it is a lot faster than the full version.

Ligands:

If you supply a single model with no alternate conformations, it may contain ligands. Any ligands are separated out and added in (one copy of each ligand for each conformer created) just before refinement.

Anisotropic refinement:

You can specify `anisotropic_refinement=True` or supply a single model previously refined with anisotropic refinement. This will instruct `create_alt_conf` to use anisotropic refinement.

Examples

Standard run of `create_alt_conf`:

Running `create_alt_conf` is easy. From the command-line you can type:

```
phenix.create_alt_conf model.pdb data.mtz
```

where `model.pdb` is the model you would like to use as a starting point and `data.mtz` is an X-ray data file. The tool will run (for a long time) and create a model named `model_overall_best.pdb` that will contain alternative conformations based on the X-ray data. If you want to use more processors, specify `nproc=32` or whatever number you would like. If you want to create more than two conformers, you can specify the number with `conformers=3` or whatever you like. If you want to use the conformers present in `model.pdb` as starting points, rearranging them as needed but not creating any new ones, you can specify `use_existing_altlocs=True` (or say nothing, this is the default). If you want to throw them away, say `use_existing_altlocs=False`.

Possible Problems

The procedure takes a long time. You would not normally want to run this on a machine with just a few processors. Running with `quick=True` is much faster, but not as comprehensive.

This procedure creates *N* full conformers (altlocs A, B, C etc), with all atoms in the macromolecule present in all the conformers. If you want to only have multiple conformations in a few places, you will need to use `phenix.pdbtools` or another method to remove multiple conformations from the rest of the structure.

This method is only suitable if you have very high-resolution data. Normally 1.5 Å is about the lowest resolution data you would want to use.

The procedure only creates alternative side-chain conformers, not main-chain. If your main chain has substantial alternative conformations (not just slight adjustments to match the side chain conformers), you will need to use another approach.

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesxray_data = None Data file with experimental data (FP SIGFP or I SIGI).
xray_data_labels = None Optional labels for X-ray data Normally these would be something like I,SIGI or F,SIGF
free_r_labels = None Optional labels for free_r flags. Normally these would be something like FreeR_flags or R_free_flags
free_r_flag_value = 0 FreeR flag value
ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
map_coeffs = None MTZ file with coefficients for a map
use_map_coeffs_if_optional = False Use map_coeffs (a map) instead of Fobs, sigFobs if both are available
map_coeffs_virtual = None Used internally
map_coeffs_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeffs_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
alt_models = None Set of alternate conformations as individual model files. You can supply alternate conformations with alt_models, or you can supply a single model file with multiple conformations and specify
use_existing_altlocs=True.
fixed_solvent_model = None Model with solvent to be used as is (no refinement). NOTE: set water_and_hydrogens will be set to always if you use a fixed_solvent_model
input_model = None Force this to be input model
restraint_files = None Restraint files for refinement
map_model_full_map = None Input full map file
half_map = None Input half map file
model = None Input model file
output_alt_conf_model = default Model with alternate conformation
overwrite = True Overwrite files with same names
file_name = None Not used
filename = None Not used
serial = None Not used
temp_dir = TEMP_CREATE_ALT_CONF Temporary directory
alt_conf_cycles = 4 Number of overall cycles
use_existing_altlocs = True Use existing conformers. Default is True (use any altlocs in the input model (if one model is supplied). If alt_models are supplied, they are used as a set of conformers.
allow_swap_fixed_solvent = True If fixed_solvent_model is supplied, try swapping conformers at each residue number to optimize geometry score.
score_this_altloc_only = None Score only this altloc if specified
sift_and_rename_waters_only = False Assign waters to altlocs if possible and stop
sift_and_refine_only = False Assign waters to altlocs if possible, refine and stop. Requires fixed_solvent_model
max_sift_cycles_fixed_solvent = None Maximum cycles of sifting with fixed solvent. Default is number of residues in solvent.
final_refinement_fixed_solvent = False Include final low-restraint-weight refinement with fixed solvent. default is False.
water_and_hydrogens = *at_end_of_cycle always never When to add hydrogens and waters in refinement and scoring. Refinement and rebuilding of side chains is done at the same time. If at_end_of_cycle, added at end of each cycle. NOTE: if fixed_solvent_model is set, water_and_hydrogens will be set to always.
max_allowed_residual = 1 Maximum contact energy allowed for adding waters
max_models_per_cycle = 72 Maximum models to consider in any cycle
max_rotamers_to_consider = None Limit the number of rotamers to consider at each site. Default is to consider all possible rotamers at each site.
min_rotamer_rmsd = 0.25 Minimum rmsd between rotamers. If side chain atoms differ by less than this, consider them the same.
conformers = 2 Number of conformers to create. If you supply conformers as alt_models or as a single model (with use_existing_altlocs=True) the number of supplied conformers must match the value of conformers. (This check is to make sure that the number of supplied conformers is what you intend.)
mask_at_large_diffs ...

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesxray_data = None Data file with experimental data (FP SIGFP or I SIGI).
xray_data_labels = None Optional labels for X-ray data Normally these would be something like I,SIGI or F,SIGF
free_r_labels = None Optional labels for free_r flags. Normally these would be something like

FreeR_flags or R_free_flags free_r_flag_value = 0 FreeR flag value ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified. map_coeffs = None MTZ file with coefficients for a map use_map_coeffs_if_optional = False Use map_coeffs (a map) instead of Fobs, sigFobs if both are available map_coeffs_virtual = None Used internally map_coeffs_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeffs_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM) alt_models = None Set of alternate conformations as individual model files. You can supply alternate conformations with alt_models, or you can supply a single model file with multiple conformations and specify use_existing_altlocs=True. fixed_solvent_model = None Model with solvent to be used as is (no refinement). NOTE: set water_and_hydrogens will be set to always if you use a fixed_solvent_model input_model = None Force this to be input model restraint_files = None Restraint files for refinement map_model full_map = None Input full map file half_map = None Input half map files model = None Input model file

- xray_data = None Data file with experimental data (FP SIGFP or I SIGI).
- xray_data_labels = None Optional labels for X-ray data Normally these would be something like I,SIGI or F,SIGF
- free_r_labels = None Optional labels for free_r flags. Normally these would be something like FreeR_flags or R_free_flags
- free_r_flag_value = 0 FreeR flag value
- ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
- map_coeffs = None MTZ file with coefficients for a map
- use_map_coeffs_if_optional = False Use map_coeffs (a map) instead of Fobs, sigFobs if both are available
- map_coeffs_virtual = None Used internally
- map_coeffs_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map_coeffs_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
- alt_models = None Set of alternate conformations as individual model files. You can supply alternate conformations with alt_models, or you can supply a single model file with multiple conformations and specify use_existing_altlocs=True.
- fixed_solvent_model = None Model with solvent to be used as is (no refinement). NOTE: set water_and_hydrogens will be set to always if you use a fixed_solvent_model
- input_model = None Force this to be input model
- restraint_files = None Restraint files for refinement
- map_model full_map = None Input full map file half_map = None Input half map files model = None Input model file
- full_map = None Input full map file

- half_map = None Input half map files
- model = None Input model file
- outputalt_conf_model = default Model with alternate conformations
 overwrite = True Overwrite files with same names
 file_name = None Not used
 filename = None Not used
 serial = None Not used
 temp_dir = TEMP_CREATE_ALT_CONF Temporary directory
- alt_conf_model = default Model with alternate conformations
- overwrite = True Overwrite files with same names
- file_name = None Not used
- filename = None Not used
- serial = None Not used
- temp_dir = TEMP_CREATE_ALT_CONF Temporary directory
- alt_confscycles = 4 Number of overall cycles
 use_existing_altlocs = True Use existing conformers. Default is True (use any altlocs in the input model (if one model is supplied). If alt_models are supplied, they are used as a set of conformers.
 allow_swap_fixed_solvent = True If fixed_solvent_model is supplied, try swapping conformers at each residue number to optimize geometry
 score_this_altloc_only = None Score only this altloc if specified
 sift_and_rename_waters_only = False Assign waters to altlocs if possible and stop
 sift_and_refine_only = False Assign waters to altlocs if possible, refine and stop. Requires fixed_solvent_model
 max_sift_cycles_fixed_solvent = None Maximum cycles of sifting with fixed solvent. Default is number of residues in solvent.
 final_refinement_fixed_solvent = False Include final low-restraint-weight refinement with fixed solvent. default is False.
 water_and_hydrogens = *at_end_of_cycle always never When to add hydrogens and waters in refinement and scoring. Refinement and rebuilding of side chains is done at the same time. If at_end_of_cycle, added at end of each cycle. NOTE: if fixed_solvent_model is set, water_and_hydrogens will be set to always.
 max_allowed_residual = 1 Maximum contact energy allowed for adding waters
 max_models_per_cycle = 72 Maximum models to consider in any cycle
 max_rotamers_to_consider = None Limit the number of rotamers to consider at each site. Default is to consider all possible rotamers at each site.
 min_rotamer_rmsd = 0.25 Minimum rmsd between rotamers. If side chain atoms differ by less than this, consider them the same.
 conformers = 2 Number of conformers to create. If you supply conformers as alt_models or as a single model (with use_existing_altlocs=True) the number of supplied conformers must match the value of conformers. (This check is to make sure that the number of supplied conformers is what you intend.)
 mask_at_large_diffs = None Mask all changes to include only residues with large differences between conformations (defined by large_rmsd). Default is False if two conformations, otherwise True
 score_confidence_ratio = 2. Score for a predicted pattern is calculated score plus score_confidence_ratio times the uncertainty. For a score_confidence ratio of 2, this is the score for which there is a 95% probability that the true score is less than this.
 use_edited_model = False Use model edited to remove poor rotamers in next cycle
 rebuild_main_chain = False Find alternate conformations for main chain. Default is False, only look for new side chain positions. Not implemented.
 diff_map_resolution = 2 Resolution for difference map (typically 2 Å)
 macro_cycles = 3 Refinement macro_cycles
 water_and_h_macro_cycles = 5 Refinement macro_cycles when placing waters and hydrogens
 rsr_macro_cycles = 3 Real-space refinement macro_cycles
 mask_atoms_atom_radius = 6 Radius for masking atoms. Set high enough to capture density in alternative conformations not present in starting model.
 large_rmsd = 0.3 Large rmsd between conformer residues indicating need for alternates
 minimum_patterns = 4 Minimum patterns to consider
 group_length = 3 Number of residues in group that is to be varied. Residues in a chain are grouped in sets of group_length and all patterns of side-chain assignments are tested.
 domain_size = 200

Maximum residues to consider when finding clashes `n_swap = 1` Add extra variation of swapping chains, `n_swap` times existing chains added on. Random locations `salt_labels = ABCDEFGHIJKLMNOPQRSTUVWXYZ` Labels to use for alt id. Include enough for all conformations. These are one-character labels only. `geometry_score_type = magnitudes` `gradients` `residue_score` `magnitudes_by_chain` `gradients_by_chain` `*residue_score_by_chain` Score type for geometry evaluation. Both are based on the geometry gradients by residue after refinement. Gradients uses the magnitudes of the mean gradient in a residue. Magnitudes uses the average magnitude of gradients in a residue. `correlated_errors = True` Assume errors in residue ge...

- `cycles = 4` Number of overall cycles
- `use_existing_altlocs = True` Use existing conformers. Default is True (use any altlocs in the input model (if one model is supplied)). If `alt_models` are supplied, they are used as a set of conformers.
- `allow_swap_fixed_solvent = True` If `fixed_solvent_model` is supplied, try swapping conformers at each residue number to optimize geometry score.
- `score_this_altloc_only = None` Score only this altloc if specified
- `sift_and_rename_waters_only = False` Assign waters to altlocs if possible and stop
- `sift_and_refine_only = False` Assign waters to altlocs if possible, refine and stop. Requires `fixed_solvent_model`
- `max_sift_cycles_fixed_solvent = None` Maximum cycles of sifting with fixed solvent. Default is number of residues in solvent.
- `final_refinement_fixed_solvent = False` Include final low-restraint-weight refinement with fixed solvent. default is False.
- `water_and_hydrogens = *at_end_of_cycle` always never When to add hydrogens and waters in refinement and scoring. Refinement and rebuilding of side chains is done at the same time. If `at_end_of_cycle`, added at end of each cycle. NOTE: if `fixed_solvent_model` is set, `water_and_hydrogens` will be set to always.
- `max_allowed_residual = 1` Maximum contact energy allowed for adding waters
- `max_models_per_cycle = 72` Maximum models to consider in any cycle
- `max_rotamers_to_consider = None` Limit the number of rotamers to consider at each site. Default is to consider all possible rotamers at each site.
- `min_rotamer_rmsd = 0.25` Minimum rmsd between rotamers. If side chain atoms differ by less than this, consider them the same.
- `conformers = 2` Number of conformers to create. If you supply conformers as `alt_models` or as a single model (with `use_existing_altlocs=True`) the number of supplied conformers must match the value of `conformers`. (This check is to make sure that the number of supplied conformers is what you intend.)
- `mask_at_large_diffs = None` Mask all changes to include only residues with large differences between conformations (defined by `large_rmsd`). Default is False if two conformations, otherwise True
- `score_confidence_ratio = 2`. Score for a predicted pattern is calculated score plus `score_confidence_ratio` times the uncertainty. For a `score_confidence_ratio` of 2, this is the score for which there is a 95% probability that the true score is less than this.
- `use_edited_model = False` Use model edited to remove poor rotamers in next cycle
- `rebuild_main_chain = False` Find alternate conformations for main chain. Default is False, only look for new side chain positions. Not implemented.
- `diff_map_resolution = 2` Resolution for difference map (typically 2 Å)

- `macro_cycles = 3` Refinement `macro_cycles`
- `water_and_h_macro_cycles = 5` Refinement `macro_cycles` when placing waters and hydrogens
- `rsr_macro_cycles = 3` Real-space refinement `macro_cycles`
- `mask_atoms_atom_radius = 6` Radius for masking atoms. Set high enough to capture density in alternative conformations not present in starting model.
- `large_rmsd = 0.3` Large rmsd between conformer residues indicating need for alternates
- `minimum_patterns = 4` Minimum patterns to consider
- `group_length = 3` Number of residues in group that is to be varied. Residues in a chain are grouped in sets of `group_length` and all patterns of side-chain assignments are tested.
- `domain_size = 200` Maximum residues to consider when finding clashes
- `n_swap = 1` Add extra variation of swapping chains, `n_swap` times existing chains added on. Random locations
- `alt_labels = ABCDEFGHIJKLMNOPQRSTUVWXYZ` Labels to use for alt id. Include enough for all conformations. These are one-character labels only.
- `geometry_score_type = magnitudes` `gradients` `residue_score` `magnitudes_by_chain` `gradients_by_chain` `*residue_score_by_chain` Score type for geometry evaluation. Both are based on the geometry gradients by residue after refinement. Gradients uses the magnitudes of the mean gradient in a residue. Magnitudes uses the average magnitude of gradients in a residue.
- `correlated_errors = True` Assume errors in residue geometry scores are correlated
- `all_single_changes = False` Include all single changes in randomization
- `geometry_mean_or_min = *mean` `min` Geometry mean or minimum. Use the mean value of scores for all examples of a particular pattern of side chains in a set of `group_length` residues.
- `clash_weight = 0` Weight on clashes within a conformer
- `allow_close = NO` `ON` `OO` `NN` atom pairs that can be close (for clashes)
- `close_dist = 2.8` Close distance for `allow_close` atoms
- `other_dist = 4.0` Close distance for non-`allow_close` atoms
- `skip_basic_optimization = False` Skip basic optimization step
- `const_shrink_donor_acceptor = 0` Allow contacts closer by `const_shrink_donor_acceptor` from normal target for H-bonding atoms. NOTE: value of 0.6 matches behavior of Phenix pre-2024.
- `keep_ligands = True` Keep one copy of each input ligand (no altlocs). Add to each model just before refinement. Requires supplying any necessary restraint files.
- `anisotropic_refinement = Auto` Anisotropic refinement. Default is to match input model file.
- `refine_args = None` Any refinement arguments to be added to default arguments. For example, `refinement.refine.occupancies.remove_selection=water` or for example, `refinement.main.number_of_macro_cycles=3`
- `h_selection = None` Selection to apply after adding H atoms
- `fixed_pattern = None` Pattern to apply. Only this pattern is used if specified. Must specify one for each `alt_conf` in this analysis
- `crystal_inforesolution = None` Nominal resolution of mapwrapping = `None` You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations). Always true for crystallographic maps. `scattering_table = *n_gaussian` `wk1995` `it1992` `electron` `neutron` Choice of

scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`. `chain_type = *None` PROTEIN RNA DNA Type of polymer (normally identified by chain automatically).

- `resolution = None` Nominal resolution of map
- `wrapping = None` You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations). Always true for crystallographic maps.
- `scattering_table = *n_gaussian` wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `chain_type = *None` PROTEIN RNA DNA Type of polymer (normally identified by chain automatically).
- `controlmultiprocessing = *multiprocessing` sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented `queue_run_command = None` run command for queue jobs. For example `qsub.nproc = 4` Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out `nproc` sets of huge files and it can be very slow with distributed queues.). You can override this with `force_nproc = True`. `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example. `use_existing_files = False` Use existing files `random_seed = 771914` random seed `verbose = False` Verbose output `quick = False` Quick run (no optimization with refinement, just find a plausible set of altlocs and finish up) `clean_up = True` Remove temporary directory when done
- `multiprocessing = *multiprocessing` sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems Not implemented
- `queue_run_command = None` run command for queue jobs. For example `qsub`.
- `nproc = 4` Number of processors to use. NOTE: by default multiple processors will only be used in the map-to-model step (this is because multiprocessing requires writing out `nproc` sets of huge files and it can be very slow with distributed queues.). You can override this with `force_nproc = True`.
- `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `use_existing_files = False` Use existing files
- `random_seed = 771914` random seed
- `verbose = False` Verbose output
- `quick = False` Quick run (no optimization with refinement, just find a plausible set of altlocs and finish up)
- `clean_up = True` Remove temporary directory when done
- `gui` GUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`

Creating maps using the Phenix GUI

create_maps.html

Contents

- The "Create Maps" GUI
- Guidelines for use
- References

Because of the differences in calculation method and input data, there is no single unified interface for creating any electron density maps; instead, separate programs are grouped together in the main GUI. For omit maps, a modified interface to AutoBuild is available; consult the AutoBuild documentation for rules on use. Isomorphous difference maps have a separate GUI, which takes two reflection files as input. For maps using phases from a model, use the "Create Maps" module. All of these programs write output as an MTZ file; the latter two may optionally output XPLOR or CCP4 maps (calculated to cover the PDB file used for phasing) if desired.

The "Create Maps" GUI

This is a frontend to phenix.maps, which shares a common set of parameters with phenix.refine. Many user-defined map types are possible, but the program defaults to the most common ones, 2mFo-DFc and mFo-DFc difference maps. Some additional predefined types are available in the drop-down menus.

The pre-defined maps include all common likelihood-weighted difference maps, anomalous difference map, likelihood-weighted F(obs) or F(calc) maps, and anomalous log-likelihood gradient (LLG) maps generated by Phaser. Custom map types can be defined by typing in an appropriate string in a format similar to those already shown; the map generator will parse the string to determine the components used.

By default the program only generates an MTZ file containing the map coefficients, but you may separately define maps to be generated in X-PLOR or CCP4 format by clicking the button labeled "CCP4 or XPLOR maps". However, if you leave these undefined, the GUI will automatically run the FFT to generate CCP4-format files if you click the "Open in PyMOL" button once map coefficients are available.

Guidelines for use

Map quality may be heavily influenced by reflections that are missing due to errors in data collection or processing. Substituting F(calc) for missing F(obs) increases model bias but often results in a more easily interpreted map; the default maps output by phenix.refine (and REFMAC) are "filled" maps. In phenix.create_maps, the default maps are filled, and we recommend using this option unless you are particularly worried about model bias. Output MTZ column labels must be unique; the predefined map types have associated predefined labels, but you can change these if you wish. Anomalous difference maps require anomalous data (i.e. F₊,F₋ or I₊,I₋); the precalculated anomalous difference DANO generated by some CCP4 programs is not suitable. If you are using neutron data, set the scattering table to "neutron"; otherwise, leave it set to "n_gaussian".

- Map quality may be heavily influenced by reflections that are missing due to errors in data collection or processing. Substituting F(calc) for missing F(obs) increases model bias but often results in a more easily interpreted map; the default maps output by phenix.refine (and REFMAC) are "filled" maps. In phenix.create_maps, the default maps are filled, and we recommend using this option unless you are particularly worried about model bias.

- Output MTZ column labels must be unique; the predefined map types have associated predefined labels, but you can change these if you wish.
- Anomalous difference maps require anomalous data (i.e. F⁺,F⁻ or I⁺,I⁻); the precalculated anomalous difference DANO generated by some CCP4 programs is not suitable.
- If you are using neutron data, set the scattering table to "neutron"; otherwise, leave it set to "n_gaussian".

References

Averaged kick maps: less noise, more signal... and probably less bias. J. Pražnikar, P.V. Afonine, G. Guncar, P.D. Adams, and D. Turk. *Acta Crystallogr D Biol Crystallogr* 65, 921-31 (2009).

- Averaged kick maps: less noise, more signal... and probably less bias. J. Pražnikar, P.V. Afonine, G. Guncar, P.D. Adams, and D. Turk. *Acta Crystallogr D Biol Crystallogr* 65, 921-31 (2009).

Cryo_fit: Fitting Structure to Cryo-EM Map Using MD Simulation

cryo_fit.html

Contents

- Overview
- Introduction
- How to Fit Cryo EM Maps
- Advantages
- How Cryo_fit Works
- How to Run Cryo_fit
- FAQ
- Reference

Overview

Cryo_fit fits biomolecular (protein and nucleic acids) structures to cryo-EM maps using molecular dynamics (MD) simulation.

Introduction

Cryo_fit is an automated cryo-EM density fitting program using MD simulation to guide models into maps. Cryo_fit produces all-atom models highly consistent with the EM density and is being used to optimize models for many functional molecular complexes including one of the first all-atom models of the human ribosome revealing the 'subunit rolling' mechanism unique to eukaryotic ribosomes. Two key advantages of Cryo_fit are: speed and stereochemistry preservation. Its fast speed allows researchers to optimize ribosome structures on a laptop instead of the need of a computing server with thousands of CPU/GPU cores. It uses a reduced model molecular dynamics potential allowing the preservation of stereochemistry information. In addition, secondary and tertiary contact potentials can be specified to place additional restraints on contacts to keep molecules intact during the fitting process.

How to Fit Cryo EM Maps

Advantages

How Cryo_fit Works

How to Run Cryo_fit

See the tutorial notes for cryo_fit

FAQ

See the cryo_fit FAQ

Reference

Journal of Structural Biology, Volume 208, 2019, Pages 1-6, Cryo_fit: Democratization of flexible fitting for cryo-EM

Cryo_fit2: Fitting to a Cryo-EM Map using Phenix Dynamics

cryo_fit2.html

Contents

- Note
- Author
- When cryo_fit1 and cryo_fit2 are useful?
- When cryo_fit2 is useful?
- When cryo_fit2 is more useful than cryo_fit1?
- Theory
- How to install
- How to Run Cryo_fit2
- FAQ
- Limitation

Note

cryo_fit2 is under active development.

Consider to use cryo_fit1 instead.

Author

Pavel Afonine, Dylan Girodat (dgirodat@lanl.gov), Jignesh Prajapati (jprajapati@lanl.gov), Doo Nam Kim (doonam.kim@gmail.com)

When cryo_fit1 and cryo_fit2 are useful?

See the When cryo_fit1 and cryo_fit2 are useful? in FAQ

When cryo_fit2 is useful?

See the When cryo_fit2 is useful? in FAQ

When cryo_fit2 is more useful than cryo_fit1?

See the When cryo_fit2 is more useful than cryo_fit1? in FAQ

Theory

- This program uses phenix.dynamics.
- Simulated annealing is carried out by default to minimize the object function T (= $T_{\text{target_map}} \cdot \text{map_weight} + T_{\text{geom}} \cdot \text{wc}$).

map_weight is cryo-EM map weight and wc is a geometry keeping weight.

- map_weight is cryo-EM map weight and wc is a geometry keeping weight.

How to install

See the installation notes for cryo_fit2

How to Run Cryo_fit2

See the tutorial notes for cryo_fit2

FAQ

See the cryo_fit2 FAQ

Limitation

- Cryo_fit2 does not use electrostatic interaction. Therefore, it does not fully capture all physical forces. See phenix.dynamics for detail. Cryo_fit version 1 also omits electrostatic interaction so that molecular dynamics simulation runs faster.
- See phenix.dynamics for detail.
- Cryo_fit version 1 also omits electrostatic interaction so that molecular dynamics simulation runs faster.

Visualizing reflection data in Phenix

data_viewer.html

Contents

- Overview
- 3D viewer
- 2D viewer

Overview

The program `phenix.data_viewer` is used to explore data in reciprocal space, as a diagnostic and learning tool. Although reciprocal space is largely an imaginary construct, it is a convenient way to describe the geometry of diffraction and data collection. Two different rendering methods are provided: a 3D view of all of reciprocal space at once, and a 2D view of planes of reflections, or "pseudo-precession photographs". These are largely complementary, although certain visualizations may be easier in one or the other view.

`phenix.data_viewer` can be launched from the GUI, under "Reflection tools", where the 2D and 3D versions are listed separately, or directly from the command line, which initially starts the 3D version. OpenGL support (and ideally a reasonably fast graphics card) is required for the 3D graphics; for this reason, we do not recommend running the 3D viewer over an X11 connection. Once the program opens it will ask you to select a reflections file, which depending on contents may prompt you to choose an array of data. Currently any array which includes amplitudes or intensities (including map coefficients) can be viewed, as well as phase angles and R-free flags. However, the examples shown here all use amplitudes or intensities.

3D viewer

For most reflections files, the data will fill some portion of a sphere, whose radius is one divided by the maximum resolution. Depending on space group symmetry and whether the data are symmetry-unique, usually only a wedge of data will be shown. If the data are anomalous this wedge will be mirrored across the origin. For instance, the intensity data for the p9-sad tutorial (included with Phenix) will appear like this:

In this case the space group is $I4$ with anomalous data, so the symmetry-unique reflections fill two 90-degree wedges of reciprocal space. Identifying features in the data is left as an exercise to the user; however, in this case (and many others) the characteristic distribution of intensities seen in the Wilson plot in Xtriage is very obvious: relatively high intensities at low resolution (at the center of the sphere), decreasing rapidly above approximately 10 Angstrom, with a large increase in mean intensity around 4-5 Angstrom due to characteristic inter-atomic distances in organic (especially macromolecular) structures. Clicking on any individual reflection will show the h,k,l index, resolution, and value in the box on the lower-left corner.

The view can be manipulated with the controls on the left-hand side of the window; this includes the option to change the symmetry or generate anomalous data if not are present in the input file. A particularly useful option is to show reflections that are missing. These will appear as white spheres:

Most datasets will have at least a handful of these at high resolution, and often the innermost low-resolution reflections will also be absent due to beamstop overlap. In this case a large contiguous region of missing data is visible for one of the quadrants; this is probably caused by imperfect orientation of the crystal during data collection. For this dataset the number of missing reflections is relatively small and because of the overall excellent quality, no impact on phasing, refinement, or map quality is observed. However, poor completeness can often lead to problems at various stages of refinement, especially if large regions of reciprocal space are

systematically missing.

2D viewer

The 2D view handles symmetry-unique data differently than the 3D version: it will always expand these data to P1 and generate anomalous pairs, covering all of reciprocal space. (For non-unique data, only those reflections present in the input file will be displayed.) This will result in a view something like this (here using the rnase-s tutorial):

In this example both missing reflections (small white open circles) and systematic absences (large purple open circles) are also displayed. The distribution of values is different than observed for the p9-sad example, partly because amplitudes are displayed instead of intensities, and partly because the data were probably not collected or processed to the maximum resolution possible for this crystal.

phenix.den_refine: DEN refinement in Phenix

den_refine.html

Deformable elastic network (DEN) refinement is a refinement protocol that uses a sparse restraint network to maintain local model geometry while allowing for larger global domain motions over the course of several macrocycles of slow-cooling simulated annealing.

The DEN restraint network is a set of atom-atom distance restraints between atoms 3 to 5 Angstrom apart and within 0 and 10 residues in sequence space. The initial target values for these distance restraints may be set to corresponding distances in a reference model (such as a higher resolution structure or homology model) or to the distances in the input model. Over the course of a simulated annealing cycle, the target distances are frequently updated to allow for improved global fit to the diffraction data. DEN restraints have been shown to be particularly useful in improving models with low resolution data.

DEN was originally described by Brunger and colleagues in:

Schroder GF, Levitt M, and Brunger AT. (2010) Super-resolution biomolecular crystallography with low-resolution data. *Nature* 464(7292), 1218-22.

Optimization

The impact of the DEN restraint network is controlled by two variable parameters, gamma, which controls the deformability of the network, and the weight, which controls how large of a contribution the DEN restraints make to the overall refinement energy term (see the above reference for details). To get the most out of a DEN refinement, it is important to run a grid search over combinations of gamma and weight values. This process is very time intensive, and is best carried out on a machine with many processors.

The default behavior is to optimize with six gamma values [0.0, 0.2, 0.4, 0.6, 0.8, 1.0] and five weights [3.0, 10.0, 30.0, 100.0, 300.0], which will run most optimally on a machine with at least 30 processor cores. The number of processors used by Phenix is controlled by the main.nproc parameter, so setting main.nproc=30 will run the optimization step on 30 processors.

Using DEN

How to run the command line version:

```
% phenix.refine model.pdb data.mtz strategy=den
```

See the full list of options below.

There is also a DEN refinement GUI for those who prefer a graphical interface.

phenix.den_refine uses a modified version of the default phenix.refine parameters optimized for DEN refinement.

The DEN-specific subset of refinement parameters:

```
refinement {
  main {
    nproc = Auto
  }
  refine {
    strategy = *individual_sites rigid_body *individual_adp \
    group_adp tls occupancies group_anomalous *den
  }
  den {
```

```

reference_file = None
gamma = 0.5
kappa = 0.1
weight = 30
sigma = 0.44
optimize = True
opt_gamma_values = 0 0.2 0.4 0.6 0.8 1
opt_weight_values = 3 10 30 100 300
num_cycles = 10
kappa_burn_in_cycles = 2
bulk_solvent_and_scale = True
refine_adp = True
final_refinement_cycle = False
verbose = False
annealing_type = *torsion cartesian
output_kinimage = False
restraint_network {
  lower_distance_cutoff = 3
  upper_distance_cutoff = 15
  sequence_separation_low = 0
  sequence_separation_limit = 10
  exclude_hydrogens = True
  ndistance_ratio = 1
  export_den_pairs = False
  den_network_file = None
}
}
tardy {
  mode = *every_macro_cycle second_and_before_last once first first_half
  start_temperature_kelvin = 3300
  final_temperature_kelvin = 300
  number_of_cooling_steps = 1440
  number_of_time_steps = 1
  time_step_pico_seconds = 0.001
  prolsq_repulsion_function_changes {
    c_rep = 100
    k_rep = 1
  }
}
simulated_annealing {
  start_temperature = 3300
  final_temperature = 300
  cool_rate = 50
  number_of_steps = 24
  time_step = 0.001
}
}

```

Processing, docking and rebuilding AlphaFold2 and other predicted models in cryo-EM maps

dock_and_rebuild.html

Author(s)

- dock_and_rebuild: Tom Terwilliger

Purpose

Dock and rebuild combines the functions of processing, docking and rebuilding predicted models produced by AlphaFold, RoseTTAFold and other prediction software into a cryo EM map.

These steps can also be carried out one at a time. There are three steps in working on an AlphaFold or other predicted model with a cryo-EM structure.

The first step is to process the predicted model by trimming off all the uncertain residues in the predicted model and breaking up the remaining structure into a best guess of rigid domains with `phenix.process_predicted_model`.

The next step is to dock each of the domains of the processed model into the map, keeping plausible connectivity. This is done with `phenix.dock_predicted_model`.

The third step is to morph the predicted model onto the docked domains and then to rebuild all the parts of the predicted model using the density in the map.

How dock_and_rebuild works:

The `dock_and_rebuild` procedure is just a way to connect all three steps in processing, docking and rebuilding a predicted model. You may want to use this procedure if you have a simple case, or if you have many models you want to process. If it does not work well, you may want to take its intermediate outputs (i.e., the docked model) and carry out the following steps individually.

The model input to `dock_predicted_model` is your starting predicted model file (i.e., an AlphaFold model)

The map input to `dock_predicted_model` is normally your best sharpened or density-modified cryo-EM map. It can also be a map generated by any other procedure (including crystallography).

If you are able to mask your map, keeping only the part representing the region where this model belongs, that can be very helpful. You can also box the map around this region. If you have a map that has many chains, this masking can greatly shorten the time for docking. If you don't know where in your map the model goes at all, you can supply the entire map. If your map has symmetry this will normally be found automatically.

The three steps described above are then carried out: processing the model to obtain domains representing the accurate parts of the model (`phenix.process_predicted_model`), docking the domains (`phenix.dock_predicted_model`), and morphing and rebuilding the predicted model to yield a rebuilt version of the predicted model.

Examples

Standard run of dock_and_rebuild:

Running dock_and_rebuild is easy. From the command-line you can type:

```
phenix.dock_and_rebuild model=my_model.pdb \  
  full_map=my_map.ccp4 \  
  resolution=3
```

This will carry out all the steps of processing, docking and rebuilding to yield my_model_rebuilt.pdb

Possible Problems

If your map has pseudo-symmetry (like a proteasome) you might need to box one subunit or try ssm_search=False to use a more thorough search in docking.

If your map is inverted (left-handed), docking and model-building will not work properly. You can often tell if your map is inverted because any helices will be left-handed. If you are unsure, you can run MapBox with invert_hand=True to invert the map and then see if docking works. Note that if your map is inverted, you will want to invert all your maps and start everything from the beginning.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_files_seq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
xray_data_file = None Input X-ray data file (MTZ format).
xray_data_label = None Label specifying which data column to use from xray_data_file
xray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_file
truncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
density_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
msa_list = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
models_are_already_placed = False You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.
model_copies_list = None List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like model_copies_list=1,2,1,1
processed_model_file = None Processed model file (e.g., from phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
docked_model_file = None Docked model (e.g., output of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model

and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model. `morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model. `previous_model_file = None` Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information. `scaffold_model = None` Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking. `scaffold_minimum_chain_cc = 0.35` Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with l...

- `job_title = None` Job title in PHENIX GUI, not used on command line
- `input_filesseq_file = None` Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines) `xray_data_file = None` Input X-ray data file (MTZ format). `xray_data_label = None` Label specifying which data column to use from `xray_data_file` `xray_test_flag_label = None` Label specifying which test flag column (if any) to use from `xray_data_file` `truncate_models_to_match_sequences = True` If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences `density_select = None` If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is `True` if map is full-size and `False` if not. `predicted_model = None` One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly. `b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein). `msa_list = None` Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly. `models_are_already_placed = False` You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model. `model_copies_list = None` List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1` `processed_model_file = None` Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed. `docked_model_file = None` Docked model (e.g., output of `dock_processed_model`). If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model. `morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed

model.previous_model_file = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If docked_model_file is also supplied, previous_model is used only as source of model information.scaffold_model = None Scaffold model. If supplied, used as target for docking of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied, model_copies_list is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.scaffold_minimum_chain_cc = 0.35 Chains in scaffold model that have scaffold_minimum_cc are not re-docked, but those with lower CC are.base_scaffold_minimum_chain_cc = -1 Value of scaffold_...

- seq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
- xray_data_file = None Input X-ray data file (MTZ format).
- xray_data_label = None Label specifying which data column to use from xray_data_file
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- b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
- msa_list = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
- models_are_already_placed = False You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.
- model_copies_list = None List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like model_copies_list=1,2,1,1
- processed_model_file = None Processed model file (e.g., from phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
- docked_model_file = None Docked model (e.g., output of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.

- `morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
- `previous_model_file = None` Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
- `scaffold_model = None` Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
- `scaffold_minimum_chain_cc = 0.35` Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
- `base_scaffold_minimum_chain_cc = -1` Value of `scaffold_minimum_chain_cc` in cases where scaffold is known to be good.
- `fragments_model_file = None` Fragments model (e.g., `map_to_model.pdb`). Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `fragments_model` is used only as source of model information.
- `model_to_rebuild_file = None` Model to be rebuilt. Must be already in place (already docked). Must match supplied sequences exactly.
- `symmetry_file = None` Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. NOTE: symmetry file applies to map file in original position. NOTE 2: only proper symmetry (point-group, helical) is allowed NOTE 3: Only applies to cryo-EM maps
- `pae_file = None` Optional input json file with matrix of inter-residue estimated errors (pae file)
- `distance_model_file = None` Distance_model_file. A PDB or mmCIF file containing the model corresponding to the PAE matrix. Only needed if `weight_by_ca_ca_distances` is True.
- `search_model_copies = None` Used internally only
- `search_model = None` used internally only
- `map_model_full_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
`half_map = None` Input half map files. Usually supply one full map or 2 half maps
`model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `full_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
- `half_map = None` Input half map files. Usually supply one full map or 2 half maps

- `model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `output_filesoutput_model_prefix = None` Output files with superposed models will begin with this prefix
- `output_seq_file = None` Sequence file (possibly edited) written to this file by PredictAndBuild
- `pdb_out = None` Used internally only
- `temp_dir = None` Temporary directory. Default is `dock_and_build_xx` where `xx` creates new directory
- `crystal_inforesolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 Å lower resolution than your data (i.e, if your data is 2.5 Å, try 5 Å).
- `scattering_table = *None n_gaussian wk1995 it1992 electron neutron` Choice of scattering table for structure factor calculations. Default for X-ray is `n_gaussian`, for cryoEM is `electron`.
- `wrapping = None` You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- `asymmetric_map = None` Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.
- `solvent_content = None` Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- `pdb70_text = None` Database of PDB entries to be used as templates (pdb70 file). Normally used internally only
- `sequence = None` Old-style sequence string (alternative to sequence file).
- `chain_type = *PROTEIN DNA RNA Chain type`

- msa = None MSA as text. Supply MSA as text or as an msa_file. Used internally
- templates_as_string = None Templates as string (single PDB file with one or more chains)
- space_group = None Space group (normally read from the data file, applies to X-ray)
- space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group
- unit_cell = None Unit Cell (normally read from the data file, applies to X-ray)
- unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""
- sequence = "Enter or edit sequence"
- copies = 1 Copies of this sequence
- label = ""
- process_predicted_modelremove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)continuous_chain = False When removing low-confidence residues, only trim from endsplit_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=Truemaximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging themminimum_sequential_residues = None Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with alt_pae_params=Trueminimum_remainder_sequence_length = 15 used to choose whether the sequence of a removed segment is written to the remainder sequence file.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factorinput_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.minimum_plddt = None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.default_maximum_rmsd = 1.5 Default value of maximum_rmsd, used if maximum_rmsd is not setsubtract_minimum_b = False If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low-confidence residues.pae_power = None If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to (1/pae**pae_power). Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 alt_pae_params=False and 2 for

alt_pae_params=True pae_cutoff = None If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with $pae < pae_cutoff$. Default is 5
alt_pae_params=False and 4 for alt_pae_params=True pae_graph_resolution = None If PAE matrix (predicted alignment error matrix) is supplied, pae_graph_resolution regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for alt_pae_params=True alt_pae_params = True If PAE matrix is supplied, use alternative set of defaults (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4 pae_graph_resolution...

- remove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)
- continuous_chain = False When removing low-confidence residues, only trim from ends
- split_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.
- maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.
- domain_size = 15 Approximate size of domains to be found (Å units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 Å).
- adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 Å and take the value that gives the smallest number of domains, but at least maximum_domains.
- minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=True
- maximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging them
- minimum_sequential_residues = None Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with alt_pae_params=True
- minimum_remainder_sequence_length = 15 used to choose whether the sequence of a removed segment is written to the remainder sequence file.
- b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor
- input_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.
- minimum_plddt = None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.
- maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.
- default_maximum_rmsd = 1.5 Default value of maximum_rmsd, used if maximum_rmsd is not set

- `subtract_minimum_b = False` If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low- confidence residues.
- `pae_power = None` If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/pae^{**pae_power})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 `alt_pae_params=False` and 2 for `alt_pae_params=True`
- `pae_cutoff = None` If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with `pae < pae_cutoff`. Default is 5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `pae_graph_resolution = None` If PAE matrix (predicted alignment error matrix) is supplied, `pae_graph_resolution` regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `alt_pae_params = True` If PAE matrix is supplied, use alternative set of defaults (`minimum_domain_length=20` `minimum_sequential_residues=10` `pae_power=2` `pae_cutoff=4` `pae_graph_resolution=4`). Standard parameters (`alt_pae_params=False`) are: (`minimum_domain_length=10` `minimum_sequential_residues=5` `pae_power=1` `pae_cutoff=5` `pae_graph_resolution=0.5`).
- `weight_by_ca_ca_distance = False` Adjust the edge weighting for each residue pair according to the distance between CA residues. If this is True, then `distance_model` can be provided, otherwise supplied model will be used. See also `distance_power`
- `distance_power = 1` If `weight_by_ca_ca_distance` is True, then edge weights will be multiplied by $1/distance^{**distance_power}$.
- `stop_if_no_residues_obtained = True` Raise Sorry and stop if processing yields no residues
- `keep_all_if_no_residues_obtained = False` Keep everything if processing yields no residues
- `vrms_from_rmsd_intercept = 0.25` Estimate of vrms (error in model) from pLDDT will be based on `vrms_from_rmsd_intercept + vrms_from_rmsd_slope * pLDDT` where mean pLDDT of non-low_confidence_residues is used.
- `vrms_from_rmsd_slope = 1.0` Estimate of vrms (error in model) from pLDDT will be based on `vrms_from_rmsd_intercept + vrms_from_rmsd_slope * pLDDT` where mean pLDDT of non-low_confidence_residues is used.
- `break_into_chunks_if_length_is = 1500` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae matrix
- `chunk_size = 600` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae_matrix
- `overlap_size = 200` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae_matrix
- `searchdock_chains_individually = False` Dock each chain (identified by `chain_id` field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with `create_unique_chain_at_end=True` for predicted models to put the model back together. `create_unique_chain_at_end = None` Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances

between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.

choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain....pick better-fitting of the two for each chain.

low_res_search = True Try to fit by searching at low resolution.

firstdock_with_mr = True Try using MR to dock.

firstssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.

refine_cycles = 3 rigid-body refinement cycles.

ssm_search_min_cc = 0.30 Stop ssm search if this cc achieved.

backup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each time.

backup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each time.

resolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.

align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similar.

max_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each other.

radius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of model.

use_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.

skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.

rigid_body_refinement = True Run rigid-body refinement on final model.

rigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains).

rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up molecule.

rigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specified.

append_to_fixed_model = True Append placed search model to fixed model (if any) after search.

min_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.

run_in_boxes = True Run on sub-boxes and combine at the end.

target_box_size = 60 Try to get boxes about this big on a side (grid units).

target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_size.

box_to_run = None Run only this box.

box_overlap_scale = 1 box overlap (ov...

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.

- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

- `weight_sequential_fragments_by_distance = None` Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if `create_unique_chain_at_end` is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.
- `choose_better_of_individual_and_group_docking = None` Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain
- `low_res_search = True` Try to fit by searching at low resolution first
- `dock_with_mr = True` Try using MR to dock first
- `ssm_search = None` Try to fit by searching for secondary structure. Default is False for `dock_in_map` and True for `dock_and_rebuild` if `dock_with_mr` is not used.
- `refine_cycles = 3` rigid-body refinement cycles
- `ssm_search_min_cc = 0.30` Stop ssm search if this cc achieved
- `backup_resolution_cutoff_searches = 2` If initial fitting does not work, try up to this many times increasing resolution by `backup_resolution_ratio` each time
- `backup_resolution_ratio = 1.67` If initial fitting does not work, try up to `backup_resolution_cutoff_searches` times increasing the resolution by `backup_resolution_ratio` each time
- `resolution_radius_scale = 0.5` Resolution for low-res search will be `resolution_radius_scale` times the radius of gyration of the search model.
- `align_moments = False` Try to fit by aligning moments of inertia if size of density region and molecule are similar
- `max_radius_ratio = 2`. Radius of gyration of molecule and density must be within `max_radius_ratio` of each other
- `radius_scale = 1.5` Mask for density and atoms will be `radius_scale` times the radius of gyration of model
- `use_symmetry = True` If `search_model_copies` or `search_map_copies` are the same for each model and greater than one and `allow_symmetry` is set, use symmetry in the map to place all copies after the first one.
- `skip_if_low_cc = True` Skip solution before rigid-body refinement if map-model CC is less than half of `min_cc`.
- `rigid_body_refinement = True` Run rigid-body refinement on final model
- `rigid_body_refinement_single_unit = True` Run rigid-body refinement with just one unit (do not break up into chains)
- `rigid_body_refinement_split_method = *chain_id segid` When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit=False`), use either `chain_id` or `segid` to split up molecule
- `rigid_body_refinement_resolution = None` Run rigid-body refinement at this resolution if specified
- `append_to_fixed_model = True` Append placed search model to fixed model (if any) after search
- `min_cc = 0.4` If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 `min_cc`.
- `run_in_boxes = True` Run on sub-boxes and combine at the end
- `target_box_size = 60` Try to get boxes about this big on a side (grid units)

- `target_boxes = None` Try to get this many boxes. Default is `nproc` unless this makes box size much smaller than `target_box_size`
- `box_to_run = None` Run only this box
- `box_overlap_scale = 1` box overlap (overlap of boxes) will be `box_overlap_scale` times the density radius
- `edge_ratio = 10` box edge `box_overlap` times `edge_ratio`
- `density_radius = None` Radius for density to be cut out and compared. Default is 6 times the resolution.
- `model_radius = 3` Radius for removing density near `fixed_model`
- `zero_value = 0` Value to set map in regions overlapping fixed model
- `density_peaks = 20` Number of NCS-related peaks of density to check
- `delta_phi = 20` Angular spacing of search
- `max_rot = None` Maximum rotations to try
- `rotz_only = None` Rotate only around Z
- `single_positions_to_try = 10` Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- `max_position_shift_frac = 0.05` Maximum fractional positional shift in `single_positions` run
- `min_relative_cc = 0.67` Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- `sieve_fit = None` Use `sieve_fit` fraction of single positions in fitting. If `None`, use all
- `ncs_copies_max = None` Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with `phenix.pdbtools placed_model.pdb keep="model 1"` etc.
- `start_rot = None` Three numbers `rotx`, `rotz`, `roty` defining the starting rotation of the search model. Normally used along with `delta_phi=1000` or `max_rot=1` to generate exactly one defined rotation.
- `search_center = None` Optional coordinates in search model for centering search. Note this is different from `target_search_center` which is the location xyz in the map to look.
- `search_center_selection = None` Optional selection defining coordinates in search model for centering search.
- `target_search_center = None` Optional coordinates in reference map where `search_center` should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to `target_search_center/density_radius`.
- `model_search_position = None` Optional coordinates (usually part of search model) for matching to `target_search_position` after transformation. These can be specified in addition to `target_search_center`. Can have multiple `model_search_position` and `target_search_position` pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- `target_search_position = None` Target positions for `model_search_position` after transformation.
- `search_position_radius = None` Radius for comparison of `target_search_position` and transformed `search_position` values. Default is `density_radius`. If specified, must be a single value or the same number of values as entries in `model_search_position` and `target_search_position`

- `rot_id_n` = None Number of rotation groups. Along with `rot_id_group`, allows defining groups of rotations to be carried out in one run. `.short_caption` = Number of rotation groups
- `rot_id_group` = None rotation group to include. See `rot_id_n`.
- `map_box` = True Run `map_box` to extract useful part of map before search
- `fix_search_position` = False You can choose to not move your model center of mass to the origin by fixing the search position
- `search_box_size` = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times `density_radius`
- `lower_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `upper_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `keep_search_order` = None Keep search order as input
- `remove_water` = False Remove waters and other hetero atoms from input files
- `iterationcycles` = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quickcycle = 1 Iteration cycle
- `cycles` = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quick
- `cycle` = 1 Iteration cycle
- `buildrebuild_strategy` = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.`refine_only` = None Refine only, no rebuilding steps (set automatically with Quick)`refine_only_resolution` = 3.5 Default cutoff for using Refine only`run_fit_loops` = True Run standard fit_loops`run_iterative_morph` = True Run iterative morphing`run_trace_loops_through_density` = True Run loop fitting with trace_through_density algorithm`run_refine` = True Refine morphed model. Required for `run_fit_loops` or `run_trace_loops_through_density``run_iterative_resolution_refine` = True Refine morphed model with iterative resolution method`run_extend` = True Extend ends of morphed model`extract_unique` = True Extract unique part of map. Applies after `density_select` (if set).`use_symmetry_in_extract_unique` = True Use symmetry from symmetry file (if available) when extracting unique part of map.`acceptable_docking_cc` = 0.5 Acceptable docking CC for a chain`minimum_docking_cc` = 0.15 Minimum docking CC for a chain`minimum_cutoff` = 0.1 Minimum cutoff for estimating density in better parts of model`reasonable_cc_ratio` = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept`reasonable_cc_diff` = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept`shift_field_distance` = None Shift field characteristic distance (default = 10 Å) for morphing`cc_sd_ratio` = 3. Keep residues with CC at least within `cc_sd_ratio` of the mean CC for good residues`cc_sd_ratio_end` = 2. Keep residues with CC at least within `cc_sd_ratio_end` of the mean CC for good residues (applies to end of fragments)`cc_sd_ratio_ok` = 2. Residues with with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of `plddt`)`max_gap_ratio` = 3. Allow CA-CA distance to be up to `max_gap_ratio` times expected`maximum_connectivity_deviation` = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus 2 \* `resolution`.`keep_fraction_of_best` = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.`keep_maximum_entries` = 10 Maximum dock positions to keep`rmsd_for_similar_placement` = None RMSD value indicating that two placements are similar. default

is resolution of maprigid_body_refine_cycles = 1 Refinement cycles for rigid-body
 refinementoverlap_ca_ca_distance = 3 Overlap distance for CA-CA atoms (or P-P)ca_ca_distance = 3.8
 CA-CA distance (or P-P)allowed_fraction_overlapping = 0.10 Allowed fraction
 overlappingmaximum_combinations = 100000 Maximum combinations of placements to
 considerproceed_with_any_symmetry = False Run even with symmetry that is not point-group or helical.
 Not recommended as symmetry may not work properlybox_cushion = 20 Size of buffer around model
 when boxingrefine_cycles = 3 Refinement cyclesloop_refine_cycles = 5 Refinement cycles for
 loopsloop_backup_residues = 3 Number of tries removing one residue at a time from each end of existing
 ends of loop if no loop is found with initial gapresidues_to_trim = 5 Residues to trim on each end of all
 trimmed fragmentsfind_ncs_from_model = True Find NCS (symmetry) from working models and apply in
 density modification. Also turns on finding NCS in any autobuild density modification (including
 density-based search)allow_split_more_than_one_chain_for_mr = False Allow splitting chains into
 domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as
 reassembly may not work with split multiple chains.acceptable_cc_ratio = 0.8 Keep segments with CC at
 least equal to average of base segments times minimum_cc_ratio minus difference between segment
 confide...

- rebuild_strategy = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
- refine_only = None Refine only, no rebuilding steps (set automatically with Quick)
- refine_only_resolution = 3.5 Default cutoff for using Refine only
- run_fit_loops = True Run standard fit_loops
- run_iterative_morph = True Run iterative morphing
- run_trace_loops_through_density = True Run loop fitting with trace_through_density algorithm
- run_refine = True Refine morphed model. Required for run_fit_loops or run_trace_loops_through_density
- run_iterative_resolution_refine = True Refine morphed model with iterative resolution method
- run_extend = True Extend ends of morphed model
- extract_unique = True Extract unique part of map. Applies after density_select (if set).
- use_symmetry_in_extract_unique = True Use symmetry from symmetry file (if available) when extracting unique part of map.
- acceptable_docking_cc = 0.5 Acceptable docking CC for a chain
- minimum_docking_cc = 0.15 Minimum docking CC for a chain
- minimum_cutoff = 0.1 Minimum cutoff for estimating density in better parts of model
- reasonable_cc_ratio = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- reasonable_cc_diff = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- shift_field_distance = None Shift field characteristic distance (default = 10 Å) for morphing
- cc_sd_ratio = 3. Keep residues with CC at least within cc_sd_ratio of the mean CC for good residues
- cc_sd_ratio_end = 2. Keep residues with CC at least within cc_sd_ratio_end of the mean CC for good residues (applies to end of fragments)

- `cc_sd_ratio_ok = 2`. Residues with with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of `plddt`)
- `max_gap_ratio = 3`. Allow CA-CA distance to be up to `max_gap_ratio` times expected
- `maximum_connectivity_deviation = 15` Maximum connectivity deviation. Reject a solution with bigger deviation than this plus $2 * \text{resolution}$.
- `keep_fraction_of_best = 0.5` Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.
- `keep_maximum_entries = 10` Maximum dock positions to keep
- `rmsd_for_similar_placement = None` RMSD value indicating that two placements are similar. default is resolution of map
- `rigid_body_refine_cycles = 1` Refinement cycles for rigid-body refinement
- `overlap_ca_ca_distance = 3` Overlap distance for CA-CA atoms (or P-P)
- `ca_ca_distance = 3.8` CA-CA distance (or P-P)
- `allowed_fraction_overlapping = 0.10` Allowed fraction overlapping
- `maximum_combinations = 100000` Maximum combinations of placements to consider
- `proceed_with_any_symmetry = False` Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
- `box_cushion = 20` Size of buffer around model when boxing
- `refine_cycles = 3` Refinement cycles
- `loop_refine_cycles = 5` Refinement cycles for loops
- `loop_backup_residues = 3` Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
- `residues_to_trim = 5` Residues to trim on each end of all trimmed fragments
- `find_ncs_from_model = True` Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)
- `allow_split_more_than_one_chain_for_mr = False` Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split multiple chains.
- `acceptable_cc_ratio = 0.8` Keep segments with CC at least equal to average of base segments times `minimum_cc_ratio` minus difference between segment confidence (`plDDT`) and confidence cutoff (typically 0.7)
- `low_res_if_multiple_solutions = 3.5` Try phaser MR at increasing lower resolutions up to this value if multiple solutions are found
- `delta_low_res = 0.5` Resolution increment for `low_res_if_multiple_solutions`
- `predictionprediction_method = *alphafold` Prediction method to use
`template_search_method = *mmseqs2` structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword `upload_msa_file=True` (`structure_search` does not generate MSAs).
`starting_alphafold_model = None` Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an `output_directory` with just an AlphaFold model named as the expected first

AlphaFold model and specifying `carry_on=True`. `input_directory` = ColabInputs Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well. `output_directory` = ColabOutputs Output directory. Copy outputs to `output_directory`. Used to restart with `carry on`. `save_outputs_in_google_drive` = False If run on Colab, copy outputs to `output_directory` in Google drive. `content_dir` = None Content directory. Default is working directory. `maxit_path` = None Path to maxit (pdb to cif) converter. Optional. `data_dir` = /mnt Data directory (location of AlphaFold parameters). `upload_file_with_jobname_resolution_sequence_lines` = None Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence. `maximum_cycles` = 10 Maximum cycles to carry out. `cycle_rmsd_to_resolution_ratio` = 0.25 Stop iteration if rmsd between subsequent AlphaFold models is less than `cycle_rmsd_to_resolution_ratio` times the resolution for two cycles in a row. `significant_increase_in_residues` = 5 Significant increase in residues in model. `password` = None Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently. `version` = dev-4502 Version of Phenix to run (Colab only). `query_sequence` = None Sequence. `resolution` = None Resolution of map (Å). Internal use only. Normally set instead `crystal_info.resolution`. `jobname` = None Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only. `use_msa` = True Use multiple sequence alignments at some points. `skip_all_msa_after_first_cycle` = False Skip multiple sequence alignments after first cycle. `include_templates_from_pdb` = True Include templates from PDB. Note special behavior when running `predict_and_build`: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept. `maximum_templates_from_pdb` = 20 Maximum templates from PDB to include. `release_date` = None release date for templates from pdb (only use up to this date. Format is 2020-05-14). `upload_msa_file` = False Supply MSA directly (.a3m format). You can supply the MSA as a file in your `input_directory` and it will be used instead of using `mmseqs2` to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target. `upload_manual_templates` = False Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only). `uploaded_tem...`

- `prediction_method` = *alphafold Prediction method to use
- `template_search_method` = *mmseqs2 structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword `upload_msa_file=True` (`structure_search` does not generate MSAs).
- `starting_alphafold_model` = None Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an `output_directory` with just an AlphaFold model named as the expected first AlphaFold model and specifying `carry_on=True`.
- `input_directory` = ColabInputs Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well.
- `output_directory` = ColabOutputs Output directory. Copy outputs to `output_directory`. Used to restart with `carry on`.

- `save_outputs_in_google_drive` = False If run on Colab, copy outputs to `output_directory` in Google drive
- `content_dir` = None Content directory. Default is working directory
- `maxit_path` = None Path to maxit (pdb to cif) converter. Optional.
- `data_dir` = /mnt Data directory (location of AlphaFold parameters)
- `upload_file_with_jobname_resolution_sequence_lines` = None Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence
- `maximum_cycles` = 10 Maximum cycles to carry out
- `cycle_rmsd_to_resolution_ratio` = 0.25 Stop iteration if rmsd between subsequent AlphaFold models is less than `cycle_rmsd_to_resolution_ratio` times the resolution for two cycles in a row
- `significant_increase_in_residues` = 5 Significant increase in residues in model
- `password` = None Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently.
- `version` = dev-4502 Version of Phenix to run (Colab only)
- `query_sequence` = None Sequence
- `resolution` = None Resolution of map (Å). Internal use only. Normally set instead `crystal_info.resolution`.
- `jobname` = None Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only
- `use_msa` = True Use multiple sequence alignments at some point
- `skip_all_msa_after_first_cycle` = False Skip multiple sequence alignments after first cycle
- `include_templates_from_pdb` = True Include templates from PDB. Note special behavior when running `predict_and_build`: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept
- `maximum_templates_from_pdb` = 20 Maximum templates from PDB to include
- `release_date` = None release date for templates from pdb (only use up to this date. Format is 2020-05-14.
- `upload_msa_file` = False Supply MSA directly (.a3m format). You can supply the MSA as a file in your `input_directory` and it will be used instead of using `mmseqs2` to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target.
- `upload_manual_templates` = False Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only).
- `uploaded_templates_are_map_to_model` = False The manual templates are models that may or may not have the sequence of the alphafold models. These are used only as suggestions for placement of the main chain.
- `upload_maps` = True Use maps (required to be True)
- `random_seed` = 7231771 Random seed. Used internally

- `random_seed_iterations = 1` Random seed iterations of AlphaFold in first cycle. The model with the highest pLDDT will be used. If running `predict_and_build` and `include_templates_from_pdb` is True this many iterations will be carried out with and without templates. In `predict_and_build` and `predict_chain` set `number_of_models` instead.
- `minimum_random_seed_iterations = 1` Random seed iterations of AlphaFold after first cycle. The model with the highest pLDDT will be used
- `big_improvement = 10` How much improvement in pLDDT is worth going through all randomization cycles
- `good_enough_plddt = 80` Value of pLDDT that is good enough to not make any more models
- `nproc = 4` Number of processors to use. Internal use only. Normally set instead `control.nproc`
- `debug = False` Debugging run (print traceback error messages)
- `get_msa_only = False` Just get MSA and save it on server, no prediction. Does not return the MSA
- `carry_on = False` Carry on from where previous run ended. Used (usually in Colab) to go on after a crash or timeout. Requires that files are saved in the output directory
- `cif_dir = None` Location of templates (normally set automatically)
- `template_hit_list = None` List of templates (normally set automatically)
- `jobnames = None` List of jobnames (normally set automatically)
- `resolutions = None` List of resolutions (normally set automatically)
- `include_templates_from_pdb_list = None` List of `include_templates_from_pdb` ((normally set automatically)
- `include_side_in_templates_list = None` List of `include_side_in_templates_list` ((normally set automatically)
- `map_filename_dict = None` Map filename dict (used internally only)
- `msa_filename_dict = None` MSA filename dict (used internally only)
- `cif_filename_dict = None` CIF filename dict (used internally only)
- `query_sequences = None` List of query sequences (one per jobname) (used internally only)
- `manual_templates_uploaded = None` List of manual templates (used internally only)
- `upload_dir = None` Upload directory (used internally only)
- `working_directory = None` Working directory (used internally only)
- `maps_uploaded = None` List of maps (used internally only)
- `msas_uploaded = None` List of msas (used internally only)
- `num_models = 1` Number of models (used internally only)
- `homooligomer = 1` Number of copies (used internally only)
- `cycle = None` Cycle number (internal use only)
- `host_url = https://api.colabfold.com` Host url for mmseqs
- `use_env = None` Use env (internal use only)
- `use_custom_msa = None` Use custom msa (internal use only)
- `use_templates = None` Use templates (internal use only)
- `template_paths = None` Template paths (internal use only)

- `mtm_file_name` = None map_to_model file name (internal use only)
- `cycle_model_file_name` = None cycle_model_file_name (internal use only)
- `previous_final_model_name` = None previous_final_model_name (internal use only)
- `msa` = None MSA (internal use only)
- `msa_is_msa_object` = None MSA info (internal use only)
- `deletion_matrix` = None Deletion matrix (internal use only)
- `structure_search_params` structure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search structure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2.get_ligand = False Use get_ligand=True to retrieve ligands.get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only.get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.use_pdb100aa = False Use PDB100 sequence database for sequence search.use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directorycustom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collectedatom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded.batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processedmin_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.pickle_hits = False Pickle blast hit results from xml output.coot_display = False (default) Display output pdb files in coot.ask_coot = True prompt for coot display optionsPDB_MIRRORDIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRRORDIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRRORDIR set in the scriptPDB_MIRROR_MMCF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be \$PDB_MIRRORDIR/data/structures/divided/mmCIF directory. PDB_MIRROR_PDB = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ##

where ## are the second and...

- `structure_searchpdb_file` = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. `output_prefix` = 'output' Provide an output prefix if needed `blastpath` = None Enter path to blastall executable `sequence_only` = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search `structure_only` = False Do only a Ramachandran-based structure search. `db_used` = 'rcsb' structure database used in search. rcsb, scop95, or AF2 `db` = 'rcsb' Database used in search. rcsb, scop95, or AF2. `get_ligand` = False Use `get_ligand=True` to retrieve ligands. `get_rama_code_only` = False Generate Rama code for input pdb/cif only. This is for developers only. `get_xml_only` = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only. `use_pdb100aa` = False Use PDB100 sequence database for sequence search. `use_custom_db` = False Use custom database specified by `custom_db_files/custom_db_dir`. `custom_db_dir` = None The directory of pdb/cif files to make custom database. Default is current directory. `custom_db_files` = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the `custom_db_dir` will be collected. `atom_selection` = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. `get_pdb` = 10 `get_pdb=N` will collect and superpose the top N homologous pdbs. Use `get_pdb=0` to disable this option. `deposited_before` = 0 Specify the latest year of matching structures to be considered for scoring. PDBs deposited after this year will be discarded. `deposited_after` = 0 Specify the earliest year of matching structures to be considered for scoring. PDBs deposited before this year will be discarded. `batch_size` = 0 Process the pdbs in batch of `<batch_size>` until `<min_match>` hits are identified or until all `<get_pdb>` pdbs are processed. `min_match` = 0 Finish structure_search when `<min_match>` matches are found. Usually uses with `<trim_ends>` to exit the search once find suitable pdbs. `keep_all_pdb` = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message. `trim_ends` = False Remove terminal residues of hit pdbs extending beyond those of the target pdb. `write_pdb` = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects. `write_results` = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects. `trim_hit_pdb` = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb. `pickle_hits` = False Pickle blast hit results from xml output. `coot_display` = False (default) Display output pdb files in coot. `ask_coot` = True prompt for coot display options. `PDB_MIRROR_DIR` = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRROR_DIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRROR_DIR` set in the script. `PDB_MIRROR_MMCIF` = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be `$PDB_MIRROR_DIR/data/structures/divided/mmcif` directory. `PDB_MIRROR_PDB` = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and third letters in the ...
- `pdb_file` = None Enter a PDB file name
- `sequence` = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.
- `output_prefix` = 'output' Provide an output prefix if needed
- `blastpath` = None Enter path to blastall executable

- `sequence_only = False` Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search
- `structure_only = False` Do only a Ramachandran-based structure search.
- `db_used = 'rcsb'` structure database used in search. rcsb, scop95, or AF2
- `db = 'rcsb'` Database used in search. rcsb, scop95, or AF2.
- `get_ligand = False` Use `get_ligand=True` to retrieve ligands.
- `get_ramcode_only = False` Generate Rama code for input pdb/cif only. This is for developers only.
- `get_xml_only = False` Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.
- `use_pdb100aa = False` Use PDB100 sequence database for sequence search.
- `use_custom_db = False` Use custom database specified by `custom_db_files/custom_db_dir`.
- `custom_db_dir = None` The directory of pdb/cif files to make custom database. Default is current directory
- `custom_db_files = None` Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the `custom_db_dir` will be collected
- `atom_selection = 'all'` Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc.
- `get_pdb = 10` `get_pdb=N` will collect and superpose the top N homologous pdbs. Use `get_pdb=0` to disable this option.
- `deposited_before = 0` Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.
- `deposited_after = 0` Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded.
- `batch_size = 0` Process the pdbs in batch of `<batch_size>` until `<min_match>` hits are identified or until all `<get_pdb>` pdbs are processed
- `min_match = 0` Finish structure_search when `<min_match>` matches are found. Usually uses with `<trim_ends>` to exit the search once find suitable pdbs.
- `keep_all_pdb = False` Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.
- `trim_ends = False` Remove terminal residues of hit pdbs extending beyond those of the the target pdb.
- `write_pdb = True` Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.
- `write_results = True` Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.
- `trim_hit_pdb = False` Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.
- `pickle_hits = False` Pickle blast hit results from xml output.
- `coot_display = False` (default) Display output pdb files in coot.
- `ask_coot = True` prompt for coot display options

- `PDB_MIRRORDIR = None` Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRRORDIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRRORDIR` set in the script
- `PDB_MIRROR_MMCIF = None` Enter the parent directory of the mmcif files in the local PDB mirror. MMCIFs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/mmcif` directory.
- `PDB_MIRROR_PDB = None` Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/pdb` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and others together. However, users may choose to specify `PDB_MIRROR_PDB` directly
- `PDB_MIRROR_STRUCTURE_FACTORS = None` Enter the parent directory of the PDB files in the local PDB mirror. structure factors s will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be the same as the `$PDB_MIRRORDIR/data/structures/divided/structure_factors` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and `DB_MIRROR_STRUCTURE_FACTORS` together. However, users may choose to specify `PDB_MIRROR_STRUCTURE_FACTORS` directly
- `local_pdb_dir = None` Enter the path directly to your local PDB repository.
- `verbose = False` verbose output
- `debug = False` debugging output
- `job_title = None` Job title in PHENIX GUI, not used on command line
- guiGUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`
- `controlstop_after_dock = None` Stop after docking step `stop_after_morph = False` Stop after morphing step `read_files = False` Read existing output files and use them if present `write_files = True` Write output files `nproc = 1` Number of processors to use on your local machine `ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example. `random_seed = 171731` Random seed `max_dirs = 1000` Maximum number of directories (dock_and_build_xxx) `verbose = False` Verbose output `quick = True` Run quickly
- `stop_after_dock = None` Stop after docking step
- `stop_after_morph = False` Stop after morphing step
- `read_files = False` Read existing output files and use them if present
- `write_files = True` Write output files
- `nproc = 1` Number of processors to use on your local machine
- `ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `random_seed = 171731` Random seed
- `max_dirs = 1000` Maximum number of directories (dock_and_build_xxx)
- `verbose = False` Verbose output

- quick = True Run quickly
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Docking processed AlphaFold2 and other predicted models in cryo-EM maps

dock_predicted_model.html

Author(s)

- dock_predicted_model: Tom Terwilliger

Purpose

Dock predicted model docks the domains from a model produced by AlphaFold, RoseTTAFold and other prediction software into a cryo EM map. It uses the connectivity of the model as a restraint in the docking process so that the docked domains normally are in a reasonable arrangement. It can take map symmetry into account.

The dock_predicted_model program is normally the second step in working on an AlphaFold or other predicted model with a cryo-EM structure.

The first step is to process the predicted model by trimming off all the uncertain residues in the predicted model and breaking up the remaining structure into a best guess of rigid domains with phenix.process_predicted_model .

The next step is to dock each of the domains of the processed model into the map, keeping plausible connectivity. This is done with phenix.dock_predicted_model .

The third step is to morph the predicted model onto the docked domains and then to rebuild all the parts of the predicted model using the density in the map.

Note that you can run all three steps sequentially with phenix.dock_and_rebuild .

How dock_predicted_model works:

The model input to dock_predicted_model is a model file that comes from your starting predicted model (AlphaFold model) but which has been split so that each chain represents one domain (fixed rigid unit) of the predicted model, and from which all uncertain residues are removed. This is normally the output of phenix.process_predicted_model .

Note that you can work with one chain at a time even in a cryo-EM map that has many chains.

The map input to dock_predicted model is normally your best sharpened or density-modified cryo-EM map. It can also be a map generated by any other procedure (including crystallography).

If you are able to mask your map, keeping only the part representing the region where this model (one chain) belongs, that can be very helpful. You can also box the map around this region. If you have a map that has many chains, this masking can greatly shorten the time for docking. If you don't know where in your map the model goes at all, you can supply the entire map. If your map has symmetry this will normally be found automatically.

The first step in docking the predicted model is to extract the unique part of the map. If the map is asymmetric, this is the region where there is density. If it is symmetric, it is the unique part of the density, chosen to make a compact unit. Note that this step may not be perfect, particularly in the case of a very asymmetric molecule. You can possibly do much better by segmenting the map by hand and selecting just the region representing

your molecule if you have the time.

The next step is to sort the domains in the processed model by size and dock them in the map. Each domain is docked and refined by rigid-body refinement. Then the transformation obtained for that domain is applied to all the other domains as a quick guess as to how they should be placed. If they match the density, they are refined as well. If a complete model can be obtained by connecting these pieces (with distances between residues in one domain and the next residues in another domain compatible with the number of residues in between that are missing), then the docking is complete. If it is not, additional domains are docked, until all have been tried.

When all domains are placed, all the residues in the domains are sorted and a single clean chain with chain ID from the original predicted model and gaps corresponding to the processed model is produced.

Examples

Standard run of dock_predicted_model:

Running dock_predicted_model is easy. From the command-line you can type:

```
phenix.dock_predicted_model model=my_model.pdb \
  processed_model_file=my_model_processed.pdb map_file=my_map.ccp4 \
  resolution=3
```

This will dock the domains (one for each chain ID) from my_model_processed.pdb into my_map.ccp4. It will rename all the chains to match the chain in my_model.pdb and arrange all the residues in order.

Possible Problems

For complicated molecules, this procedure may not work well because it does not examine every possible docking position of every domain, and a complex structure might have more than one copy of a particular chain. You can improve the success a lot if you are able to mask the map around one copy of the chain you are looking for.

If that doesn't work, you can try to dock the domains from your AlphaFold model using phenix.dock_in_map and asking for multiple copies of the domains, then you could manually try to choose which domains go together. You can then create one PDB file with one copy of each domain (you will need to create a PDB file that has just one chain ID and all residues arranged sequentially, but with gaps allowed) and go on to phenix.dock_and_rebuild with that as your docked model.

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesseq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
xray_data_file = None Input X-ray

data file (MTZ format).xray_data_label = None Label specifying which data column to use from xray_data_filexray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_filetruncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequencesdensity_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).msa_list = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.models_are_already_placed = False You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.model_copies_list = None List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like model_copies_list=1,2,1,1processed_model_file = None Processed model file (e.g., from phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.docked_model_file = None Docked model (e.g., output of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.morphed_model_file = None Morphed model (e.g., full model morphed to match output of dock_processed_model). If supplied, skip the process_predicted_model and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.previous_model_file = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If docked_model_file is also supplied, previous_model is used only as source of model information.scaffold_model = None Scaffold model. If supplied, used as target for docking of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied, model_copies_list is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.scaffold_minimum_chain_cc = 0.35 Chains in scaffold model that have scaffold_minimum_cc are not re-docked, but those with l...

- job_title = None Job title in PHENIX GUI, not used on command line

- input_filesseq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)xray_data_file = None Input X-ray data file (MTZ format).xray_data_label = None Label specifying which data column to use from xray_data_filexray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_filetruncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequencesdensity_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file

exactly.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factor. Only applies to protein chains (always B-factor for non-protein).msa_list = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.models_are_already_placed = False You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.model_copies_list = None List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like model_copies_list=1,2,1,1processed_model_file = None Processed model file (e.g., from phenix.process_predicted_model). If supplied, skip the process_predicted_model step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.docked_model_file = None Docked model (e.g., output of dock_processed_model). If supplied, use this file as the docked model. (Skip process_predicted_model and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.morphed_model_file = None Morphed model (e.g., full model morphed to match output of dock_processed_model). If supplied, skip the process_predicted_model and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.previous_model_file = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If docked_model_file is also supplied, previous_model is used only as source of model information.scaffold_model = None Scaffold model. If supplied, used as target for docking of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied, model_copies_list is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.scaffold_minimum_chain_cc = 0.35 Chains in scaffold model that have scaffold_minimum_cc are not re-docked, but those with lower CC are.base_scaffold_minimum_chain_cc = -1 Value of scaffold_...

- seq_file = None Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
- xray_data_file = None Input X-ray data file (MTZ format).
- xray_data_label = None Label specifying which data column to use from xray_data_file
- xray_test_flag_label = None Label specifying which test flag column (if any) to use from xray_data_file
- truncate_models_to_match_sequences = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
- density_select = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
- predicted_model = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
- b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factor. Only applies to protein chains (always B-factor for

non-protein).

- `msa_list` = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
- `models_are_already_placed` = False You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model.
- `model_copies_list` = None List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
- `processed_model_file` = None Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
- `docked_model_file` = None Docked model (e.g., output of `dock_processed_model`). If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.
- `morphed_model_file` = None Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
- `previous_model_file` = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
- `scaffold_model` = None Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
- `scaffold_minimum_chain_cc` = 0.35 Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
- `base_scaffold_minimum_chain_cc` = -1 Value of `scaffold_minimum_chain_cc` in cases where scaffold is known to be good.
- `fragments_model_file` = None Fragments model (e.g., `map_to_model.pdb`). Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `fragments_model` is used only as source of model information.
- `model_to_rebuild_file` = None Model to be rebuilt. Must be already in place (already docked). Must match supplied sequences exactly.
- `symmetry_file` = None Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. NOTE: symmetry file applies to map file in original position. NOTE 2: only proper symmetry (point-group, helical) is allowed NOTE 3: Only applies to cryo-EM maps

- `pae_file` = None Optional input json file with matrix of inter-residue estimated errors (pae file)
- `distance_model_file` = None Distance_model_file. A PDB or mmCIF file containing the model corresponding to the PAE matrix. Only needed if `weight_by_ca_ca_distances` is True.
- `search_model_copies` = None Used internally only
- `search_model` = None used internally only
- `map_modelfull_map` = None Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map` = False. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.`half_map` = None Input half map files. Usually supply one full map or 2 half maps`model` = None Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `full_map` = None Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map` = False. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
- `half_map` = None Input half map files. Usually supply one full map or 2 half maps
- `model` = None Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `output_filesoutput_model_prefix` = None Output files with superposed models will begin with this prefix`output_seq_file` = None Sequence file (possibly edited) written to this file by PredictAndBuild`pdb_out` = None Used internally only`temp_dir` = None Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `output_model_prefix` = None Output files with superposed models will begin with this prefix
- `output_seq_file` = None Sequence file (possibly edited) written to this file by PredictAndBuild
- `pdb_out` = None Used internally only
- `temp_dir` = None Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `crystal_inforesolution` = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).`scattering_table` = *None `n_gaussian` wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is `n_gaussian`, for cryoEM is `electron`.`wrapping` = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).`asymmetric_map` = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.`solvent_content` = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.`pdb70_text` = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally only`sequence` = None Old-style sequence string (alternative to sequence file).`chain_type` = *PROTEIN DNA RNA Chain type`msa` = None MSA as text. Supply MSA as text or as an `msa_file`. Used internally`templates_as_string` = None Templates as string (single PDB file with one or more chains)`space_group` = None Space group (normally read from the data file, applies to X-ray)`space_group_alternatives` = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group`unit_cell` = None Unit Cell (normally read from the data file,

applies to X-ray)unique_sequencesFile names and data labels.sequence = "Enter or edit sequence"
copies = 1 Copies of this sequencelabel = ""

- resolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).

- scattering_table = *None n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.

- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).

- asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.

- solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.

- pdb70_text = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally only

- sequence = None Old-style sequence string (alternative to sequence file).

- chain_type = *PROTEIN DNA RNA Chain type

- msa = None MSA as text. Supply MSA as text or as an msa_file. Used internally

- templates_as_string = None Templates as string (single PDB file with one or more chains)

- space_group = None Space group (normally read from the data file, applies to X-ray)

- space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group

- unit_cell = None Unit Cell (normally read from the data file, applies to X-ray)

- unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""

- sequence = "Enter or edit sequence"

- copies = 1 Copies of this sequence

- label = ""

- process_predicted_modelremove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)continuous_chain = False When removing low-confidence residues, only trim from endsplit_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=Truemaximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging

themminimum_sequential_residues = None Minimum length of a short segment to keep (reject at end).
 Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with
 alt_pae_params=Trueminimum_remainder_sequence_length = 15 used to choose whether the sequence
 of a removed segment is written to the remainder sequence file.b_value_field_is = *plddt rmsd b_value
 The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a
 B-factorinput_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are
 fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.minimum_plddt =
 None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd,
 whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd
 of 1.5. Minimum plddt values are fractional or not depending on the value of
 input_plddt_is_fractional.maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is
 defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum
 plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not
 depending on the value of input_plddt_is_fractional.default_maximum_rmsd = 1.5 Default value of
 maximum_rmsd, used if maximum_rmsd is not setsubtract_minimum_b = False If set, subtract the lowest
 B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low-
 confidence residues.pae_power = None If PAE matrix (predicted alignment error matrix) is supplied, each
 edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number
 of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 alt_pae_params=False and 2 for
 alt_pae_params=Truepae_cutoff = None If PAE matrix (predicted alignment error matrix) is supplied,
 graph edges will only be created for residue pairs with $\text{pae} < \text{pae_cutoff}$. Default is 5
 alt_pae_params=False and 4 for alt_pae_params=Truepae_graph_resolution = None If PAE matrix
 (predicted alignment error matrix) is supplied, pae_graph_resolution regulates how aggressively the
 clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and
 values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for
 alt_pae_params=Truealt_pae_params = True If PAE matrix is supplied, use alternative set of defaults
 (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4
 pae_graph_resolution...

- remove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)
- continuous_chain = False When removing low-confidence residues, only trim from ends
- split_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.
- maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.
- domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).
- adjust_domain_size = True If more that maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.
- minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=True
- maximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before merging them

- `minimum_sequential_residues = None` Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or `alt_pae_params=False` and 4 for pae_matrix with `alt_pae_params=True`
- `minimum_remainder_sequence_length = 15` used to choose whether the sequence of a removed segment is written to the remainder sequence file.
- `b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor
- `input_plddt_is_fractional = None` You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.
- `minimum_plddt = None` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `maximum_rmsd = 1.5` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `default_maximum_rmsd = 1.5` Default value of `maximum_rmsd`, used if `maximum_rmsd` is not set
- `subtract_minimum_b = False` If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low- confidence residues.
- `pae_power = None` If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/pae^{**pae_power})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 `alt_pae_params=False` and 2 for `alt_pae_params=True`
- `pae_cutoff = None` If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with `pae < pae_cutoff`. Default is 5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `pae_graph_resolution = None` If PAE matrix (predicted alignment error matrix) is supplied, `pae_graph_resolution` regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `alt_pae_params = True` If PAE matrix is supplied, use alternative set of defaults (`minimum_domain_length=20` `minimum_sequential_residues=10` `pae_power=2` `pae_cutoff=4` `pae_graph_resolution=4`). Standard parameters (`alt_pae_params=False`) are: (`minimum_domain_length=10` `minimum_sequential_residues=5` `pae_power=1` `pae_cutoff=5` `pae_graph_resolution=0.5`).
- `weight_by_ca_ca_distance = False` Adjust the edge weighting for each residue pair according to the distance between CA residues. If this is True, then `distance_model` can be provided, otherwise supplied model will be used. See also `distance_power`
- `distance_power = 1` If `weight_by_ca_ca_distance` is True, then edge weights will be multiplied by $1/distance^{**distance_power}$.
- `stop_if_no_residues_obtained = True` Raise Sorry and stop if processing yields no residues
- `keep_all_if_no_residues_obtained = False` Keep everything if processing yields no residues
- `vrms_from_rmsd_intercept = 0.25` Estimate of vrms (error in model) from pLDDT will be based on `vrms_from_rmsd_intercept + vrms_from_rmsd_slope * pLDDT` where mean pLDDT of

non-low_confidence_residues is used.

- `vrms_from_rmsd_slope = 1.0` Estimate of vrms (error in model) from pLDDT will be based on $\text{vrms_from_rmsd_intercept} + \text{vrms_from_rmsd_slope} * \text{pLDDT}$ where mean pLDDT of non-low_confidence_residues is used.

- `break_into_chunks_if_length_is = 1500` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae matrix

- `chunk_size = 600` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae_matrix

- `overlap_size = 200` If a sequence is at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size` for domain identification using `split_model_by_compact_regions` without a pae_matrix

- `searchdock_chains_individually = False` Dock each chain (identified by `chain_id` field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with `create_unique_chain_at_end=True` for predicted models to put the model back together. `create_unique_chain_at_end = None` Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if `dock_chains_individually = True` and False otherwise. `minimum_cc_to_keep_domain = 0.2` Do not keep domains in `create_unique_chain_at_end` if cc is less than

`minimum_cc_to_keep_domain`. `weight_sequential_fragments_by_distance = None` Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if `create_unique_chain_at_end` is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain. `choose_better_of_individual_and_group_docking = None` Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain. `low_res_search = True` Try to fit by searching at low resolution first. `dock_with_mr = True` Try using MR to dock first. `ssm_search = None` Try to fit by searching for secondary structure. Default is False for `dock_in_map` and True for `dock_and_rebuild` if `dock_with_mr` is not used. `refine_cycles = 3` rigid-body refinement cycles. `ssm_search_min_cc = 0.30` Stop ssm search if this cc

achieved. `backup_resolution_cutoff_searches = 2` If initial fitting does not work, try up to this many times increasing resolution by `backup_resolution_ratio` each time. `backup_resolution_ratio = 1.67` If initial fitting does not work, try up to `backup_resolution_cutoff_searches` times increasing the resolution by

`backup_resolution_ratio` each time. `resolution_radius_scale = 0.5` Resolution for low-res search will be `resolution_radius_scale` times the radius of gyration of the search model. `align_moments = False` Try to fit by aligning moments of inertia if size of density region and molecule are similar.

`max_radius_ratio = 2`. Radius of gyration of molecule and density must be within `max_radius_ratio` of each other.

`radius_scale = 1.5` Mask for density and atoms will be `radius_scale` times the radius of gyration of model. `use_symmetry = True` If `search_model_copies` or `search_map_copies` are the same for each model and greater than one and `allow_symmetry` is set, use symmetry in the map to place all copies after the first one. `skip_if_low_cc = True` Skip solution before rigid-body refinement if map-model CC is less than half of

`min_cc`. `rigid_body_refinement = True` Run rigid-body refinement on final

model. `rigid_body_refinement_single_unit = True` Run rigid-body refinement with just one unit (do not

break up into chains). `rigid_body_refinement_split_method = *chain_id segid` When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit=False`), use either `chain_id` or `segid` to

split up molecule rigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specified append_to_fixed_model = True Append placed search model to fixed model (if any) after search min_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.run_in_boxes = True Run on sub-boxes and combine at the end target_box_size = 60 Try to get boxes about this big on a side (grid units) target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_size box_to_run = None Run only this box box_overlap_scale = 1 box overlap (ov...

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.

- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

- weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.

- choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain

- low_res_search = True Try to fit by searching at low resolution first

- dock_with_mr = True Try using MR to dock first

- ssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.

- refine_cycles = 3 rigid-body refinement cycles

- ssm_search_min_cc = 0.30 Stop ssm search if this cc achieved

- backup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each time

- backup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each time

- resolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.

- align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similar

- max_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each other

- radius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of model

- use_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first

one.

- `skip_if_low_cc = True` Skip solution before rigid-body refinement if map-model CC is less than half of `min_cc`.
- `rigid_body_refinement = True` Run rigid-body refinement on final model
- `rigid_body_refinement_single_unit = True` Run rigid-body refinement with just one unit (do not break up into chains)
- `rigid_body_refinement_split_method = *chain_id segid` When splitting up molecule for rigid-body refinement (if `rigid_body_refinement_single_unit=False`), use either `chain_id` or `segid` to split up molecule
- `rigid_body_refinement_resolution = None` Run rigid-body refinement at this resolution if specified
- `append_to_fixed_model = True` Append placed search model to fixed model (if any) after search
- `min_cc = 0.4` If quick run, stop if minimum CC is achieved in local search. Also always skip if starting `CC_mask` is less than $1/4$ `min_cc`.
- `run_in_boxes = True` Run on sub-boxes and combine at the end
- `target_box_size = 60` Try to get boxes about this big on a side (grid units)
- `target_boxes = None` Try to get this many boxes. Default is `nproc` unless this makes box size much smaller than `target_box_size`
- `box_to_run = None` Run only this box
- `box_overlap_scale = 1` box overlap (overlap of boxes) will be `box_overlap_scale` times the density radius
- `edge_ratio = 10` box edge `box_overlap` times `edge_ratio`
- `density_radius = None` Radius for density to be cut out and compared. Default is 6 times the resolution.
- `model_radius = 3` Radius for removing density near `fixed_model`
- `zero_value = 0` Value to set map in regions overlapping fixed model
- `density_peaks = 20` Number of NCS-related peaks of density to check
- `delta_phi = 20` Angular spacing of search
- `max_rot = None` Maximum rotations to try
- `rotz_only = None` Rotate only around Z
- `single_positions_to_try = 10` Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- `max_position_shift_frac = 0.05` Maximum fractional positional shift in `single_positions` run
- `min_relative_cc = 0.67` Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- `sieve_fit = None` Use `sieve_fit` fraction of single positions in fitting. If `None`, use all
- `ncs_copies_max = None` Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with `phenix.pdbtools placed_model.pdb keep="model 1"` etc.
- `start_rot = None` Three numbers `rotx`, `rotz`, `roty` defining the starting rotation of the search model. Normally used along with `delta_phi=1000` or `max_rot=1` to generate exactly one defined rotation.

- `search_center` = None Optional coordinates in search model for centering search. Note this is different from `target_search_center` which is the location xyz in the map to look.
- `search_center_selection` = None Optional selection defining coordinates in search model for centering search.
- `target_search_center` = None Optional coordinates in reference map where `search_center` should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to `target_search_center/density_radius`.
- `model_search_position` = None Optional coordinates (usually part of search model) for matching to `target_search_position` after transformation. These can be specified in addition to `target_search_center`. Can have multiple `model_search_position` and `target_search_position` pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- `target_search_position` = None Target positions for `model_search_position` after transformation.
- `search_position_radius` = None Radius for comparison of `target_search_position` and transformed `search_position` values. Default is `density_radius`. If specified, must be a single value or the same number of values as entries in `model_search_position` and `target_search_position`
- `rot_id_n` = None Number of rotation groups. Along with `rot_id_group`, allows defining groups of rotations to be carried out in one run. `short_caption` = Number of rotation groups
- `rot_id_group` = None rotation group to include. See `rot_id_n`.
- `map_box` = True Run `map_box` to extract useful part of map before search
- `fix_search_position` = False You can choose to not move your model center of mass to the origin by fixing the search position
- `search_box_size` = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times `density_radius`
- `lower_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `upper_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `keep_search_order` = None Keep search order as input
- `remove_water` = False Remove waters and other hetero atoms from input files
- `buildrebuild_strategy` = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
`refine_only` = None Refine only, no rebuilding steps (set automatically with Quick)
`refine_only_resolution` = 3.5 Default cutoff for using Refine only
`run_fit_loops` = True Run standard fit loops
`run_iterative_morph` = True Run iterative morphing
`run_trace_loops_through_density` = True Run loop fitting with trace_through_density algorithm
`run_refine` = True Refine morphed model. Required for `run_fit_loops` or `run_trace_loops_through_density`
`run_iterative_resolution_refine` = True Refine morphed model with iterative resolution method
`run_extend` = True Extend ends of morphed model
`extract_unique` = True Extract unique part of map. Applies after `density_select` (if set).
`use_symmetry_in_extract_unique` = True Use symmetry from symmetry file (if available) when extracting unique part of map.
`acceptable_docking_cc` = 0.5 Acceptable docking CC for a chain
`minimum_docking_cc` = 0.15 Minimum docking CC for a chain
`minimum_cutoff` = 0.1 Minimum cutoff for estimating density in better parts of model
`reasonable_cc_ratio` = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to

make sure a chain entirely in very bad density is not kept
 reasonable_cc_diff = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
 shift_field_distance = None Shift field characteristic distance (default = 10 Å) for morphing
 cc_sd_ratio = 3. Keep residues with CC at least within cc_sd_ratio of the mean CC for good residues
 cc_sd_ratio_end = 2. Keep residues with CC at least within cc_sd_ratio_end of the mean CC for good residues (applies to end of fragments)
 cc_sd_ratio_ok = 2. Residues with CC at least within cc_sd_ratio_ok of the mean CC for good residues are ok (do not delete on the basis of plddt)
 max_gap_ratio = 3. Allow CA-CA distance to be up to max_gap_ratio times expected
 maximum_connectivity_deviation = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus 2 * resolution.
 keep_fraction_of_best = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found is at least acceptable_docking_cc.
 keep_maximum_entries = 10 Maximum dock positions to keep
 rmsd_for_similar_placement = None RMSD value indicating that two placements are similar. default is resolution of map
 rigid_body_refine_cycles = 1 Refinement cycles for rigid-body refinement
 overlap_ca_ca_distance = 3 Overlap distance for CA-CA atoms (or P-P)
 ca_ca_distance = 3.8 CA-CA distance (or P-P)
 allowed_fraction_overlapping = 0.10 Allowed fraction overlapping
 maximum_combinations = 100000 Maximum combinations of placements to consider
 proceed_with_any_symmetry = False Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
 box_cushion = 20 Size of buffer around model when boxing
 refine_cycles = 3 Refinement cycles
 loop_refine_cycles = 5 Refinement cycles for loops
 loop_backup_residues = 3 Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
 residues_to_trim = 5 Residues to trim on each end of all trimmed fragments
 find_ncs_from_model = True Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)
 allow_split_more_than_one_chain_for_mr = False Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split
 multiple_chains.acceptable_cc_ratio = 0.8 Keep segments with CC at least equal to average of base segments times minimum_cc_ratio minus difference between segment confide...

- rebuild_strategy = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if X-ray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
- refine_only = None Refine only, no rebuilding steps (set automatically with Quick)
- refine_only_resolution = 3.5 Default cutoff for using Refine only
- run_fit_loops = True Run standard fit_loops
- run_iterative_morph = True Run iterative morphing
- run_trace_loops_through_density = True Run loop fitting with trace_through_density algorithm
- run_refine = True Refine morphed model. Required for run_fit_loops or run_trace_loops_through_density
- run_iterative_resolution_refine = True Refine morphed model with iterative resolution method
- run_extend = True Extend ends of morphed model
- extract_unique = True Extract unique part of map. Applies after density_select (if set).
- use_symmetry_in_extract_unique = True Use symmetry from symmetry file (if available) when extracting unique part of map.
- acceptable_docking_cc = 0.5 Acceptable docking CC for a chain

- `minimum_docking_cc` = 0.15 Minimum docking CC for a chain
- `minimum_cutoff` = 0.1 Minimum cutoff for estimating density in better parts of model
- `reasonable_cc_ratio` = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `reasonable_cc_diff` = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `shift_field_distance` = None Shift field characteristic distance (default = 10 Å) for morphing
- `cc_sd_ratio` = 3. Keep residues with CC at least within `cc_sd_ratio` of the mean CC for good residues
- `cc_sd_ratio_end` = 2. Keep residues with CC at least within `cc_sd_ratio_end` of the mean CC for good residues (applies to end of fragments)
- `cc_sd_ratio_ok` = 2. Residues with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of plddt)
- `max_gap_ratio` = 3. Allow CA-CA distance to be up to `max_gap_ratio` times expected
- `maximum_connectivity_deviation` = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus $2 * \text{resolution}$.
- `keep_fraction_of_best` = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.
- `keep_maximum_entries` = 10 Maximum dock positions to keep
- `rmsd_for_similar_placement` = None RMSD value indicating that two placements are similar. default is resolution of map
- `rigid_body_refine_cycles` = 1 Refinement cycles for rigid-body refinement
- `overlap_ca_ca_distance` = 3 Overlap distance for CA-CA atoms (or P-P)
- `ca_ca_distance` = 3.8 CA-CA distance (or P-P)
- `allowed_fraction_overlapping` = 0.10 Allowed fraction overlapping
- `maximum_combinations` = 100000 Maximum combinations of placements to consider
- `proceed_with_any_symmetry` = False Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
- `box_cushion` = 20 Size of buffer around model when boxing
- `refine_cycles` = 3 Refinement cycles
- `loop_refine_cycles` = 5 Refinement cycles for loops
- `loop_backup_residues` = 3 Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
- `residues_to_trim` = 5 Residues to trim on each end of all trimmed fragments
- `find_ncs_from_model` = True Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)
- `allow_split_more_than_one_chain_for_mr` = False Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split multiple chains.
- `acceptable_cc_ratio` = 0.8 Keep segments with CC at least equal to average of base segments times `minimum_cc_ratio` minus difference between segment confidence (pLDDT) and confidence cutoff

(typically 0.7)

- `low_res_if_multiple_solutions = 3.5` Try phaser MR at increasing lower resolutions up to this value if multiple solutions are found
- `delta_low_res = 0.5` Resolution increment for `low_res_if_multiple_solutions`
- `controlstop_after_dock = True` Stop after docking step
`stop_after_morph = False` Stop after morphing
- `stepread_files = False` Read existing output files and use them if present
`write_files = True` Write output files
- `nproc = 1` Number of processors to use on your local machine
- `ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `random_seed = 171731` Random seed
- `max_dirs = 1000` Maximum number of directories (`dock_and_build_xxx`)
- `verbose = False` Verbose output
- `quick = True` Run quickly
- `gui` GUI-specific parameter required for output directory
`output_dir = None`
- `output_dir = None`

Build water into cryo-EM map using phenix.douse

douse.html

phenix.douse is a tool to build water into cryo-EM maps.

Facts about the procedure

- 1) The water is built into density map also ensuring each water is linked to the protein via H-bond, directly or indirectly (via another water).
- 2) Atomic model of the macro-molecule is assumed to be as complete and accurate as possible (for example, B factors are refined individually).
- 3) Water is looked for in a band of 6Å width extending out from the protein surface. This allows to build about two solvation layers around protein. Any unmodeled density in this region is treated as potential water.
- 4) Minimal (dist_min) and maximal (dist_max) distances between any water and protein or other water are 2 and 3.2Å, correspondingly.
- 5) Not all protein atoms are used to assess contacts with prospective water molecules but only those that can participate in H-bonds. These atoms are additionally filtered to leave a subset of atoms that are confidently placed in the map. The confidence is quantified by CCmask (cc_mask_threshold_interacting_atoms) - the correlation between model-calculated and experimental maps for the atom in question. The default is set to 0.5. The CCmask is calculated in a sphere of 2Å radius (atom_radius) around the atom in question.
- 6) Only strong enough density peaks are considered as potential water-candidates. "Strong enough" is defined as the average of map density values interpolated at atomic centers of protein atoms times the attenuation scale factor map_threshold_scale which is set to 0.25 by default.
- 7) The remaining peaks are filtered by two additional criteria.

One is CCmask (cc_mask_filter) with the cutoff value of 0.5 (cc_mask_filter_threshold). The CCmask is calculated similarly to "5)". Since CCmask strongly depends on atomic B factors and B factors of newly placed water-candidates are not known, unrestrained isotropic B factor refinement is performed for newly placed water prior to CCmask calculation.

Another selection criterion is peak sphericity (scc). The peak sphericity is evaluated as minimal correlation between any combination of two sets of map values interpolated along vectors going out from the peak center and having length of 1Å. For a perfectly spherical peak the correlation between any "pair of vectors" (any pair of sets of map values along these vectors) must be 1. The water is rejected if scc is less than 0.5.

Comments

- 1) Geometric (water-protein or water-water distance) and chemical (types of interacting atoms) criteria are rather strong and rigid criteria and do not allow much room for variation. The set default values are taken from standard literature.
- 2) Map-based criteria, such as various correlations, are more heuristics and rule-of-thumb based. For correlations, 1 means perfect similarity, 0 is no similarity and -1 is anti-similarity. So a choice like 0.5 means some similarity, half-way between none and perfect. Choosing a larger value will mean choosing more confident waters.

3) Map sphericity reflects the observation that water density appears to be approximately spherical most of the time. This observation breaks down in a number of special cases that need a special treatment and that can be a subject of future improvement of the tool. For example, at high resolution water with high directional mobility typically yield elliptical peak shapes. Water coordinated by heavier ions may not even resolved as individual peak or even if resolved as individual peak its sphericity can be largely perturbed.

4) Peak height threshold for selecting peak candidates is another rather ad-hoc criterion. It uses peak heights of confidently placed protein atoms as a guide. However, since water molecules are more mobile and often partially occupied, their peaks are expected to be weaker than peaks of better resolved protein atoms. By how much weaker? That is unknown. Thus some arbitrary attenuation scale is used.

5) Building 'single-atom' entities, such as water or ions, is a much more difficult task to do compared to building more feature-rich molecules such as proteins or multi-atom ligands. This is because it is much easier to confuse a true peak with a noise peak, while a constellation of many peaks (corresponding to protein chain, for example) contains very important information about shape and connectivity, making interpretation such peaks in terms of atomic model easier. Therefore, water building procedure is unlikely to be ever fully automated.

Availability

Command line, GUI.

Required inputs

Atomic model (PDB or mmCIF format) and map (MRC format).

resolution is required if `cc_mask_filter` is set to True.

Optional inputs

`mode = whole or per_chain`: work on the whole model or per chain. Working on the whole model may require a lot of memory.

`sphericity_filter = True`: use peak sphericity criteria. Disable to add more water.

`scc = 0.5`: map peak sphericity characteristic. Decrease to add more water.

`keep_input_water = False`: keep or remove existing water.

`dist_min = 2.0`: minimal peak-peak or peak-atom distance. Decrease to add more water.

`dist_max = 3.2`: maximal peak-peak or peak-atom distance. Increase to add more water.

`step = 0.3`: map re-sampling step.

`map_threshold_scale = 0.5`: map threshold for peak selection = (mean map value at atom centers) * `map_threshold_scale`. Decrease to add more water.

`cc_mask_filter = True`: reject water with `CC_mask < cc_mask_filter_threshold`.

`cc_mask_filter_threshold = 0.5`: `CC_mask` threshold for rejecting water.

`resolution`: map resolution. Required if `cc_mask_filter=True`

Limitations

P1 symmetry is the only supported symmetry.

Assumptions

Macromolecular part of the atomic model is as complete as possible (no missing chains or other unmodelled parts present). The model is fully refined including individual B-factors.

Map resolution is sufficiently high to expect water oxygens be visible in the map.

Example

```
phenix.douse model.pdb map.mrc resolution=2.5
```

resolution is required if `cc_mask_filter` is set to `True`.

More information

https://phenix-online.org/phenixwebsite_static/mainsite/files/presentations/water.pdf

This document reflects the procedure at the time of making the document. The procedure in most current version of Phenix can be (slightly) different.

Author

Pavel Afonine (PAfonine@lbl.gov)

phenix.dynamics: simple model perturbation

dynamics.html

This program performs very crude molecular dynamics on a model, usually for the purpose of removing any bias in R-free due to previous refinement against a different test set. It is not suitable for generating physically realistic trajectories, due to the lack of attractive forces such as electrostatics, and lengthy runs will result in the model falling apart. However, it is very effective for de-biasing while maintaining approximately correct geometry, and is also very fast (unlike simulated annealing in phenix.refine).

The inputs for the program are very similar to those for phenix.geometry_minimization; the only required files are a model and any necessary restraints (CIF) files for non-standard ligands. By default the dynamics is run at approximately room temperature; higher temperatures will result in faster movement. Note however that unlike simulated annealing, the model will tend to unfold rapidly at high temperatures.

Alternatives and complementary programs

Simulated annealing has some advantages due to the ability to simultaneously de-bias and optimize the model against the data, which prevents the structure from unfolding at high temperatures. The main disadvantage is the much longer runtime. phenix.pdbtools can be used to reset B-factors, or shake coordinates without geometry restraints. The latter is redundant if dynamics is run, but setting the B-factors to suitably low starting values (approximately 10 for most resolutions) will help remove bias. Both of the above steps can also be run at the start of phenix.refine; click the "Modify start model" button in the GUI to access options. To perturb structures for use as molecular replacement models, there is a separate program providing an interface to the normal modes method in Phaser (also available in the GUI under "Model tools"). This does a better job conserving local structure.

- Simulated annealing has some advantages due to the ability to simultaneously de-bias and optimize the model against the data, which prevents the structure from unfolding at high temperatures. The main disadvantage is the much longer runtime.
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List of all available keywords

silent = False write_geo_file = True file_name = None show_states = False restraints = None
restraints_directory = None output_file_name_prefix = None directory = None job_title = None Job title in
PHENIX GUI, not used on command line fix_rotamer_outliers = True Remove outliers allow_allowed_rotamers
= True More strict fixing outliers stop_for_unknowns = True dynamics_type = *cartesian stop_at_diff = None
stop after reaching specified cutoff value pdb_interpretationsort_atoms = True use_ncs_to_build_restraints =
True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand =
*None all SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False
allow_polymer_cross_special_position = False correct_hydrogens = True c_beta_restraints = True

use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
 disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
 link_distance_cutoff = 3 Length of link between the linked residues
 disulfide_distance_cutoff = 3
 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2
 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1
 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only**nonbonded_weight = None
 Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
 const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
 translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False
 restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
 restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry
 restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpd1 = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry
 restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraints
 action = *pre post Choice to look here before GeoStd, the default, or after
 secondary_structures_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
 from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
 max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
 use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
 max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
 enabled = False Turn on secondary structure restraints (main switch)
 proteinenabled = True Turn on secondary structure restraints for protein
 search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
 distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
 distance_cut_n_o...

- silent = False
- write_geo_file = True
- file_name = None
- show_states = False
- restraints = None
- restraints_directory = None
- output_file_name_prefix = None
- directory = None
- job_title = None Job title in PHENIX GUI, not used on command line
- fix_rotamer_outliers = True Remove outliers
- allow_allowed_rotamers = True More strict fixing outliers

- stop_for_unknowns = True
- dynamics_type = *cartesian
- stop_at_diff = None stop after reaching specified cutoff value
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT
 disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False
 correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H
 distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None
 exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it
 may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between
 the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides =
 True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
 chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None
 default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers
 only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16
 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are
 missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
 translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False
 restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
 restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL)
 for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False
 Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl =
 False rdl_selection = *all TRP hpdl = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and
 RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of
 user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or
 aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary
 structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much
 slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many
 fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make
 from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if
 search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same
 for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a
 representative of all chains with the same sequence. Alternative is to examine each chain individually.
 Can be much slower with use_representative_of_chain=False if there are many symmetry copies.
 Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method =
 from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure
 restraints (main switch)proteinenabled = True Turn on secondary structure restraints for
 proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary
 structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5
 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true,
 h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True
 helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection =
 None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10
 helices are used for hydrogen-bond res...
- sort_atoms = True
- use_ncs_to_build_restraints = True

- `show_restraints_histograms = True`
- `flip_symmetric_amino_acids = True`
- `superpose_ideal_ligand = *None all SF4 F3S DVT`
- `disable_uc_volume_vs_n_atoms_check = False`
- `allow_polymer_cross_special_position = False`
- `correct_hydrogens = True`
- `c_beta_restraints = True`
- `use_neutron_distances = False` Use neutron X-H distances (which are longer than X-ray ones)
- `disulfide_bond_exclusions_selection_string = None`
- `exclusion_distance_cutoff = 3` If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- `link_distance_cutoff = 3` Length of link between the linked residues
- `disulfide_distance_cutoff = 3`
- `add_angle_and_dihedral_restraints_for_disulfides = True`
- `dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic`
- `chir_volume_esd = 0.2`
- `max_reasonable_bond_distance = 50.0`
- `nonbonded_distance_cutoff = None`
- `default_vdw_distance = 1`
- `min_vdw_distance = 1`
- `nonbonded_buffer = 1` **EXPERIMENTAL, developers only**
- `nonbonded_weight = None` Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- `const_shrink_donor_acceptor = 0` **EXPERIMENTAL, developers only**
- `vdw_1_4_factor = 0.8`
- `min_distance_sym_equiv = 0.5`
- `custom_nonbonded_symmetry_exclusions = None`
- `translate_cns_dna_rna_residue_names = None`
- `proceed_with_excessive_length_bonds = False`
- `restraints_librarycdl = True` Use Conformation Dependent Library (CDL) for geometry
- `restraintsinclude_modified_amino_acid_in_cdl = False` `mcl = True` Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ...`cis_pro_eh99 = True` `omega_cdl = False` Use Omega Conformation Dependent Library (omega-CDL) for geometry `restraintscdl_svl = False` `rdl = False` `rdl_selection = *all TRP` `hpdI = False` `cdl_nucleotidesenable = False` Use RestraintsLib for DNA and RNA for geometry `restraintsesd = *phenix` `csd factor = 2.0` `user_suppliedpath = None` Root directory of user supplied restraints`action = *pre` post Choice to look here before GeoStd, the default, or after
- `cdl = True` Use Conformation Dependent Library (CDL) for geometry restraints
- `include_modified_amino_acid_in_cdl = False`

- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ...
- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuresss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic

acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond

- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
- restrain_hbond_angles = True
- helixserial_number = None helix_identifier = None enabled = True Restrain this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None
- serial_number = None
- helix_identifier = None
- enabled = True Restrain this particular helix
- selection = None
- helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- sheetenabled = True Restrain this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None
- enabled = True Restrain this particular sheet
- first_strand = None
- sheet_id = None
- sigma = 0.05
- slack = 0
- top_out = False
- angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
- angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
- strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None

- selection = None
- sense = parallel antiparallel *unknown
- bond_start_current = None
- bond_start_previous = None
- hbonddonor = None acceptor = None
- donor = None
- acceptor = None
- nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenable = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residue
- saenger_class = 0 Type of base-pairing
- restrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0
- parallelity_sigma = 0.0335 stacking_pairenable = True
- Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Turn on secondary structure restraints for nucleic acids
- hbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- scale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- base_pairenable = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residue
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- restrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176
- restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True
- restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0
- parallelity_sigma = 0.0335

- stacking_pairenable = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1.0 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- enabled = False
- exclude_outliers = True
- selection = all
- sigma = 0.2
- limit = 1.0
- top_out = False
- reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- automatic_restraintsside_chain_parallelity = False
- side_chain_parallelity = False
- automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3.0 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False
- link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.
- link_none = False
- link_metals = Auto
- link_residues = True
- link_amino_acid_rna_dna = False
- link_carbohydrates = True
- link_ligands = True
- link_small_molecules = False

- metal_coordination_cutoff = 3.0
- amino_acid_bond_cutoff = 1.9
- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!!
 Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.cis_threshold = 45
 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False
 omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield
 emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran
 plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or
 restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using

oldfield potential.oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027

- ramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False
- omega_esd_override_value = None
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- esd = 10.0
- weight_scale = 1.0
- dist_weight_max = 10.0
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.
inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use botholdfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))$
weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027
emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0
weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest

highest_probability random weighted_random

- enabled = False
- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min (=2.), \min(distance_weight_max (=10.), current_distance_to_allowed))$ weight_scale = 0.01 distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region. distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region. plot_cutoff = 0.027
- weight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \cdot 1 / esd^2 \cdot \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) \cdot weight_scale \cdot \max(2.0, current_distance_to_allowed) \cdot 1 / esd^2 \cdot weight_scale \cdot \max(distance_to_allowed_cutoff, current_distance_to_allowed) \cdot weight_scale (=0.01) \cdot \max(distance_weight_min (=2.), \min(distance_weight_max (=10.), current_distance_to_allowed))$
- weight_scale = 0.01
- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0

delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9
o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable =
True

- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5
bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5
chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5
nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5
dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3
planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions
= 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will
be checked and filtered if needed.reference = None Residue selection string for the complete master
NCS copyselection = None Residue selection string for each NCS copy location in ASU

- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU
- ncs_searchSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 2. limit of rms difference between chains to be considered as copiesresidue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residuestry_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atomsvalidate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- enabled = False Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- chain_max_rmsd = 2. limit of rms difference between chains to be considered as copies
- residue_match_radius = 4.0 Maximum allowed distance difference between pairs of matching atoms of two residues
- try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
- validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.
- clash_guardnonbonded_distance_threshold = 0.5 max_number_of_distances_below_threshold = 100 max_fraction_of_distances_below_threshold = 0.1
- nonbonded_distance_threshold = 0.5
- max_number_of_distances_below_threshold = 100
- max_fraction_of_distances_below_threshold = 0.1
- geometry_restraintseditsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None

angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete
change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 =
None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals =
None chiralities = None planarities = None parallelities = None

- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None
atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and
symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example:
symmetry_operation = -x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out =
False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None
atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change
atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None
angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete
change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 =
None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0

- excessive_bond_distance_limit = 10

- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None
symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with
atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation =
-x-1,-y,zdistance_ideal = None sigma = None slack = None limit = -1.0 top_out = False

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with
atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z

- distance_ideal = None

- sigma = None

- slack = None

- limit = -1.0

- top_out = False

- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None
atom_selection_3 = None angle_ideal = None sigma = None

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None

- angle_ideal = None

- sigma = None

- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None
atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma
= None periodicity = 1

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None
- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1.0 limit = 15.0 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common

residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copies
residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- enabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.

- file = None

- use_starting_model_as_reference = False

- sigma = 1.0

- limit = 15.0

- hydrogens = False Include dihedrals with hydrogen atoms

- main_chain = True Include dihedrals formed by main chain atoms

- side_chain = True Include dihedrals formed by side chain atoms

- fix_outliers = True Try to fix rotamer outliers in refined model

- strict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliers

- auto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraints

- secondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)

- reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- reference = None

- selection = None

- file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- search_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.
exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
chain_max_rmsd = 100. limit of rms difference between chains to be considered as copies
residue_match_radius = 1000 Maximum allowed distance difference between pairs of matching atoms of two residues
try_shortcuts = False Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
minimum_number_of_atoms_in_copy = 3 Do not create ncs groups where master and copies would contain less than specified amount of atoms
validate_user_supplied_groups = True Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 100`. limit of rms difference between chains to be considered as copies
- `residue_match_radius = 1000` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `cartesian_dynamicstemperture = 300` `number_of_steps = 200` `time_step = 0.0005`
`initial_velocities_zero_fraction = 0` `n_print = 100` `verbose = -1` `random_seed = None` `n_collect = 10`
`stop_cm_motion = True`
- `temperature = 300`
- `number_of_steps = 200`
- `time_step = 0.0005`
- `initial_velocities_zero_fraction = 0`
- `n_print = 100`
- `verbose = -1`
- `random_seed = None`
- `n_collect = 10`
- `stop_cm_motion = True`

Docking a model into a cryo-EM map with em_placement

em_placement.html

Authors

- Randy J Read, Claudia Millán, Airlie J McCoy, Thomas C Terwilliger

Purpose

The em_placement tool docks the input model in a cryo-EM reconstruction, using the half-maps in an adaptive search strategy.

How em_placement works

The half-maps are compared and analysed to determine signal and noise power as smooth functions in Fourier space. This information is used to predict whether or not a rotation search with a structural model over the full map volume is likely to succeed. If not, the maps are divided into sub-volumes small enough that the noise will be reduced to a level where signal from the rotation search can be detected. Following this, translation searches in the full map volume or the sub-volumes are carried out. Finally, rigid-body refinement is carried out using spherical sub-volumes centered on potential solutions to give the final solutions and confidence indicators.

A log-likelihood-gain (LLG) score greater than 60 suggests that a solution is significantly better than random.

NB: the statistical analysis requires the half-maps to correspond to the full volume of the cryo-EM reconstruction, so cut-out boxes cannot be used. Half-maps are required for a search over the whole reconstruction volume, whereas just a single full map can be used optionally for a search over a local volume.

Running em_placement

There is an EMplacement GUI to run em_placement, but it can also be controlled by an input script using the phil syntax. A basic phil script would look like the following, in which a model derived from the membrane domain of PDB entry 4cof is docked into the reconstruction of the GABA receptor in EMDB entry 11657.

```
em_placement
{
  remove_phaser_tng_folder = True

  map_model
  {
    half_map = emd_11657_half_map_1.map
    half_map = emd_11657_half_map_2.map
    best_resolution = 1.7
    point_group_symmetry = C5
    sequence_composition = 7a5v.fa
  }

  biological_unit {
    molecule
    {
      molecule_name = 4cofA_membrane
      model_file = 4cofA_membrane.pdb
      starting_model_vrms = 1.2
    }
  }
}
```

```
}
```

If the phil script is in a file named `docking_script.phil`, the search would be run from the command-line as follows.

```
phenix.voyager.em_placement docking_script.phil
```

Most parameters specified in this script have been named in a way intended to convey the purpose of that parameter.

The `remove_phasertng_folder` parameter is activated to clean up the graph database produced by phasertng, which could be used in other circumstances as part of a larger automation framework or for debugging.

If the `point_group_symmetry` is provided, the symmetry detected in the reconstruction is applied to generate a full assembly from a single copy.

The `sequence_composition` parameter specifies the name of a file containing the sequences of all the components in the reconstruction.

By default, `em_placement` carries out an adaptive search, which is normally recommended. Aspects of this search can be over-ridden with expert user keywords specified below, for example to force a full 6D search or to force only a single rotation search over the whole ordered volume regardless of the estimated signal.

Literature

Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023). Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023).

- Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023).
- Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023).

List of all available keywords

`em_placementremove_phasertng_folder = True` Removes the phasertng folder with the databases at the end of the run
`level = process concise *summary logfile verbose testing` Set the verbosity level
`job_title = None` Job title in PHENIX GUI, not used on command line
`map_modelfull_map = None` Input full map file as alternative to half-map
`half_map = None` Input half map files
`point_group_symmetry = C1` Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
`best_resolution = None` Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically
`cell_rescale = None` Rescaling factor for map cell or magnification factor
`sequence_composition = None` Sequence file for entire EM reconstruction. Ignored if `fixed_scattering_ratio` is set, in which case total composition is set to model composition times `fixed_scattering_ratio`
`fixed_scattering_ratio = None` If set, total composition is model composition times this factor
`region_approach = *automatic manual rigid_body` Method for specifying search regions
`region_for_docking` Manual specification for region(s)
`name = None` A name for the region
`radius = None` radius for the sphere for anisotropy correction and docking
`centerx = None` x coord center of sphere for docking search
`chy = None` y coord center of sphere for docking search
`chz = None` z coord center of sphere for docking search
`dockingfrf_rescoring = True` use frfr after frf or not
`frf_clustering = True` use clustering or not for frf (or frfr if `frf_rescoring`)
`frf_target_ellg = 225`. ideal target eLLG for rotation search with strong signal
`minimum_frf_ellg = 7.5` minimum target eLLG for rotation search: controls sub-volume size
`refine_scale = False` flag for whether or not to refine cell scale
`top_files = 5` number of top potential solutions to save for

output and focused docking sigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score) save_output_folder = True create output folder to save docked models and statistics output_folder = None non-default path for output folder brute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute) angle_around alpha = None alpha for brute rotation function angle beta = None beta for brute rotation function angle gamma = None gamma for brute rotation function angle range = None angular range for brute rotation function biological_unit molecule molecule_name = model A name for the search model model_file = None Coordinate in .pdb or .cif starting_model_vrms = 1.2 Starting RMSD estimate for the model molecule_fixed fixed_molecule_name = "fixed" A name for the fixed model fixed_model_file = None Fixed model coordinates in .pdb or .cif gui GUI-specific parameter required for output directory output_dir = None

- em_placement remove_phaser_tng_folder = True Removes the phaser_tng folder with the databases at the end of the run level = process concise *summary logfile verbose testing Set the verbosity level job_title = None Job title in PHENIX GUI, not used on command line map_model full_map = None Input full map file as alternative to half-map half_map = None Input half map file point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry best_resolution = None Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically cell_rescale = None Rescaling factor for map cell or magnification factor sequence_composition = None Sequence file for entire EM reconstruction. Ignored if fixed_scattering_ratio is set, in which case total composition is set to model composition times fixed_scattering_ratio. fixed_scattering_ratio = None If set, total composition is model composition times this factor region_approach = *automatic manual rigid_body Method for specifying search regions region_for_docking Manual specification for region(s) name = None A name for the region radius = None radius for the sphere for anisotropy correction and docking center_x = None x coord center of sphere for docking search center_y = None y coord center of sphere for docking search center_z = None z coord center of sphere for docking search docking_frf_rescoring = True use frf after frf or not frf_clustering = True use clustering or not for frf (or frf if frf_rescoring) frf_target_ellg = 225. ideal target eLLG for rotation search with strong signal minimum_frf_ellg = 7.5 minimum target eLLG for rotation search: controls sub-volume size refine_scale = False flag for whether or not to refine cell scale top_files = 5 number of top potential solutions to save for output and focused docking sigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score) save_output_folder = True create output folder to save docked models and statistics output_folder = None non-default path for output folder brute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute) angle_around alpha = None alpha for brute rotation function angle beta = None beta for brute rotation function angle gamma = None gamma for brute rotation function angle range = None angular range for brute rotation function biological_unit molecule molecule_name = model A name for the search model model_file = None Coordinate in .pdb or .cif starting_model_vrms = 1.2 Starting RMSD estimate for the model molecule_fixed fixed_molecule_name = "fixed" A name for the fixed model fixed_model_file = None Fixed model coordinates in .pdb or .cif gui GUI-specific parameter required for output directory output_dir = None

- remove_phaser_tng_folder = True Removes the phaser_tng folder with the databases at the end of the run

- level = process concise *summary logfile verbose testing Set the verbosity level

- job_title = None Job title in PHENIX GUI, not used on command line

- map_model full_map = None Input full map file as alternative to half-map half_map = None Input half map file point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is

C1, no symmetry
 best_resolution = None Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically
 cell_rescale = None Rescaling factor for map cell or magnification
 factorsequence_composition = None Sequence file for entire EM reconstruction. Ignored if fixed_scattering_ratio is set, in which case total composition is set to model composition times fixed_scattering_ratio.
 fixed_scattering_ratio = None If set, total composition is model composition times this factor
 region_approach = *automatic manual rigid_body Method for specifying search regions
 region_for_dockingManual specification for region(s)
 name = None A name for the region
 radius = None radius for the sphere for anisotropy correction and docking
 centerx = None x coord center of sphere for docking search
 y = None y coord center of sphere for docking search
 z = None z coord center of sphere for docking search

- full_map = None Input full map file as alternative to half-maps
- half_map = None Input half map files
- point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
- best_resolution = None Best resolution (smallest d-spacing) found in input map. If zero it will be set automatically
- cell_rescale = None Rescaling factor for map cell or magnification factor
- sequence_composition = None Sequence file for entire EM reconstruction. Ignored if fixed_scattering_ratio is set, in which case total composition is set to model composition times fixed_scattering_ratio.
- fixed_scattering_ratio = None If set, total composition is model composition times this factor
- region_approach = *automatic manual rigid_body Method for specifying search regions
- region_for_dockingManual specification for region(s)
 name = None A name for the region
 radius = None radius for the sphere for anisotropy correction and docking
 centerx = None x coord center of sphere for docking search
 y = None y coord center of sphere for docking search
 z = None z coord center of sphere for docking search
- name = None A name for the region
- radius = None radius for the sphere for anisotropy correction and docking
- centerx = None x coord center of sphere for docking search
 y = None y coord center of sphere for docking search
 z = None z coord center of sphere for docking search
- x = None x coord center of sphere for docking search
- y = None y coord center of sphere for docking search
- z = None z coord center of sphere for docking search
- dockingfrf_rescoring = True use frfr after frf or not
 frf_clustering = True use clustering or not for frf (or frfr if frf_rescoring)
 frf_target_ellg = 225. ideal target eLLG for rotation search with strong signal
 minimum_frf_ellg = 7.5 minimum target eLLG for rotation search: controls sub-volume size
 refine_scale = False flag for whether or not to refine cell scale
 top_files = 5 number of top potential solutions to save for output and focused dockings
 sigma_cut = 7. threshold for carrying out focused docking is top score - sigma_cut*sqrt(top score)
 save_output_folder = True create output folder to save docked models and statistics
 output_folder = None non-default path for output folder
 brute_approach = *search six_dimensional_full six_dimensional_local search strategy (frf or brute)
 angle_aroundalpha = None alpha for brute rotation function
 anglebeta = None beta for brute rotation function
 anglegamma = None gamma for brute rotation function
 anglerange = None angular range for brute rotation function

- `frf_rescoring` = True use frfr after frf or not
- `frf_clustering` = True use clustering or not for frf (or frfr if `frf_rescoring`)
- `frf_target_ellg` = 225. ideal target eLLG for rotation search with strong signal
- `minimum_frf_ellg` = 7.5 minimum target eLLG for rotation search: controls sub-volume size
- `refine_scale` = False flag for whether or not to refine cell scale
- `top_files` = 5 number of top potential solutions to save for output and focused docking
- `sigma_cut` = 7. threshold for carrying out focused docking is top score - $\sigma_cut \times \sqrt{\text{top score}}$
- `save_output_folder` = True create output folder to save docked models and statistics
- `output_folder` = None non-default path for output folder
- `brute_approach` = *search six_dimensional_full six_dimensional_local search strategy (frf or brute)
- `angle_aroundalpha` = None alpha for brute rotation function `anglebeta` = None beta for brute rotation function `anglegamma` = None gamma for brute rotation function `anglerange` = None angular range for brute rotation function
- `alpha` = None alpha for brute rotation function angle
- `beta` = None beta for brute rotation function angle
- `gamma` = None gamma for brute rotation function angle
- `range` = None angular range for brute rotation function
- `biological_unitmoleculename` = model A name for the search model `model_file` = None Coordinate in .pdb or .cif `starting_model_vrms` = 1.2 Starting RMSD estimate for the model `molecule_fixedfixed_molecule_name` = "fixed" A name for the fixed model `fixed_model_file` = None Fixed model coordinates in .pdb or .cif
- `moleculename` = model A name for the search model `model_file` = None Coordinate in .pdb or .cif `starting_model_vrms` = 1.2 Starting RMSD estimate for the model
- `molecule_name` = model A name for the search model
- `model_file` = None Coordinate in .pdb or .cif
- `starting_model_vrms` = 1.2 Starting RMSD estimate for the model
- `molecule_fixedfixed_molecule_name` = "fixed" A name for the fixed model `fixed_model_file` = None Fixed model coordinates in .pdb or .cif
- `fixed_molecule_name` = "fixed" A name for the fixed model
- `fixed_model_file` = None Fixed model coordinates in .pdb or .cif
- `gui` GUI-specific parameter required for output directory `output_dir` = None
- `output_dir` = None

Euclidian Model Matching

emma.html

phenix.emma

EMMA stands for Euclidean Model Matching and is the algorithm used by HySS to superimpose two putative solutions and to derive the consensus model. The same algorithm is also available through the external phenix.emma command-line interface. Enter phenix.emma without arguments to obtain the help page:

```
usage: phenix.emma [options] reference_coordinates reference_coordinates other_coordinates
```

options:

```
-h, --help show this help message and exit
--unit_cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space_group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--tolerance=FLOAT match tolerance
--diffraction_index_equivalent
Use only if models are diffraction-index equivalent.
```

Example: phenix.emma model1.pdb model2.sdb

The command takes two coordinate files in various formats (.pdb, CNS .sdb, SOLVE output, SHELX .ins) and compares the structures taking the space group symmetry, the allowed origin shifts and the hand ambiguity into account. The output is similar to the Match summary shown above in the example HySS output.

The match tolerance defaults to 3 Angstrom. For structures obtained with very low resolution data it may be necessary to specify a different tolerance, e.g. --tolerance=5.

The --symmetry option works just like it does for phenix.hyss. It can be used to extract symmetry information from external files such as input files for other programs (CNS, SHELX, SOLVE, ...) or reflection files.

However, the --symmetry option is only required if the information about the unit cell and the space group is missing in both coordinate files given to phenix.emma.

phenix.emma conducts an exhaustive search and, in contrast to HySS, displays all possible matches. The match with the largest number of matching sites is shown first, the match with the smallest number of matching sites is shown last (often just one site). Therefore you have to look at the beginning of the output to see the best match. I.e. if the output goes to the screen don't let yourself get distracted if you see a large number of Singles near the end of the output. Scroll back to see the best match.

Emma is also available via a web interface.

Docking a model into a local region of density with `emplace_local`

`emplace_local.html`

Authors

- Randy J Read, Claudia Millán, Airlie J McCoy, Thomas C Terwilliger
- Eric F Petterson, Tristan I Croll, Billy K Poon, Elaine C Meng, Dorothee Liebschner, Paul D Adams (ChimeraX interface)

Purpose

The `emplace_local` tool docks the input model within a sphere of density centered on a defined point in the reconstruction, returning the docked model, a local weighted map, and quality of fit statistics.

How `emplace_local` works

Ideally, half-maps are provided, then the chosen region of density is analysed for local signal and noise (both of which are resolution and direction-dependent) to define a likelihood target for rotation and translation searches, followed by rigid-body refinement.

If half-maps are unavailable, just a full map can be provided and then the likelihood target is defined by assuming that signal-to-noise drops off isotropically to the stated resolution limit.

The quality of fit can be judged by three factors:

- 1) whether there is a unique solution;
- 2) the log-likelihood-gain (LLG) score, which should preferably be greater than 60;
- 3) the map correlation coefficient to the weighted map used for docking.

The basic inputs are:

```
model file
two unmasked half-map files (preferred) or full-map file
nominal resolution (optional if half-maps provided, but recommended)
specification of center of search region
```

Note that the search center can be specified explicitly with the `sphere_center` keyword (the usual approach) or by translating the search model to the desired search region and not providing a `sphere_center` keyword.

It is also possible to provide a fixed model as prior information, accounting for the contributions of the fixed model to the sphere.

If the symmetry of a symmetric reconstruction is specified, the symmetry is applied to the output structure. Note that in the ChimeraX interface, ChimeraX functions are used to apply the symmetry.

Standard run of `emplace_local`

```
phenix.voyager.emplace_local map1=half1.map map2=half2.map
d_min=3.3 model_file=model.pdb sphere_center=100,140,170
```

Literature

Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023). Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023). Likelihood-based interactive local docking into cryo-EM maps in ChimeraX. R.J. Read, E.F. Pettersen, A.J. McCoy, T.I. Croll, T.C. Terwilliger, B.K. Poon, E.C. Meng, D. Liebschner, and P.D. Adams. *Acta Cryst.* D80, 588-598 (2024).

- Likelihood-based signal and noise analysis for docking of models into cryo-EM maps. R.J. Read, C. Millán, A.J. McCoy, and T.C. Terwilliger. *Acta Cryst.* D79, 271-280 (2023).
- Likelihood-based docking of models into cryo-EM maps. C. Millán, A.J. McCoy, T.C. Terwilliger, and R.J. Read. *Acta Cryst.* D79, 281-289 (2023).
- Likelihood-based interactive local docking into cryo-EM maps in ChimeraX. R.J. Read, E.F. Pettersen, A.J. McCoy, T.I. Croll, T.C. Terwilliger, B.K. Poon, E.C. Meng, D. Liebschner, and P.D. Adams. *Acta Cryst.* D80, 588-598 (2024).

List of all available keywords

model_file = None Search model file
 fixed_model_file = None Fixed model file
 sequence_composition = None Optional sequence file with composition for full reconstruction, overrides fixed_scattering_ratio
 map = None Map file for full map (optional, not used if half-maps provided)
 map1 = None Map file for half-map 1 (optional if full map is provided)
 map2 = None Map file for half-map 2 (optional if full map is provided)
 point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
 d_min = None Nominal local resolution. Preferred but optional if half-maps provided, required otherwise
 sphere_center = None Centre of sphere for docking
 model_vrms = 1.5 estimated RMSD for model
 fixed_scattering_ratio = 1.5 Estimated scattering ratio between sphere and model
 brute_rotation = False Test all orientations instead of fast rotation function
 refine_scale = False Turn on cell scale (voxel size) refinement
 cell_rescale = None Rescaling factor for map cell or magnification factor
 level = *process concise summary logfile verbose testing Output level from process (minimum output) to testing
 top_files = 5 Maximum number of top docking solutions to save and report
 outputfileroot = top_model Manually set file root

- model_file = None Search model file
- fixed_model_file = None Fixed model file
- sequence_composition = None Optional sequence file with composition for full reconstruction, overrides fixed_scattering_ratio
- map = None Map file for full map (optional, not used if half-maps provided)
- map1 = None Map file for half-map 1 (optional if full map is provided)
- map2 = None Map file for half-map 2 (optional if full map is provided)
- point_group_symmetry = C1 Schonflies symbol plus order (e.g. D2, C7, I) or helical. Default is C1, no symmetry
- d_min = None Nominal local resolution. Preferred but optional if half-maps provided, required otherwise
- sphere_center = None Centre of sphere for docking
- model_vrms = 1.5 estimated RMSD for model
- fixed_scattering_ratio = 1.5 Estimated scattering ratio between sphere and model
- brute_rotation = False Test all orientations instead of fast rotation function
- refine_scale = False Turn on cell scale (voxel size) refinement

- cell_rescale = None Rescaling factor for map cell or magnification factor
- level = *process concise summary logfile verbose testing Output level from process (minimum output) to testing
- top_files = 5 Maximum number of top docking solutions to save and report
- outputfileroot = top_model Manually set file root
- fileroot = top_model Manually set file root

Explore Metric Symmetry

explore_metric_symmetry.html

Purpose

iotbx.explore_metric_symmetry is a program that allows a user to quickly determine the symmetry of the lattice, given a unit cell, and determine relations between various possible point groups.

Another use of iotbx.explore_metric_symmetry is in the comparison of related unit cells, that are related by a linear recombination of their basis vectors.

Keywords

A list of keywords and concise help can be obtained by typing: iotbx.explore_metric_symmetry

```
options:
-h, --help show this help message and exit
--unit_cell=10,10,20,90,90,120|FILENAME
External unit cell parameters
--space_group=P212121|FILENAME
External space group symbol
--symmetry=FILENAME External file with symmetry information
--max_delta=FLOAT Maximum delta/obliquity used in determining the
lattice symmetry, using a modified Le-Page algorithm.
Default is 5.0 degrees
--start_from_p1 Reduce to Niggli cell and forget the input space group
before higher metric symmetry is sought.
--graph=GRAPH A graphical representation of the graph will be
written out. Requires Graphviz to be installed and in
path.
--centring_type=CENTRING_TYPE
Centring type, choose from P,A,B,C,I,R,F
--other_unit_cell=10,20,30,90,103.7,90
Other unit cell, for unit cell comparison
--other_space_group=OTHER_SPACE_GROUP
space group for other_unit_cell, for unit cell
comparison
--other_centring_type=OTHER_CENTRING_TYPE
Centring type, choose from P,A,B,C,I,R,F
--no_point_group_graph
Do not carry out the construction of a point group
graph.
--relative_length_tolerance=FLOAT
Tolerance for unit cell lengths to be considered
equal-ish.
--absolute_angle_tolerance=FLOAT
Angular tolerance in unit cell comparison
--max_order=INT Maximum volume change for target cell
```

Explore Metric Symmetry. A list of possible unit cells and space groups is given for the given specified unit cell and space group combination.

The keywords unit_cell, space_group (or centring_type) define the crystal symmetry for which a point group graph is constructed.

The keyword max_delta sets the tolerance used in the determination of the lattice symmetry.

the keyword start_from_p1 in combination with the space group is equivalent to specification of the centring_type only.

If graphviz is installed, an a png file with the point group graph can be constructed by specifying the filename of the png graph with the keyword graph.

If a second crystal is specified by the keywords other_unit_cell, other_space_group (or other_centring_type) the unit cells will be compared. Using linear combinations of the smallest unit cell, possible matches for the large unit cell are sought. If desired, the larger unit cell can be expanded as well using the keyword max_order.

The tolerances in the unit cell comparison can be changed from their defaults (10% on the lengths and 20 degrees on the angles) using the keywords relative_length_tolerance and absolute_angle_tolerance.

Construction of a point group graph can be skipped using the key no_point_group_graph.

Examples

Constructing a point group graph given some basic information:

```
iotbx.explore_metric_symmetry --unit_cell="20,30,40,90,90,90" \  
--centring_type=P
```

All point groups between P 1 and P 2 2 2 will be listed

Comparing two related unit cells can be done using:

```
iotbx.explore_metric_symmetry --unit_cell="20,30,40,90,90,90" \  
--centring_type=P --other_unit_cell="40,80,60,90,90,90" \  
--other_centring_type=F
```

phenix.f000: given PDB file and mean bulk-solvent density estimate F000 reflection

f000.html

Description

The program reads PDB model and optionally provided mean bulk-solvent density value (default is 0.35).
Output: estimate of F000 value and solvent content

The program reads PDB model and optionally provided mean bulk-solvent density value (default is 0.35).
Output: estimate of F000 value and solvent content

Command line usage examples:

```
% phenix.f000 model.pdb  
% phenix.f000 model.pdb mean_solvent_density=0.44
```

phenix.find_alt_orig_sym_mate: Identify equivalent MR solutions of the same dataset irrespective of origin and symmetry operation

find_alt_orig_sym_mate.html

Contents

- Author(s)
- Purpose
- Usage
- Using from the command-line
- Examples of PHIL input
- Output
- Do the MR solutions match?
- Algorithm
- Also move HETATM (hetero atoms)
- Debug mode
- Caveats
- Changes
- Literature
- List of all available keywords

Author(s)

- phenix.find_alt_orig_sym_mate: Robert D. Oeffner
- PHENIX GUI: Nat Echols

Purpose

phenix.find_alt_orig_sym_mate (or phenix.famos) attempts to find the best superposition of two different molecular replacement solutions of the same dataset termed moving.pdb and fixed.pdb with respect to all symmetry operations and alternate origin shifts permitted by the spacegroup of the crystal. If either of the pdb files contain more chains each chain will be tested for their best match against chains in the other pdb file. It does so by calculating a score value, MLAD (see Algorithm), for all possible symmetry operations and alternate origin shifts of each chain in moving.pdb compared with chains in fixed.pdb. The transformation with the smallest MLAD is retained for that particular pair of chains.

Usage

phenix.find_alt_orig_sym_mate can be run from the GUI by clicking on the Find alternate-origin symmetry mates button under the Model tools category on the right hand side of the main Phenix window.

Once the run button has been pressed the program will execute for a minute or two depending on the size of the molecules and space group. Afterwards the run status page features a table of the best matches between

pairs of chains in the two files. This includes the MLAD values, chain IDs, symmetry transformations and alternate origins for the two structures. If the same origin shift is not applied to all pairs of chains a warning will be printed. If the spacegroup of the crystal has a floating origin this generally results in an origin offset between chains that is not a rational number.

In the case illustrated above the chains A, B, C in the file 1tfx.pdb all have unique good matches with chains H, A, I in the file test42.pdb. This is stated in the log file as well as printed with green text on the GUI. One pair of chains does not make a good match. A warning is then printed with red text. The second column in the table in the end of the log file contains the RMSDs of the pairs of matched chains. From the results tab the superposed structure can be visually inspected in Coot or Pymol. Matches with low MLAD scores will have correspondingly good superpositions.

Using from the command-line

phenix.find_alt_orig_sym_mate can also be run from the command-line with PHIL input in either a text file or as keywords like:

```
phenix.famos moving.pdb=pdbfile1 fixed.pdb=pdbfile2
```

or:

```
phenix.find_alt_orig_sym_mate my_phil_input.txt.
```

The PHIL input scopes, moving and fixed, specifies the MR solutions. Both scopes are programmatically equivalent and must be non-empty. This means that a scope should either specify the parameter xyzfname or the sub-scope mrsolution or the sub-scope pickle.solution.

Examples:

- Use the pdb file from one of the MR solutions in a scope. This is useful for simple cases such as when the solution comes from the PHENIX MRage GUI or when it is obtained from another MR program than Phaser.
- Specify a Phaser MR solution by assigning the .sol file name of the solution to the mrsolution.solfname parameter of the sub-scope as well as assigning the IDs of the ensembles and the corresponding MR models to the parameters mrsolution.ensembles.name and mrsolution.ensembles.xyzfname. Multiple search components are specified as multiple ensembles. This way of specifying solutions is useful for solution files produced when Phaser is run from the command line and it produces several solutions of which one is the correct solution.
- Use a solution from the Phaser-MR GUI in PHENIX. In that case the parameter pickle.solution.pkifname in the sub-scope should be assigned to the solution file that is produced by PHENIX after an MR calculation. The pickle.solution.philfname should then be assigned to the input file for the MR calculation. This way of specifying solutions is useful when MR solutions are available from having run Phaser through the PHENIX interface.
- If space group and the unit cell dimensions is not available from any of the input files then these need to be specified by assigning the parameter, spacegroupname, either to a PDB file with a CRYST1 record or to an MTZ file with that information. This should be the data file used for the molecular replacement calculation.

Examples of PHIL input

Unless the input just constitutes of two PDB files a PHIL file is the easiest way to enter input. A few examples of PHIL for phenix.find_alt_orig_sym_mate are given below.

Testing a command-line Phaser MR solution file against a solution specified as a PDB file:

```
AltOrigSymMates.fixed.mrsolution
{
  solfname = "testdata/MR_3ECI_A0_2P82_A0.sol"
  ensembles
  {
    name = "MR_2P82_A0"
    xyzfname = "testdata/sculpt_2P82_A0.pdb"
  }
  ensembles
  {
    name = "MR_3ECI_A0"
    xyzfname = "testdata/sculpt_3ECI_A0.pdb"
  }
}
AltOrigSymMates.moving.pdb="testdata/2z0d.pdb"
```

Testing two set of solution files from the PHENIX Phaser-MR GUI against one another:

```
AltOrigSymMates.fixed.pickle_solution
{
  philfname = "testdata/phaser_mr_13.eff"
  pklfname = "testdata/phaser_mr_13.pkl"
}

AltOrigSymMates.moving.pickle_solution
{
  philfname = "testdata/phaser_mr_11.eff"
  pklfname = "testdata/phaser_mr_11.pkl"
}
```

Testing two solution files from the command-line version of Phaser against one another:

```
AltOrigSymMates.fixed.mrsolution
{
  solfname = "testdata/MR_3ECI_A0_2P82_A0.sol"
  ensembles
  {
    name = "MR_2P82_A0"
    xyzfname = "testdata/sculpt_2P82_A0.pdb"
  }
  ensembles
  {
    name = "MR_3ECI_A0"
    xyzfname = "testdata/sculpt_3ECI_A0.pdb"
  }
}

AltOrigSymMates.moving.mrsolution
{
  solfname = "testdata/MR_2ZPN_A0_2P82_A0.sol"
  ensembles
  {
    name = "MR_2P82_A0"
    xyzfname = "testdata/sculpt_2P82_A0.pdb"
  }
  ensembles
  {
    name = "MR_2ZPN_A0"
    xyzfname = "testdata/sculpt_2ZPN_A0.pdb"
  }
}
AltOrigSymMates.spacegroupfname = "testdata/2Z0D.mtz"
```

For more information on the PHIL input see the bottom of this page.

Output

The closest match between chains in the moving section to chains in the fixed section is saved with the name of the pdb file in the fixed scope, concatenated with the name of the pdb file in the moving scope but prepended with "MinMLAD_". A log file with the name of the pdb file in the fixed scope, concatenated with the name of the pdb file in the moving scope but prepended with "AltOrigSymMLAD_" is written containing standard output. All files are saved in the current working directory. If MR solutions specified in the PHIL contains multiple solutions then phenix.find_alt_orig_sym_mate will output multiple log files corresponding to each MR solution.

Do the MR solutions match?

A good match between two chains usually have a MLAD value below 1.5 whereas a bad match usually have a value above 2.0. The RMSD value for each match is also computed using the sequence alignment that the SSM superpose program uses. But without superposing the chains. If a match is good the RMSD will be small just like the MLAD value. It is advisable to visually inspect that the chains of structures have been transformed to the same origins in a molecular graphics viewing program.

Algorithm

phenix.find_alt_orig_sym_mate computes configurations by looping over all symmetry operations and alternative origin shifts. An alignment between C-alpha atoms from moving.pdb and fixed.pdb is computed using secondary structure matching (SSM). If that fails or if the MLAD score achieved with SSM is larger than 2.0 an alignment is computed using MMTBX alignment functions which is part of the CCTBX. To estimate the best match a distance measure between the aligned C-alpha atoms is computed for each configuration. The mean log absolute deviation (MLAD) is defined as:

$$\text{MLAD}(\text{dR}) = \Sigma(\log(\text{dr} \cdot \text{dr} / (|\text{dr}| + 0.9) + \max(0.9, \min(\text{dr} \cdot \text{dr}, 1))) - \log(0.9)),$$

where dr is the difference vector between a pair of aligned C-alpha atoms and the sum is taken over all atom pairs in the alignment. The factor $\log(0.9)$ is subtracted to ensure that $\text{MLAD}(0) = 0.0$, i.e. that two identical structures produces the value zero.

MLAD can loosely be interpreted as a distance measure between structures. But it is not a metric in a strict mathematical sense since the triangle inequality is not fulfilled. Unlike a plain root mean square deviation the logarithm in the MLAD formula will downplay contributions of atom pairs where the atoms are spatially distant. If an RMSD were employed such atom pairs would contribute on an equal footing with those groups of atom pairs that can be superposed perfectly. This in turn may lead to non-optimal superpositions when the structures tested against one another consists of multiple domains where one domain has undergone a domain motion, i.e. where a subset of atoms in one chain are bound to be spatially distant from the atoms in the other chain they have been paired with.

phenix.find_alt_orig_sym_mate.py will for each chain in the fixed scope find the smallest MLAD with a copy of each chain in the moving scope for a given symmetry operation and alternative origin. When all chains in the fixed scope have been tested these copies will be saved to a file. For spacegroups with floating origin the minimum MLAD is found by doing a Golden Sectioning minimization along the polar axis for each copy of chains in moving.pdb.

Also move HETATM (hetero atoms)

Invoking this flag will move hetero atoms (ligands, waters, metals, etc.) in conjunction with their associated peptide chain. The program first invokes phenix.sort_hetatms as to associate hetero atoms sensibly with adjacent peptide chains.

After having identified the transformation for a chain yielding the smallest MLAD score it then subjects the associated hetero atoms to the same transformation.

Debug mode

Invoking the debug flag produces individual pdb files of the C-alpha atoms used for each SSM alignment of the moving scope for each permitted symmetry operation and alternative origin. A gold atom is placed at the centroid of the C-alpha atoms. Similar files are produced for the fixed scope. These files are stored in a subfolder named "AltOrigSymMatesFiles".

If a floating origin is present in the space group a table of MLAD values is produced by sliding a copy of the chains in the moving scope along the polar axis in the fractional interval [-0.5, 0.5] for each permitted symmetry operation and alternative origin.

Caveats

The program tests all solutions present in solution files entered as fixed scope against all solutions present in solution files entered as moving scope. Consequently the execution time is proportional to the number of solutions in fixed scope times the number of solutions in moving scope.

The execution time is proportional to the number of SSM alignments being tested; if SSM identifies 12 alignments the program will take 12 times as long.

Changes

The command-line syntax mentioned in the Computational Crystallography Newsletter 2012 January has been replaced by PHIL syntax.

Literature

Algorithms for deriving crystallographic space-group information R.W. Große-Kunstleve Acta Cryst. A55, 383-395 (1999)

phenix.find_alt_orig_sym_mate Robert D. Oeffner, Gábor Bunkóczi and Randy J. Read Computational Crystallography Newsletter 2012 January, 5-10 (2012)

Secondary-structure matching (SSM), a new tool for fast protein structure alignment in three dimensions. E. Krissinel and K. Henrick Acta Cryst. D60, 2256-2268 (2004)

List of all available keywords

AltOrigSymMatespacegroupfname = None A model file or a reflection data file providing the space group and cell dimension. This is needed in the uncommon situation both moving and fixed model files do not already contain these, (CRYST1 record in a pdb file or _cell and _symmetry blocks in a cif file). Must be based on the data set used for the MR calculation.chosenorigin = "Any" By default all allowed origin shifts are probed. Specifying '2/3, 1/3, 1/2' tells the program to only explore symmetry operations for the unit cells displaced at fractional positions (2/3, 1/3, 1/2)use_all_SSM = False By default only the best sequence alignment from SSM is used. Setting this value to True will include other suboptimal alignments for

testing.no_symmetry_operations = False When set to False all symmetry operations are probed during the computation.movehetatms = False Move hetatoms close to a chain in conjunction with that chain. Hetatoms are first associated with individual chains with phenix.sort_hetatms before undergoing the same spatial transformation as the chains.writepdbfiles = True Save final transformed pdb file and any intermediate pdb files realised from MR solution filesoutputfileprefix = None Alternative prefix for output files .pdb, .log and .sol instead of AltOrigSymMLAD_*.nproc = 1 Number of CPU cores to use when submitting MR solution files with multiple solutionsquiet = False Set to True for silencing output when using multiple cpu coresdebug = False verbose = False write_cif = False goodmatch = 1.5 Weed out poor matches with MLAD values larger than goodmatch.strictly_protein = True Relax the condition that structures must only be protein chains with conventional residue names This might help solving cases where one or both pdb files are flagged as having no chainsworkdir = None Specify if output should be written elsewhere than the current working directory.gui_output_dir = None For PHENIX GUI only.job_title = None Job title in PHENIX GUI, not used on command linemovingSpecify either a PDB file, an MR solution or a pickle_solution.pdb = None Deprecated, use model in the future. Path to a PDB file from an MR solution. Constituent chains will be moved separately.model = None Path to a PDB file from an MR solution. Constituent chains will be moved separately.mrsolutionMR solution produced by command-line version of Phasersolfname = None path to the solution file, i.e. the .sol fileensemblesEnsembles constituent in this MR solution. Typically these are chains in the PDB file.name = None ID for ensemble in the MR solution, i.e. the name after the 'SOLU 6DIM ENSE' string in the .sol file.xyzfname = None Deprecated, use model in the future Path to the PDB file with atoms representing this ensemble This is the search model used by Phaser for this ensemble prior to solving the structure. It is not the solved structure.model = None Path to the PDB file with atoms representing this ensemble. This is the search model used by Phaser for this ensemble prior to solving the structure. It is not the solved structure.fixedSpecify either a PDB file, an MR solution or a pickle_solution.pdb = None Deprecated, use model in the future. Path to a PDB file from an MR solution. Constituent chains will remain fixed.model = None Path to a PDB file from an MR solution. Constituent chains will remain fixed.mrsolutionMR solution produced by command-line version of Phasersolfname = None path to the solution file, i.e. the .sol fileensemblesEnsembles constituent in this MR solution. Typically these are chains in the PDB file.name = None ID for ensemble in the MR solution, i.e. the name after the 'SOLU 6DIM ENSE' string in the .sol file.xyzfname = None Deprecated, use model in the future. Path to the PDB file with atoms representing this ensemble. This is the search model used by Phaser for this ensemble prior to solving the structure. It is not the solved structure.model = None Path to the PDB file with atoms representing this ensembl...

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- movehetatms = False Move hetatoms close to a chain in conjunction with that chain. Hetatoms are first associated with individual chains with phenix.sort_hetatms before undergoing the same spatial transformation as the chains.
- writepdbfiles = True Save final transformed pdb file and any intermediate pdb files realised from MR solution files
- outputfileprefix = None Alternative prefix for output files .pdb, .log and .sol instead of AltOrigSymMLAD_*.*
- nproc = 1 Number of CPU cores to use when submitting MR solution files with multiple solutions
- quiet = False Set to True for silencing output when using multiple cpu cores
- debug = False
- verbose = False

- `write_cif = False`
- `goodmatch = 1.5` Weed out poor matches with MLAD values larger than `goodmatch`.
- `strictly_protein = True` Relax the condition that structures must only be protein chains with conventional residue names This might help solving cases where one or both `pdb` files are flagged as having no chains
- `workdir = None` Specify if output should be written elsewhere than the current working directory.
- `gui_output_dir = None` For PHENIX GUI only.
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `movingSpecify` either a PDB file, an MR solution or a `pickle_solution.pdb = None` Deprecated, use `model` in the future. Path to a PDB file from an MR solution. Constituent chains will be moved separately.
- `model = None` Path to a PDB file from an MR solution. Constituent chains will be moved separately.
- `mrsolution` MR solution produced by command-line version of Phaser
- `solfname = None` path to the solution file, i.e. the `.sol` file
- `ensembles` Ensembles constituent in this MR solution. Typically these are chains in the PDB file.
- `name = None` ID for ensemble in the MR solution, i.e. the name after the `"SOLU 6DIM ENSE` string in the `.sol` file.
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- `pdb = None` Deprecated, use `model` in the future. Path to a PDB file from an MR solution. Constituent chains will be moved separately.
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- `fixedSpecify` either a PDB file, an MR solution or a `pickle_solution.pdb = None` Deprecated, use `model` in the future. Path to a PDB file from an MR solution. Constituent chains will remain fixed.
`model = None` Path to a PDB file from an MR solution. Constituent chains will remain fixed.
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Rapid fitting of secondary structure to a map with `find_helices_strands`

`find_helices_strands.html`

Contents

- Author(s)
- Purpose
- Running the program
- Methods
- Using `find_helices_strands` to bootstrap `phenix.autobuild`
- Command-line use
- Specific limitations and problems
- References
- List of all available keywords

Author(s)

`find_helices_strands`: Tom Terwilliger PULCHRA: Piotr Rotkiewicz

- `find_helices_strands`: Tom Terwilliger
- PULCHRA: Piotr Rotkiewicz

Purpose

`find_helices_strands` is a command line tool for finding helices and strands in a map and building an model of the parts of a structure that have regular secondary structure. It can be used for protein, RNA, and DNA. An option is to use a rapid chain-tracing algorithm to build CA of proteins, followed by reconstruction of a full model.

Running the program

The program is available in the Phenix GUI under "Model building". The only required input is an electron density map, such as the output of density modification. A sequence file is optional.

The default behavior is to search for protein helices and strands, but you may instead look for nucleic acid helices, or use the `trace_chain` method (describe below) to find C-alpha positions independently of secondary structure motifs. Using the default options without a sequence, the model will be a poly-alanine backbone; with the `trace_chain` method it will be a C-alpha trace, but you may optionally have PULCHRA (Rotkiewicz & Skolnick 2008) complete the backbone.

The program usually only requires a few minutes at most to run. Output will be to a new directory, with a single PDB file containing a partial model.

Methods

How `find_helices_strands` finds helices and strands in maps:

`find_helices_strands` first identifies helical segments as rods of density at 5-8 Å. Then it identifies helices at higher resolution keeping the overall locations of the helices fixed. Then it identifies the directions and CA positions of helices by noting the helical pattern of high-density points offset slightly along the helix axis from the main helical density (as used in "O" to identify helix direction). Finally model helices are fit to the density using the positions and orientations identified in the earlier steps. A similar procedure is used to identify strands. Then the helices and strands are combined into a single model.

How `find_helices_strands` finds RNA and DNA helices in maps:

`find_helices_strands` finds RNA and DNA helices differently than it finds helices in proteins. It uses a convolution search to find places in the asymmetric unit where an A-form RNA or B-form DNA helix can be placed. These are assembled into contiguous helical segments if possible. The resolution of this search is 4.5 Å if you have resolution beyond 4.5 Å, and the resolution of your data otherwise.

How `trace_chain` finds CA positions in maps:

The RESOLVE `trace_chain` algorithm places dummy atoms down the middle of all the tubes of density in a map, then it attempts to find sets of these atoms that may be CA atoms, where the atoms are spaced by 3.8 Å and where there is strong density between each pair. This yields segments represented by CA atoms. Next PULCHRA (Rotkiewicz & Skolnick 2008) can optionally be used to reconstruct a full main-chain model. Finally RESOLVE is used to assemble all the resulting fragments into a model.

Using `find_helices_strands` to bootstrap `phenix.autobuild`

If you run `phenix.autobuild` at low resolution (3.5 Å or lower) then your model may have strands built instead of helices. You can use `find_helices_strands` to help bootstrap autobuild model-building by providing the helical model from `find_helices_strands` to `phenix.autobuild`. Just run `phenix.find_helices_strands` with your best map `map_coeffs.mtz`. Then take the helical model pass it to `phenix.autobuild` with the keyword (in addition to your usual keywords for autobuild):

```
consider_main_chain_list=map_coeffs.mtz_helices.pdb
```

Then the AutoBuild wizard will treat your helical model just like one of the models that it builds, and merge it into the model as it is being assembled.

Command-line use

From the command-line you can type:

```
phenix.find_helices_strands map_coeffs.mtz quick=True
```

If you want a more thorough run, then skip the "quick=True" flag. If you want (or need) to specify the column names from your mtz file, you will need to tell `find_helices_strands` what FP and PHIB are, in this format:

```
phenix.find_helices_strands map_coeffs.mtz \  
  labin="LABIN FP=2FOFCWT PHIB=PH2FOFCWT"
```

If you want to specify a sequence file, then in the last step `find_helices_strands` will try to align your sequence with the map and model:

```
phenix.find_helices_strands map_coeffs.mtz seq_file=seq.dat
```

If you want to use the `trace_chain` algorithm, then specify:

```
phenix.find_helices_strands map_coeffs.mtz seq_file=seq.dat trace_chain=True
```

Here is an example using data from the PHENIX examples library:

```
phenix.find_helices_strands $PHENIX/phenix_examples/p9-build/p9-resolve.mtz \
  labin="FP=FP PHIB=PHIM FOM=FOMM" trace_chain=True
```

That should build a model using sample data in a few seconds.

Specific limitations and problems

References

Fast procedure for reconstruction of full-atom protein models from reduced representations. P. Rotkiewicz, and J. Skolnick. J Comput Chem 29, 1460-5 (2008).

- Fast procedure for reconstruction of full-atom protein models from reduced representations. P. Rotkiewicz, and J. Skolnick. J Comput Chem 29, 1460-5 (2008).

Rapid model building of alpha-helices in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 268-75 (2010).

- Rapid model building of alpha-helices in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 268-75 (2010).

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- Rapid model building of beta-sheets in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 276-84 (2010).

Rapid chain tracing of polypeptide backbones in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 285-94 (2010).

- Rapid chain tracing of polypeptide backbones in electron-density maps. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 66, 285-94 (2010).

List of all available keywords

use_pdb_in_directly = False Use PDB input model directly
output_dir = None Output directory
seq_file = None Sequence file for sequence alignment
compare_file = None PDB file for comparison only
res_convolution = 4.5 high-resolution limit for convolution calculation. (Applies to nucleic acids only)
temp_dir = "temp_dir" Optional temporary work directory
helices_only = False Find only helices (Applies only if trace_chain=False)
strands_only = False Find only strands (Applies only if trace_chain=False)
trace_chain = False Use resolve trace_chain algorithm
helices_before_trace = None Find helices before running resolve trace_chain algorithm
strands_before_trace = None Find strands before running resolve trace_chain algorithm
pulchra = False Use PULCHRA to create all-atom model from CA model
reverse_if_better = True Reverse chain direction in trace_chain if pulchra=True and reversed chain has improved geometry.
remove_side = False Try to remove density for side chains in trace_chain
rho_cut_min = None Minimum rho/sigma at potential CA positions
rat_pair_min = None Target minimum ratio of rho/sigma at midpoint between potential CA positions to the average at the two CA positions
minimum_length_angstroms_helices_strands = None Minimum length (A CA start - CA end) for helix/strand to keep
dist_ca_target = 3.8 Target CA-CA distance
dist_ca_sep = 4.0 Max separation of CA atoms
dist_ca_tol_max = None Maximum tolerance for CA-CA distances. Normally 0.8 A for medium and 1.3 A for quick
dist_ca_start = None Guess of tolerance for CA-CA distances. Set automatically by default.
dist_ca_tol = None Tolerance for CA-CA distances. Normally set automatically Typical values are 0.3-1.0 A A high number can be used to force a more thorough search. Compare with target_p_ratio which adjusts CA-CA tolerance to achieve a targeted ratio of nonamers to atoms.
cutoff_trace = 0.000 The top

cutoff_trace fraction of peaks in trace_chain will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0.00)ncut_trace_min = 0 The top ncut_trace_min peaks in trace_chain will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0)target_p_ratio = None Target ratio of nonamers found to atoms in a.u default=3 for quick, 4 otherwise. This can be used as an alternate method to adjust the thoroughness of trace_chain searches. Differs from setting dist_ca_tol by adjusting CA-CA tolerance to achieve the desired target ratio, while dist_ca_tol=tol sets the tolerance directly to toltrace_ratio_long = 0.5 When setting tolerances for CA-CA distances in trace_chain the upper bound will increase trace_ratio_long as fast as the lower bound. (Default=0.5)ratio_trace_extra = None Minimum ratio of dist between extra atoms added in trace_chain to rad_sep_trace Default is 1.25 for standard and 1.5 for quickrad_sep_trace = None Dummy atom separation in trace_chain Default is 0.6 A for standard and 0.75 for quick Increased if resolution is greater than 3 A Value of rad_mask_trace in resolve will be rad_sep_trace*2atom_target_ratio = 1.0 Ratio of target number of atoms to estimated actual number_n_atoms_total_scale = 1.0 Ratio of estimated atoms in au to standard estimatefill_gaps = True Try to fill in gaps in trace_chaintarget_angle = None Target angle for CA-CA-CA or P-P-P use_any_side = False Use any side chain that fits density in assemblycc_helix_min = None Minimum CC of low-res helical density to map to keep. group_ca_length = None Minimum length of a segment of helix or strand to keep. (only applies if trace_chain=False) cc_strand_min = None Minimum CC of strand density to map to keep. (only applies if trace_chain=False) quick = False Try to find structure quicklytarget_time_for_trace_chain = 15 Target length of time to work on trace_chain (sec)recycle = False Recycle CA positions in trace_chainoptimize = True Try to optimize CA-CA tolerance in trace_chain to obtain nonamer ratio equal to target_p_ratioassemble = False Assemble model wi...

- use_pdb_in_directly = False Use PDB input model directly
- output_dir = None Output directory
- seq_file = None Sequence file for sequence alignment
- compare_file = None PDB file for comparison only
- res_convolution = 4.5 high-resolution limit for convolution calculation. (Applies to nucleic acids only)
- temp_dir = "temp_dir" Optional temporary work directory
- helices_only = False Find only helices (Applies only if trace_chain=False)
- strands_only = False Find only strands (Applies only if trace_chain=False)
- trace_chain = False Use resolve trace_chain algorithm
- helices_before_trace = None Find helices before running resolve trace_chain algorithm
- strands_before_trace = None Find strands before running resolve trace_chain algorithm
- pulchra = False Use PULCHRA to create all-atom model from CA model
- reverse_if_better = True Reverse chain direction in trace_chain if pulchra=True and reversed chain has improved geometry.
- remove_side = False Try to remove density for side chains in trace_chain
- rho_cut_min = None Minimum rho/sigma at potential CA positions
- rat_pair_min = None Target minimum ratio of rho/sigma at midpoint between potential CA positions to the average at the two CA positions
- minimum_length_angstroms_helices_strands = None Minimum length (A CA start - CA end) for helix/strand to keep
- dist_ca_target = 3.8 Target CA-CA distance

- `dist_ca_sep` = 4.0 Max separation of CA atoms
- `dist_ca_tol_max` = None Maximum tolerance for CA-CA distances. Normally 0.8 Å for medium and 1.3 Å for quick
- `dist_ca_start` = None Guess of tolerance for CA-CA distances. Set automatically by default.
- `dist_ca_tol` = None Tolerance for CA-CA distances. Normally set automatically Typical values are 0.3-1.0 Å A high number can be used to force a more thorough search. Compare with `target_p_ratio` which adjusts CA-CA tolerance to achieve a targeted ratio of nonamers to atoms.
- `cutoff_trace` = 0.000 The top `cutoff_trace` fraction of peaks in `trace_chain` will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0.00)
- `ncut_trace_min` = 0 The top `ncut_trace_min` peaks in `trace_chain` will be assumed to be non-protein, and all peaks near them will be ignored. (Default = 0)
- `target_p_ratio` = None Target ratio of nonamers found to atoms in a.u default=3 for quick, 4 otherwise. This can be used as an alternate method to adjust the thoroughness of `trace_chain` searches. Differs from setting `dist_ca_tol` by adjusting CA-CA tolerance to achieve the desired target ratio, while `dist_ca_tol=tol` sets the tolerance directly to `tol`
- `trace_ratio_long` = 0.5 When setting tolerances for CA-CA distances in `trace_chain` the upper bound will increase `trace_ratio_long` as fast as the lower bound. (Default=0.5)
- `ratio_trace_extra` = None Minimum ratio of dist between extra atoms added in `trace_chain` to `rad_sep_trace` Default is 1.25 for standard and 1.5 for quick
- `rad_sep_trace` = None Dummy atom separation in `trace_chain` Default is 0.6 Å for standard and 0.75 for quick Increased if resolution is greater than 3 Å Value of `rad_mask_trace` in `resolve` will be `rad_sep_trace*2`
- `atom_target_ratio` = 1.0 Ratio of target number of atoms to estimated actual number
- `n_atoms_total_scale` = 1.0 Ratio of estimated atoms in au to standard estimate
- `fill_gaps` = True Try to fill in gaps in `trace_chain`
- `target_angle` = None Target angle for CA-CA-CA or P-P-P
- `use_any_side` = False Use any side chain that fits density in assembly
- `cc_helix_min` = None Minimum CC of low-res helical density to map to keep.
- `group_ca_length` = None Minimum length of a segment of helix or strand to keep. (only applies if `trace_chain=False`)
- `cc_strand_min` = None Minimum CC of strand density to map to keep. (only applies if `trace_chain=False`)
- `quick` = False Try to find structure quickly
- `target_time_for_trace_chain` = 15 Target length of time to work on `trace_chain` (sec)
- `recycle` = False Recycle CA positions in `trace_chain`
- `optimize` = True Try to optimize CA-CA tolerance in `trace_chain` to obtain nonamer ratio equal to `target_p_ratio`
- `assemble` = False Assemble model with `resolve` after `trace_chain`
- `resolve_size` = 12 Size of `resolve` to use. You may need a bigger size than in other `resolve` applications
- `coarse_grid` = False `Coarse_grid` allows the use of a smaller `resolve` size
- `verbose` = False Verbose output

- `raise_sorry` = False Raise sorry if problems
- `debug` = False Debugging output
- `dry_run` = False Just read in and check parameter names
- `mtz_label_prefix` = None Prefix for column names, used by Coot. The GUI will detect and set this automatically. It should not be changed directly by users.
- `delete_tmp_dir` = False GUI setting, does not apply to command-line version
- `job_title` = None Job title in PHENIX GUI, not used on command line
- `input_filesmtz_in` = None MTZ file with coefficients for a map
`map_coeff_labels` = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
`labin` = "" Labin line for MTZ file with map coefficients. Normally use instead `map_coeff_labels`. This is available for backward compatibility. You can specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
`map_in` = None CCP4 or MRC-style map file
`pdb_in` = None Optional PDB file to be used for seeding trace_chain with CA
- `mtz_in` = None MTZ file with coefficients for a map
- `map_coeff_labels` = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
- `labin` = "" Labin line for MTZ file with map coefficients. Normally use instead `map_coeff_labels`. This is available for backward compatibility. You can specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
- `map_in` = None CCP4 or MRC-style map file
- `pdb_in` = None Optional PDB file to be used for seeding trace_chain with CA
- `output_filetarget_output_format` = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
`output_model` = None Output PDB file
`output_log` = None Output log file name. If you want to specify a directory to put this file in then please use "`output_dir=myoutput_dir`"
`gui_output_dir` = None GUI parameter only
- `target_output_format` = *None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- `output_model` = None Output PDB file
- `output_log` = None Output log file name. If you want to specify a directory to put this file in then please use "`output_dir=myoutput_dir`"
- `gui_output_dir` = None GUI parameter only
- `crystal_inforesolution` = None high-resolution limit for map calculation
`chain_type` = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms)
`solvent_fraction` = 0.5 You can specify the solvent fraction
- `resolution` = None high-resolution limit for map calculation

- chain_type = *PROTEIN DNA RNA Chain type (for identifying main-chain and side-chain atoms)
- solvent_fraction = 0.5 You can specify the solvent fraction

Finding and analyzing NCS from heavy-atom sites or a model with `find_ncs`

`find_ncs.html`

Contents

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Authors:

- `find_ncs`: Tom Terwilliger
- `simple_ncs_from_pdb` : Tom Terwilliger
- Phil command interpreter: Ralf W. Grosse-Kunstleve
- `find_domain`: Peter Zwart

Purpose

The `find_ncs` method identifies NCS in (a) the chains in a PDB file or (b) a set of heavy-atom sites, or (c) from the density in a map and writes out the NCS operators in forms suitable for phenix.refine, resolve, and the AutoSol and AutoBuild Wizards, or (d) from an existing symmetry file (`.ncs_spec`), or (e) it can generate NCS for a helical array from helical parameters.

How `find_ncs` works

The basic steps that the `find_ncs` carries out are:

- Decide whether to use `simple_ncs_from_pdb` (used if the input file contains chains from a PDB file) or RESOLVE NCS identification (used if the input file contains heavy-atom sites) or `find_ncs_from_density` (used if a map is supplied and sites are not found from either of the above)
- call either `simple_ncs_from_pdb` or RESOLVE or `find_ncs_from_density` to identify NCS

- Evaluate the NCS by calculating the correlation of NCS-related electron density based on the input map coefficients mtz file.
- Report the NCS operators and correlations

Output files from find_ncs

The output files that are produced are:

- NCS operators written in format for phenix.refine: find_ncs.ncs
- NCS operators written in format for the PHENIX Wizards: find_ncs.ncs_spec

What find_ncs needs

- find_ncs needs a file with map coefficients and optionally a second file containing NCS information.

The file with NCS information can be...

- a PDB file with a model (find_ncs will call simple_ncs_from_pdb to extract NCS operators from the chains in your model)
- a PDB file with heavy-atom sites (find_ncs will call RESOLVE to find NCS operators from your heavy-atom sites)
- an NCS definitions file written by a PHENIX wizard (e.g., AutoSol_1.ncs_spec, produced by AutoSol)
- a RESOLVE log file containing formatted NCS operators

Map coefficients file

The file with map coefficients can be any MTZ file with coefficients for a map. If find_ncs does not choose the correct columns automatically, then you can specify them with a command like:

```
labin="labin FP=FP PHIB=PHIB FOM=FOM "
```

If you have no map coefficients yet (you just have some sites and want to get operators, for example), you can tell find_ncs to ignore the map with:

```
ncs_parameters.force_ncs=True
```

You still need to supply a map coefficients file (the map itself will be ignored though).

Format of NCS operators

The NCS operators look like this:

```
rota_matrix 0.7955 -0.5660 0.2164
rota_matrix -0.5511 -0.8242 -0.1300
rota_matrix 0.2520 -0.0159 -0.9676
tran_orth 18.4021 5.3569 -23.3621

center_orth 15.9430 11.8822 0.6609
```

rota_matrix and tran_orth are the rotation and translation information, and center_orth is (approximately) the center of the region where the operator applies.

The standard in the .ncs_spec files is that the NCS operator is applied to copy j to map copy j on to copy 1. If you want the inverse operators you can specify invert_matrices=True and then you will get operators that map copy 1 on to copy j.

Examples

Standard run of find_ncs

If you have just a map coefficients file you can say:

```
phenix.find_ncs mlt.mtz
```

and find_ncs will look for NCS-related density in mlt.mtz (see find_ncs_from_density for more details on how this works)

If you have a model and a map file, from the command-line you can type:

```
phenix.find_ncs anb.pdb mlt.mtz
```

This will produce the following output:

```
Getting column labels from mlt.mtz for input map file
FILE TYPE: ccp4_mtz
All labels: ['FP', 'SIGFP', 'PHIC', 'FOM']
Labin line will be: labin FP=FP PHIB=PHIC FOM=FOM
To change it modify this: params.ncs.labin="labin FP=FP PHIB=PHIC FOM=FOM "
This is the map that will be used to evaluate NCS

Reading NCS information from: anb.pdb
Copying mlt.mtz to temp_dir/mlt.mtz
This PDB file contains 2 chains and 636 total residues
and 636 C-alpha or P atoms and 4740 total atoms
NCS will be found using the chains in this PDB file
Chains in this PDB file: ['M', 'Z']
Two chains were found in the file anb.pdb, chain M and chain Z

GROUPS BASED ON QUICK COMPARISON: []
Looking for invariant domains for ...: ['M', 'Z'] [[[2, 138], [193, 373]], [[2, 138], [193, 373]]]
Residues 2-138, 193-373, matched between the two chains

Copying mlt.mtz to temp_dir/mlt.mtz
Copying temp_dir/NCS_correlation.log to NCS_correlation.log
Log file for NCS correlation is in NCS_correlation.log
List of refined NCS correlations: [1.0, 0.80000000000000004]
There were two separate groups of residues that had different NCS
relationships. Residues 193-373 of each chain were in one group, and
residues 2-138 in each chain were in the other group.

The electron density map had a correlation between the two NCS-related
chains of 1.0 for the first group, and 0.8 for the second

The NCS operators for each are listed.

GROUP 1
Summary of NCS group with 2 operators:
ID of chain/residue where these apply: [['M', 'Z'], [[[193, 373]], [[193, 373]]]]
RMSD (A) from chain M: 0.0 0.0
Number of residues matching chain M:[181, 181]
Source of NCS info: anb.pdb
Correlation of NCS: 1.0

OPERATOR 1
CENTER: 69.1058 -9.5443 59.4674

ROTA 1: 1.0000 0.0000 0.0000
ROTA 2: 0.0000 1.0000 0.0000
ROTA 3: 0.0000 0.0000 1.0000
TRANS: 0.0000 0.0000 0.0000

OPERATOR 2
```

CENTER: 37.5004 -37.0709 -62.5441

ROTA 1: 0.7751 -0.6211 -0.1162
ROTA 2: -0.3607 -0.5859 0.7256
ROTA 3: -0.5188 -0.5205 -0.6782
TRANS: 9.7485 27.6460 17.2076

GROUP 2

Summary of NCS group with 2 operators:

ID of chain/residue where these apply: [['M', 'Z'], [[2, 138]], [[2, 138]]]

RMSD (A) from chain M: 0.0 0.0

Number of residues matching chain M:[137, 137]

Source of NCS info: anb.pdb

Correlation of NCS: 0.8

OPERATOR 1

CENTER: 66.6943 -13.3128 21.6769

ROTA 1: 1.0000 0.0000 0.0000
ROTA 2: 0.0000 1.0000 0.0000
ROTA 3: 0.0000 0.0000 1.0000
TRANS: 0.0000 0.0000 0.0000

OPERATOR 2

CENTER: 39.0126 -53.7392 -13.4457

ROTA 1: 0.3702 -0.9275 -0.0516
ROTA 2: -0.8933 -0.3402 -0.2938
ROTA 3: 0.2549 0.1548 -0.9545
TRANS: 1.7147 -0.6936 7.2172

Creating a model containing unique set of chains and nearby NCS-related chains

If you want to create a model that surrounds a unique set of chains with all the NCS-related ones, for now you need to carry out a few extra steps. In this example the symmetry is helical with 5 unique chains; a similar approach can be used for point-group symmetry.

For helical assemblies, NCS is not always simple to apply. The general idea is to generate more than enough copies, pare them down to the ones you want, get the NCS file for these copies, and apply it to your original 5 unique copies

- Make a PDB file with one set of the unique 5 copies
- Make a PDB file with a single monomer from these 5
- use `apply_ncs` on the file with one chain to generate more copies
- if you didn't get enough copies to cover all the locations you want, extend your NCS operators:

```
phenix.find_ncs my_helical_ncs.ncs_spec helix_extend_range=30
```

This will extend the operators +/-30 Å and write out a new `.ncs_spec` file

- Once you get enough copies from step 2, edit that PDB file to remove all the chains that you don't want.
- Run `find_ncs` on the PDB file with your desired chains to get the NCS relationships

6. Now (finally) run `apply_ncs` with the new `.ncs_spec` file and your PDB file with one set of 5 copies

Additional information

References

List of all available keywords

`find_ncsjob_title` = None Job title in PHENIX GUI, not used on command line
`input_filesncs_in_type` = *None chains sites `ncs_file` Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS
`ncs_file`: an ncs object file from PHENIX. `ncs_in` = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS object
`mtz_in` = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom sites
`labin` = "" Labin line for MTZ file with map coefficients. This is optional if `find_ncs` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
`map_in` = None CCP4 or MRC-style map file (instead of `mtz_in`)
`output_filesparams_out` = `find_ncs_params.eff` Parameters file to rerun `find_ncs`
`ncs_out` = `find_ncs.ncs_spec` Output file with NCS information. If set, then files will also be written for resolve and phenix.refine and as BIOMT records
`ncs_format` = *`ncs_spec` biomt Format of output NCS information. `ncs_spec` is standard phenix ncs format. biomt is BIOMT records.
`ncs_mask_file` = None If defined and `mtz_in` is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use `ncs_mask_as_mtz=False` to write map coefficients for the mask.
`ncs_mask_as_mtz` = True If True, the `ncs_mask_file` will be map coefficients. If False a CCP4-style map will be written.
`ncs_au_file` = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.
`crystal_inforesolution` = None high-resolution limit for map calculation
`directorytemp_dir` = "temp_dir" Temporary work directory
`output_dir` = "" Output directory where files are to be written
`gui_output_dir` = None control
`verbose` = True Verbose output
`debug` = False Debugging output
`raise_sorry` = False Raise sorry if problems
`dry_run` = False Just read in and check parameter names
`require_nonzero` = True Require non-zero values in data columns to consider reading in.
`ncs_parametersinvert_matrices` = False Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set `invert_matrices=True`
`chain_max_rmsd` = 2 limit of rms difference between chains to be considered as copies
`ncs_restrict` = 0 You can specify the number of NCS operators to look for
`force_ncs` = False You can tell `find_ncs` to ignore the map. This is useful if you only have FP but no phases yet...
`optimize_ncs` = False You can tell `find_ncs` to optimize the NCS by making as compact a molecule as possible.
`no_ncs_sg_mismatch` = True If one or more NCS operators is the same as a SG operator, skip the NCS
`stry_density_after_ha` = True If search for NCS using ha sites does not yield NCS correlation of at least `target_ncs_cc` or the number of operators is less than `target_ncs_operators` then try finding NCS directly from density
`target_ncs_operators` = None Number of expected NCS operators (used to decide whether to try density search)
`ncs_copies_max` = None Max number of allowed NCS operators in density search
`target_ncs_cc` = 0.80 Expected NCS correlation
`minimum_ncs_cc` = 0.30 Minimum NCS correlation. If the overall NCS correlation is less than `minimum_ncs_cc` it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter `minimum_ncs_cc_from_density` is applied in finding NCS from density as well.
`fraction_ncs_min` = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
`helix_extend_range` = None Range in z to extend helical operators (A)
`helix_trans_along_z` = None Along with `helix_theta`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the ...

- `find_ncsjob_title` = None Job title in PHENIX GUI, not used on command line
`input_filesncs_in_type` = *None chains sites `ncs_file` Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS
`ncs_file`: an ncs object file from PHENIX. `ncs_in` = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS object
`mtz_in` = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom sites
`labin` = "" Labin line for MTZ file with map coefficients. This is optional

if find_ncs can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP map_in = None CCP4 or MRC-style map file (instead of mtz_in)output_filesparams_out = find_ncs_params.eff Parameters file to rerun find_ncsncs_out = find_ncs.ncs_spec Output file with NCS information. If set, then files will also be written for resolve and phenix.refine and as BIOMT recordsncs_format = *ncs_spec biomt Format of output NCS information. ncs_spec is standard phenix ncs format. biomt is BIOMT records.ncs_mask_file = None If defined and mtz_in is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use ncs_mask_as_mtz=False to write map coefficients for the mask.ncs_mask_as_mtz = True If True, the ncs_mask_file will be map coefficients. If False a CCP4-style map will be written.ncs_au_file = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.crystal_inresolution = None high-resolution limit for map calculationdirectorytemp_dir = "temp_dir" Temporary work directoryoutput_dir = "" Output directory where files are to be writtengui_output_dir = None controlverbose = True Verbose outputdebug = False Debugging outputraise_sorry = False Raise sorry if problemsdry_run = False Just read in and check parameter namesrequire_nonzero = True Require non-zero values in data columns to consider reading in.ncs_parametersinvert_matrices = False Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set invert_matrices=Truechain_max_rmsd = 2 limit of rms difference between chains to be considered as copiesncs_restrict = 0 You can specify the number of NCS operators to look forforce_ncs = False You can tell find_ncs to ignore the map. This is useful if you only have FP but no phases yet...optimize_ncs = False You can tell find_ncs to optimize the NCS by making as compact a molecule as possible. no_ncs_sg_mismatch = True If one or more NCS operators is the same as a SG operator, skip the NCstry_density_after_ha = True If search for NCS using ha sites does not yield NCS correlation of at least target_ncs_cc or the number of operators is less than target_ncs_operators then try finding NCS directly from densitytarget_ncs_operators = None Number of expected NCS operators (used to decide whether to try density search)ncs_copies_max = None Max number of allowed NCS operators in density search target_ncs_cc = 0.80 Expected NCS correlationminimum_ncs_cc = 0.30 Minimum NCS correlation . If the overall NCS correlation is less than minimum_ncs_cc it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter minimum_ncs_cc_from_density is applied in finding NCS from density as well.fraction_ncs_min = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.helix_extend_range = None Range in z to extend helical operators (A)helix_trans_along_z = None Along with helix_theta, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the ...

- job_title = None Job title in PHENIX GUI, not used on command line

- input_filesncs_in_type = *None chains sites ncs_file Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS ncs_file: an ncs object file from PHENIX. ncs_in = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS objectmtz_in = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom siteslabin = "" Labin line for MTZ file with map coefficients. This is optional if find_ncs can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP map_in = None CCP4 or MRC-style map file (instead of mtz_in)

- ncs_in_type = *None chains sites ncs_file Type of ncs information. Choices are: chains: a PDB file with two or more chains that have a consistent residue-numbering system. sites: a PDB file or fractional-coordinate file with atomic positions of heavy-atoms that show NCS ncs_file: an ncs object file from PHENIX.

- `ncs_in` = None File with NCS information (PDB file with heavy-atom sites or with NCS-related chains) or NCS object
- `mtz_in` = None MTZ file with coefficients for a map that can be used to assess NCS. Required for finding NCS from heavy-atom sites
- `labin` = "" Labin line for MTZ file with map coefficients. This is optional if `find_ncs` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: LABIN FP=myFP PHIB=myPHI FOM=myFOM where myFP is your column label for FP
- `map_in` = None CCP4 or MRC-style map file (instead of `mtz_in`)
- `output_filesparams_out` = `find_ncs_params.eff` Parameters file to rerun `find_ncs`
`ncs_out` = `find_ncs.ncs_spec` Output file with NCS information. If set, then files will also be written for `resolve` and `phenix.refine` and as BIOMT records
`ncs_format` = `*ncs_spec biomt` Format of output NCS information. `ncs_spec` is standard phenix ncs format. `biomt` is BIOMT records.
`ncs_mask_file` = None If defined and `mtz_in` is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use `ncs_mask_as_mtz=False` to write map coefficients for the mask.
`ncs_mask_as_mtz` = True If True, the `ncs_mask_file` will be map coefficients. If False a CCP4-style map will be written.
`ncs_au_file` = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.
- `params_out` = `find_ncs_params.eff` Parameters file to rerun `find_ncs`
- `ncs_out` = `find_ncs.ncs_spec` Output file with NCS information. If set, then files will also be written for `resolve` and `phenix.refine` and as BIOMT records
- `ncs_format` = `*ncs_spec biomt` Format of output NCS information. `ncs_spec` is standard phenix ncs format. `biomt` is BIOMT records.
- `ncs_mask_file` = None If defined and `mtz_in` is defined, map coefficients will be written to this file that show the asymmetric unit of NCS. By default the file contains map coefficients. To write a CCP4-style map. Use `ncs_mask_as_mtz=False` to write map coefficients for the mask.
- `ncs_mask_as_mtz` = True If True, the `ncs_mask_file` will be map coefficients. If False a CCP4-style map will be written.
- `ncs_au_file` = None If defined, map coefficients showing the asymmetric unit of NCS will be written to this file.
- `crystal_inforesolution` = None high-resolution limit for map calculation
- `resolution` = None high-resolution limit for map calculation
- `directories`temp_dir = "temp_dir" Temporary work directory
output_dir = "" Output directory where files are to be written
gui_output_dir = None
- `temp_dir` = "temp_dir" Temporary work directory
- `output_dir` = "" Output directory where files are to be written
- `gui_output_dir` = None
- `control`verbose = True Verbose output
debug = False Debugging output
raise_sorry = False Raise sorry if problems
dry_run = False Just read in and check parameter names
require_nonzero = True Require non-zero values in data columns to consider reading in.
- `verbose` = True Verbose output
- `debug` = False Debugging output
- `raise_sorry` = False Raise sorry if problems

- `dry_run = False` Just read in and check parameter names
- `require_nonzero = True` Require non-zero values in data columns to consider reading in.
- `ncs_parameters`
 - `invert_matrices = False` Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set `invert_matrices=True`
 - `chain_max_rmsd = 2` limit of rms difference between chains to be considered as copies
 - `ncs_restrict = 0` You can specify the number of NCS operators to look for
 - `force_ncs = False` You can tell `find_ncs` to ignore the map. This is useful if you only have FP but no phases yet...
 - `optimize_ncs = False` You can tell `find_ncs` to optimize the NCS by making as compact a molecule as possible.
 - `no_ncs_sg_mismatch = True` If one or more NCS operators is the same as a SG operator, skip the NCS
 - `try_density_after_ha = True` If search for NCS using ha sites does not yield NCS correlation of at least `target_ncs_cc` or the number of operators is less than `target_ncs_operators` then try finding NCS directly from density
 - `target_ncs_operators = None` Number of expected NCS operators (used to decide whether to try density search)
 - `ncs_copies_max = None` Max number of allowed NCS operators in density search
 - `target_ncs_cc = 0.80` Expected NCS correlation
 - `minimum_ncs_cc = 0.30` Minimum NCS correlation . If the overall NCS correlation is less than `minimum_ncs_cc` it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter `minimum_ncs_cc_from_density` is applied in finding NCS from density as well.
 - `fraction_ncs_min = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
 - `helix_extend_range = None` Range in z to extend helical operators (A)
 - `helix_trans_along_z = None` Along with `helix_theta`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
 - `helix_theta = None` Along with `helix_trans_along_z`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
 - `n_try_ncs = 3` Number of tries to find ncs from heavy-atom sites
 - `ncs_thorough = 8` ha_ds_window = None Used to specify window of atom-atom lengths in heavy-atom search. None uses default
 - `sha_window_max = 30` Used to specify max atom-atom length in heavy-atom search. None uses default. Specify a longer distance to look more thoroughly, a shorter distance to speed up the process
 - `dist_cut_ncs = None` Thoroughness for looking for heavy-atom sites (high=more thorough)
 - `coordinate_offset = None` Origin offset to apply to `ncs_intol_r`
 - `intol_r = 0.02` tolerance in rotations for point group or helical symmetry
 - `abs_tol_t = 2` tolerance in translations (A) for point group or helical symmetry
 - `rel_tol_t = 0.05` tolerance in translations (fractional) for point group or helical symmetry
- `invert_matrices = False` Matrices normally describe how to map each operator on to the first one. To get operators for mapping the first one on the others, set `invert_matrices=True`
- `chain_max_rmsd = 2` limit of rms difference between chains to be considered as copies
- `ncs_restrict = 0` You can specify the number of NCS operators to look for
- `force_ncs = False` You can tell `find_ncs` to ignore the map. This is useful if you only have FP but no phases yet...
- `optimize_ncs = False` You can tell `find_ncs` to optimize the NCS by making as compact a molecule as possible.
- `no_ncs_sg_mismatch = True` If one or more NCS operators is the same as a SG operator, skip the NCS
- `try_density_after_ha = True` If search for NCS using ha sites does not yield NCS correlation of at least `target_ncs_cc` or the number of operators is less than `target_ncs_operators` then try finding NCS directly from density
- `target_ncs_operators = None` Number of expected NCS operators (used to decide whether to try density search)
- `ncs_copies_max = None` Max number of allowed NCS operators in density search

- `target_ncs_cc = 0.80` Expected NCS correlation
- `minimum_ncs_cc = 0.30` Minimum NCS correlation . If the overall NCS correlation is less than `minimum_ncs_cc` it will be ignored. NOTE: This is applied at the very end of NCS identification. The parameter `minimum_ncs_cc_from_density` is applied in finding NCS from density as well.
- `fraction_ncs_min = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
- `helix_extend_range = None` Range in z to extend helical operators (A)
- `helix_trans_along_z = None` Along with `helix_theta`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
- `helix_theta = None` Along with `helix_trans_along_z`, defines helical parameters. Note: matrices are for mapping position k on to identity, this may be the opposite of what you would expect.
- `n_try_ncs = 3` Number of tries to find ncs from heavy-atom sites
- `ncs_thorough = 8`
- `ha_ds_window = None` Used to specify window of atom-atom lengths in heavy-atom search. None uses defaults
- `ha_window_max = 30` Used to specify max atom-atom length in heavy-atom search. None uses defaults. Specify a longer distance to look more thoroughly, a shorter distance to speed up the process
- `dist_cut_ncs = None` Thoroughness for looking for heavy-atom sites (high=more thorough)
- `coordinate_offset = None` Origin offset to apply to `ncs_in`
- `tol_r = 0.02` tolerance in rotations for point group or helical symmetry
- `abs_tol_t = 2` tolerance in translations (A) for point group or helical symmetry
- `rel_tol_t = 0.05` tolerance in translations (fractional) for point group or helical symmetry
- `find_ncs_from_densityinput_filesmtz_in = None` MTZ file with coefficients for a map `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP `center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model `density_mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in` `density_labin = ""` Labin line for MTZ file `density_mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `output_filescenters_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers
- `log = find_ncs_from_density.log` Output log file
- `ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
- `params_out = find_ncs_from_density_params.eff` Parameters file to rerun `find_ncs_from_density`
- `ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
- `directorytemp_dir = "temp_dir"` Temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `density_searchdensity_radius = 10.` Radius for density to be cut out and compared
- `peak_separation = 15` Minimum distance between centers. Use about $1.5 * \text{density_radius}$
- `density_peaks = 20` Number of NCS-related peaks of density to output
- `delta_phi = 20` Angular spacing of search
- `rotz_only = False` Rotate around Z only
- `ncs_copies_max = None` Maximum number of NCS copies to look for
- `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first
- `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high
- `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than

`this.dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`)
`map_operators_inside_unit_cell = False` If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal
`data.find_centers`
`smoothing_radius = None` Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`.
`n_center_find = None` Target number of centers to find Smoothing radius will be varied from 2 to $2 \times \text{smoothing_radius}$ and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to
`Nonen_center_use = 20` Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS.
`try_all_centers = False` If set, try all `n_center_use` centers. Normally stop if a good solution is found.
`ok_ncs_cc = 0.6` NCS correlation good enough to terminate search
`crystal_info.resolution = 4` High-resolution limit for map calculation It is useful to cut the resolution at 3-5 Å If you use higher resolution, use a finer `delta`
`phisolvent_fraction = 0.5` solvent fraction
`control.verbose = True` Verbose output
`debug = False` Debugging output
`raise_sorry = False` Raise sorry if problems
`dry_run = False` Just read in and check parameter names
`resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: `resolve_command_list="'no_build...'`

- `input_files.mtz_in = None` MTZ file with coefficients for a map `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where myFP is your column label for FP
`center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model
`density.mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in`
`labin = ""` Labin line for MTZ file `density.mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where myFP is your column label for FP
- `mtz_in = None` MTZ file with coefficients for a map
- `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where myFP is your column label for FP
- `center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model
- `density.mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in`
- `density_labin = ""` Labin line for MTZ file `density.mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where myFP is your column label for FP
- `output_files.centers_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers
`log = find_ncs_from_density.log` Output log file
`ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
`params_out = find_ncs_from_density.params.eff` Parameters file to rerun `find_ncs_from_density`
`ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
- `centers_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers
- `log = find_ncs_from_density.log` Output log file

- `ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
- `params_out = find_ncs_from_density_params.eff` Parameters file to rerun `find_ncs_from_density`
- `ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
- `directoriestemp_dir = "temp_dir"` Temporary work directory `output_dir = ""` Output directory where files are to be written
- `temp_dir = "temp_dir"` Temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `density_searchdensity_radius = 10` Radius for density to be cut out and compared `peak_separation = 15` Minimum distance between centers. Use about $1.5 \times \text{density_radius}$ `density_peaks = 20` Number of NCS-related peaks of density to output `delta_phi = 20` Angular spacing of search `rotz_only = False` Rotate around Z only `ncs_copies_max = None` Maximum number of NCS copies to look for `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this. `dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`) `map_operators_inside_unit_cell = False` If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data.
- `density_radius = 10` Radius for density to be cut out and compared
- `peak_separation = 15` Minimum distance between centers. Use about $1.5 \times \text{density_radius}$
- `density_peaks = 20` Number of NCS-related peaks of density to output
- `delta_phi = 20` Angular spacing of search
- `rotz_only = False` Rotate around Z only
- `ncs_copies_max = None` Maximum number of NCS copies to look for
- `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first
- `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high
- `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
- `dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`)
- `map_operators_inside_unit_cell = False` If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data.
- `find_centerssmoothing_radius = None` Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`. `n_center_find = None` Target number of centers to find Smoothing radius will be varied from 2 to $2 \times \text{smoothing_radius}$ and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to `None` `n_center_use = 20` Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS. `try_all_centers = False` If set, try all `n_center_use` centers.

Normally stop if a good solution is found.
ok_ncs_cc = 0.6 NCS correlation good enough to terminate search

- smoothing_radius = None Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as density_radius.

- n_center_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2*smoothing_radius and the value giving the number of peaks closest to n_center_find will be used. Normally leave this set to None

- n_center_use = 20 Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS.

- try_all_centers = False If set, try all n_center_use centers. Normally stop if a good solution is found.

- ok_ncs_cc = 0.6 NCS correlation good enough to terminate search

- crystal_inforesolution = 4. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 A If you use higher resolution, use a finer delta_phi

- resolution = 4. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 A If you use higher resolution, use a finer delta_phi

- solvent_fraction = 0.5 solvent fraction

- controlverbose = True Verbose output
debug = False Debugging output
raise_sorry = False Raise sorry if problems
dry_run = False Just read in and check parameter names
resolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this:
resolve_command_list="'no_build' 'b_overall 23' ";

- verbose = True Verbose output

- debug = False Debugging output

- raise_sorry = False Raise sorry if problems

- dry_run = False Just read in and check parameter names

- resolve_command_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (") and each individual command in single quotes (') like this: resolve_command_list="'no_build' 'b_overall 23' ";

Finding and analyzing difference map peaks

find_peaks_holes.html

Contents

- Overview
- Configuration
- Output
- List of all available keywords

Overview

phenix.find_peaks_holes is a utility for identifying points of interest in difference maps, typically at the end of refinement. It will calculate an mFo-DFc map and locate peaks and holes, plus anomalous map peaks (if anomalous data are provided) and flagging of waters with suspiciously high mFo-DFc values. This information may be used to improve the model, especially in the addition of solvent and small ions.

Configuration

The program requires only a PDB file and data (including R-free flags). The default options should be sufficient for most cases, although for very large structures or especially noisy maps, you may find it helpful to increase the sigma cutoff.

By default, peaks which overlap with the input atoms (within a cutoff distance) will not be included. If you want to find all peaks, set the minimum peak-model distance to zero. Additional options are available by clicking the "Peak search" button.

On the command line, the minimal input is simply the paths to your model and data files (column labels may need to be specified if the choice is ambiguous). Running the program without arguments will print out a summary of available options, which are also listed below.

Output

A summary tab will appear at the end of the run with peak counts, plus output files and buttons to view them. By default the program will write a PDB file containing up to five "chains", all with atoms labeled UNK:

A: mFo-DFc peaks (positive). B: mFo-DFc holes (negative) C: anomalous map peaks D: waters in mFo-DFc peaks E: waters in anomalous peaks

- A: mFo-DFc peaks (positive).
- B: mFo-DFc holes (negative)
- C: anomalous map peaks
- D: waters in mFo-DFc peaks
- E: waters in anomalous peaks

The program will also write the 2mFo-DFc, mFo-DFc, and anomalous (if available) maps. Additional tabs will display each type of result in a list; clicking these items will recenter the view in Coot and/or PyMOL on the selected point.

List of all available keywords

map_cutoff = 3.0 anom_map_cutoff = 3.0 include_peaks_near_model = False By default, the program will only display peaks that map to points outside of the current model, ignoring those that overlap with atoms. Setting this option to True is equivalent to specifying a distance cutoff of zero for the filtering step. wavelength = None Optional parameter, if defined this will cause all atoms to be treated as anomalous scatterers using the standard Sasaki table to obtain theoretical fp and fpp values. Only really useful if the Phaser LLG map is being used for the anomalous map. filter_peaks_by_2fofc = None If this is set, peaks outside 2mFo-DFc density at the cutoff will be discarded. (This does not apply to the analysis of solvent atoms.) Holes will not be changed. use_phaser_if_available = True If True, and Phaser is installed and configured, an anomalous LLG map will be used in place of the simple anomalous difference map. The wavelength should be specified for this to be maximally useful. write_pdb = True write_maps = True output_file_prefix = peaks_holes job_title = None Job title in PHENIX GUI, not used on command line. input_sequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = False skip_twin_detection = False twin_law = Auto Enter twin law if known. xray_dataScope of X-ray data and free-R flags. file_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers, extreme wilson outliers, and beamstop shadow outliers. french_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilson_max_bins = 60 Maximum number of resolution bins. min_bin_size = 40 Minimum number of reflections per bin. free_flags. file_name = None This is normally the same as the file containing Fobs and is usually selected automatically. label = None required = True Specify if free-r flags are must be present (or else generated). test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing! ignore_r_free_flags = False Use all reflections in refinement (work and test). disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. generate = False Generate R-free flags (if not available in input files). fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. n_shells = 20 pdbfile_name = None Model file(s) name (PDB). monomersfile_name = None Monomer file(s) name (CIF). mapsmap_file_name = None A CCP4-formatted map. d_min = None Resolution of map. map_coefficients_file_name = None MTZ file containing map. map_coefficients_label = None Data label for complex map coefficients in MTZ. file. find_peaks. use_sigma_scaled_maps = True Default is sigma scaled map, map in absolute scale is used otherwise. grid_step = 0.6 Grid step for map sampling. max_number_of_peaks = None map_next_to_model. min_model_peak_dist = 1.8 max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8 use_hydrogens = False peak_search. peak_search_level = 1 max_peaks = 0 interpolate = True min_distance_sym_equiv = None general_positions_only = False min_cross_distance = 1.8 min_cubicle_edge = 5

- map_cutoff = 3.0
- anom_map_cutoff = 3.0
- include_peaks_near_model = False By default, the program will only display peaks that map to points outside of the current model, ignoring those that overlap with atoms. Setting this option to True is equivalent to specifying a distance cutoff of zero for the filtering step.
- wavelength = None Optional parameter, if defined this will cause all atoms to be treated as anomalous scatterers using the standard Sasaki table to obtain theoretical fp and fpp values. Only really useful if the Phaser LLG map is being used for the anomalous map.

- `filter_peaks_by_2focf` = None If this is set, peaks outside 2mFo-DFc density at the cutoff will be discarded. (This does not apply to the analysis of solvent atoms.) Holes will not be changed.
- `use Phaser if available` = True If True, and Phaser is installed and configured, an anomalous LLG map will be used in place of the simple anomalous difference map. The wavelength should be specified for this to be maximally useful.
- `write_pdb` = True
- `write_maps` = True
- `output_file_prefix` = peaks_holes
- `job_title` = None Job title in PHENIX GUI, not used on command line
- `inputsequence` = None `scattering_table` = wk1995 it1992 *n_gaussian neutron electron wavelength = None `energy` = None `use_experimental_phases` = False `skip_twin_detection` = False `twin_law` = Auto Enter twin law if known. `xray_dataScope` of X-ray data and free-R flags `file_name` = None `labels` = None `high_resolution` = None `low_resolution` = None `twin_law` = None `outliers_rejection` = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers `french_wilson_scale` = True `sigma_fobs_rejection_criterion` = None `sigma_jobs_rejection_criterion` = None `ignore_all_zeros` = True `force_anomalous_flag_to_be_equal_to` = None `convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold` = 0.5 `french_wilsonmax_bins` = 60 Maximum number of resolution bins `min_bin_size` = 40 Minimum number of reflections per `binr_free_flagsfile_name` = None This is normally the same as the file containing Fobs and is usually selected automatically. `label` = None `required` = True Specify if free-r flags are must be present (or else generated) `test_flag_value` = None This value is usually selected automatically - do not change unless you really know what you're doing! `ignore_r_free_flags` = False Use all reflections in refinement (work and test) `disable_suitability_test` = False `ignore_pdb_hexdigest` = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. `generate` = False Generate R-free flags (if not available in input files) `fraction` = 0.1 `max_free` = 2000 `lattice_symmetry_max_delta` = 5 `use_lattice_symmetry` = True `use_dataman_shells` = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. `n_shells` = 20 `pdbfile_name` = None Model file(s) name (PDB) `monomersfile_name` = None Monomer file(s) name (CIF) `mapsmap_file_name` = None A CCP4-formatted map `d_min` = None Resolution of map `map_coefficients_file_name` = None MTZ file containing map `map_coefficients_label` = None Data label for complex map coefficients in MTZ file
- `sequence` = None
- `scattering_table` = wk1995 it1992 *n_gaussian neutron electron
- `wavelength` = None
- `energy` = None
- `use_experimental_phases` = False
- `skip_twin_detection` = False
- `twin_law` = Auto Enter twin law if known.
- `xray_dataScope` of X-ray data and free-R flags `file_name` = None `labels` = None `high_resolution` = None `low_resolution` = None `twin_law` = None `outliers_rejection` = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers `french_wilson_scale` = True `sigma_fobs_rejection_criterion` = None `sigma_jobs_rejection_criterion` = None `ignore_all_zeros` = True `force_anomalous_flag_to_be_equal_to` = None `convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold` = 0.5 `french_wilsonmax_bins` = 60 Maximum number of resolution bins `min_bin_size` = 40 Minimum number of reflections per

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- file_name = None
- labels = None
- high_resolution = None
- low_resolution = None
- twin_law = None
- outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- french_wilson_scale = True
- sigma_fobs_rejection_criterion = None
- sigma_iobs_rejection_criterion = None
- ignore_all_zeros = True
- force_anomalous_flag_to_be_equal_to = None
- convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5
- french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per bin
- max_bins = 60 Maximum number of resolution bins
- min_bin_size = 40 Minimum number of reflections per bin
- r_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20
- file_name = None This is normally the same as the file containing Fobs and is usually selected automatically.
- label = None
- required = True Specify if free-r flags are must be present (or else generated)
- test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!
- ignore_r_free_flags = False Use all reflections in refinement (work and test)

- `disable_suitability_test = False`
- `ignore_pdb_hexdigest = False` If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- `generate = False` Generate R-free flags (if not available in input files)
- `fraction = 0.1`
- `max_free = 2000`
- `lattice_symmetry_max_delta = 5`
- `use_lattice_symmetry = True`
- `use_dataman_shells = False` Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- `n_shells = 20`
- `pdbfile_name = None` Model file(s) name (PDB)
- `file_name = None` Model file(s) name (PDB)
- `monomersfile_name = None` Monomer file(s) name (CIF)
- `file_name = None` Monomer file(s) name (CIF)
- `mapsmap_file_name = None` A CCP4-formatted map
- `mapmap_min = None` Resolution of map
- `mapmap_coefficients_file_name = None` MTZ file containing map
- `mapmap_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `map_file_name = None` A CCP4-formatted map
- `d_min = None` Resolution of map
- `map_coefficients_file_name = None` MTZ file containing map
- `map_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `find_peaksuse_sigma_scaled_maps = True` Default is sigma scaled map, map in absolute scale is used otherwise.
- `grid_step = 0.6` Grid step for map sampling
- `max_number_of_peaks = None`
- `map_next_to_modelmin_model_peak_dist = 1.8` max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8
- `use_hydrogens = False` peak_searchpeak_search_level = 1 max_peaks = 0 interpolate = True
- `min_distance_sym_equiv = None` general_positions_only = False min_cross_distance = 1.8
- `min_cubicle_edge = 5`
- `use_sigma_scaled_maps = True` Default is sigma scaled map, map in absolute scale is used otherwise.
- `grid_step = 0.6` Grid step for map sampling
- `max_number_of_peaks = None`
- `map_next_to_modelmin_model_peak_dist = 1.8` max_model_peak_dist = 3.2 min_peak_peak_dist = 1.8
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- `use_hydrogens = False`
- `peak_searchpeak_search_level = 1` max_peaks = 0 interpolate = True min_distance_sym_equiv = None
- `general_positions_only = False` min_cross_distance = 1.8 min_cubicle_edge = 5
- `peak_search_level = 1`

- max_peaks = 0
- interpolate = True
- min_distance_sym_equiv = None
- general_positions_only = False
- min_cross_distance = 1.8
- min_cubicle_edge = 5